Gandaki University

Bachelor of Information Technology

Detailed Course Bulletine

October 29, 2025

Cirriculum Development Committee
Gandaki University

Part I
General Introduction

Gandaki University

Gandaki University (G.U) was established in 2019 with the objective of producing world class skilled human resources required for building a prosperous province and making all its citizen happy by 2043. The university nurtures students with knowledge and skills necessary to face the challenges of the 21st century. The university aims to establish itself as a centre of excellence for academia and research, thereby helping the nation by producing highly skilled human resources in various domains.

Mission

The mission of G.U is to develop highly qualified and committed technical professionals and development ambassadors for Nepal and South Asia. It will offer student-centric and market-driven quality education for building prosperous nation and the region. Inspired and guided by above clear vision and mission, the dedicated students, faculties, research scientists and staff members of G.U will steer the University towards the path of becoming:

- An innovative and pedagogic center that produces high-quality, market-driven technical professionals and development leaders.
- A technology generation center for transformation of economy and society.
- An exemplary academic center for building partnership with governments, industries, private sectors, other universities and research institutes for research, innovation and new technology generation.
- A center of entrepreneur production with lifelong education and lasting impact among its graduates.

GU is committed to excellence, service, collaboration, engagement and impact in everything it does.

Vision

Gandaki University aspires – to become a center of excellence in research, teaching, innovation and new technology generation in higher education in Nepal and South Asia Region. It will be distinguished by its commitment and actions as a center of excellence and be admired for its contribution to local, national, and regional needs and making a transformative and sustainable impact on the society.

Undergraduate Programs in Gandaki University

Gandaki University is currently running 5 academic programs. They are:

1. Bachelor in Sports Management (BSM)
2. Bachelor of Information Technology (BIT)
3. Bachelor of Pharmacy (B-Pharm)
4. Bachelor of Business Administration (BBA)
5. Bachelor of Arts, Bachelor of Laws (BALLB)

Research Center

Research Center, Invention and Innovation Center (RC-II) was established in 2021 under the Gandaki University Act-2019. The main aim of establishing RC-II was to support Gandaki University's vision of academic excellence through a combinational approach of teaching and research to meet the growing demands of global context. Research is an integral part of academia that fosters scientific knowledge and awareness among the university graduates and faculty members. Research, invention and innovation are prime requirements for the development of a society, nation and the whole world. RC-II thus promotes research, invention and innovation at Gandaki University through facilitating funds and grants for the researchers (students and faculty members) and promoting collaborations with different national and international research centers, institutions, and agencies. Furthermore, RC-II promotes research-related infrastructure development by establishing a core research facility within the university.

International Relation and Development Affairs

International Relations and Development Affairs (IRDA)supports and facilitates the international mission of Gandaki University. It provides services for

faculty and students and offers international educational and cultural experiences for the University and community, and is a major contributor to the globalization process of the University and growing its reputation as a major international educational and research center.
IRDA focuses on

- Global Engagement
- International Events
- International Researchers/Scholars
- International Partnerships
- Collaborations with national and international institutions
- MoU with International, National Universities and Institutions
- International Research Project

Part II
Bachelor of Information Technology

BIT Program

Bachelor of Information Technology (BIT) course is one of the most demanding and popular degrees worldwide. The BIT program has been developed by the Gandaki University in order to provide an excellent IT education platform for students as well as to avail ample skilled graduates for the professional business, industries and research. The BIT course is designed with the objective to hone the student's ability with world class knowledge in advance theoretical and real practical aspects.

Key Learning Outcomes

Bachelor of Information Technology degree program is aimed at producing highly skilled quality IT human resource to deliver information technology solutions required by the current digital society. For this purpose, students will be honed with intensive knowledge and latest practical skills which will enable them to globally compete and develop IT solutions. The students will be able to:

1. Coordinate, plan, and lead IT-related activities in an organization. They analyze, evaluate and estimate the IT needs and deploy IT systems to fulfil them.
2. Define IT infrastructure strategy and processes.
3. Illustrate a substantial understanding of concepts in key areas of information system, computer system, computer applications and its management.
4. Understand, configure, integrate, use, demonstrate and develop computer and IT applications such as web, mobile, network, security, and multimedia.
5. Illustrate thorough practical skills to identify and troubleshoot computer and IT related problems in an organization.

Admission Process and Eligibility

Interested candidates will have to adhere to the G.U Admission process for enrolling into the BIT program. Applicants who have passed Intermediate level with minimum grade 2 (out of 4) or equivalent from any institutions recognized by G.U are eligible for application, provided that they have undertaken a 100 marks course in Mathematics and English during their intermediate study.

Career Prospective

BIT graduates will have job opportunities in various fields in computer science and information technology domain. Careers at different national and international, government and non-government organizations, software companies, Telecommunication and computer networking companies, private and public organizations at different roles are available for our graduates. Further, this course enables graduates to work as Freelancer or start their own startups and ventures. The graduates of BIT program will have ample opportunities to work as IT Officer, IT Project Managers, Software Developer, Mobile App Developer, Web Developer, System Analyst, Quality Analyst, System Administrator, Database and Network Administrators and so on. Besides, graduates will have enough opportunities to work in Academia and also for higher studies across the globe.

Education System

The academic session comprises two semesters per year and a total of eight semesters in four years. The fall semester generally starts around September-October and the spring semester generally starts around March-April. All the courses are designed to be completed in a normal duration of four years (i.e., eight semesters). All the course requirements must be accomplished within a maximum duration of eight years.
The university requires every student to maintain a minimum of 80% attendance during a semester. The university will follow continuous evaluation as well as final end semester evaluation. The student evaluation will generally be based on various aspects such as assignments, assessments, practical, projects, exams, etc. It may vary for some subjects. The final grading for a subject will be done based on the students' consolidated performance during the entire semester and the final examinations.

Course Structure of Bachelor of Information Technology (BIT)

BIT is a four-year full time study program spread over eight semesters, consisting of 47 courses with the total academic load of 138 credit hours.

Year 1, Sem 1			Credit
1	ELX111	Basic Electronics	3
2	BSM111	Mathematics -I	3
3	CIT111	Computer Programming, I (C)	3
4	CIT 112	Basics of IT	3
5	CIT113	Workshop: Problem Solving and Logic	1
6	BCT111	Business Communication Techniques	12
			15

Year 1, Sem 2			Credit
1	CIT121	Discrete Mathematics	3
2	CIT122	Computer Programming II (Java)	3
3	BSM121	Mathematics -II	3
4	ELX121	Digital Logic	3
5	CIT123	Web Technology I	3
6	CIT124	Project I	2
			17

Year 2, Sem 3			Credit
1	CIT 211	Web Technology II	3
2	ELX 211	Microprocessor and Computer Architecture	4
3	CIT 212	Software Engineering	3
4	CIT 213	Database Management System	3
5	CIT 214	Data structures and Algorithms	3
6	BCT 211	Principles of Organization and Management	3
			19

Year 2, Sem 4			Credit
1	CIT221	Operating Systems	3
2	CIT222	Management Information System	3
3	CIT223	Data Communication and Computer Networks	4
4	BSM221	Fundamentals of Probability and Statistics	3
5	CIT224	Computer Graphics Technology	3
6	CIT225	Project II	2
			18

Year 3, Sem 5			**Credit**
1	CIT311	Mobile Application Development	3
2	BCT311	Economics	3
3	CIT312	Object oriented Analysis and Design with UML	3
4	BSM311	Numerical Methods	3
5	BCT312	Research Methodology	3
6	BCT313	Technical Proposal Writing	1
			16
Year 3, Sem 6			**Credit**
1	BCT321	IT Project Management	3
2	CIT321	Human Computer Interface and UI Design	3
3	CIT322	Digital Forensic Security Technologies	3
4	CIT323	Artificial Intelligence	3
5	BCT322	Financial Accounting	3
6	CIT324	Project III	3
			18
Year 4, Sem 7			**Credit**
1	CIT411	Data Mining and Warehousing	3
2	CIT412	Software Development and Operations (DevOps)	3
3	BCT411	Technology Entrepreneurship	3
4	CIT413	Cloud Computing	3
5	CIT414	Wireless Communication Systems	3
6	ELE	Elective 1	3
			18
Year 4, Sem 8			**Credit**
1	CIT421	Big Data Technologies	3
2	BCT421	Society, IT and Law	3
3	CIT422	Internship	2
4	ELE	Elective 2	3
5	CIT423	Project IV	6
			17
		Total Credit	138

Electives

Cluster 1:	Good Governance E-Governance E-Business and E-Commerce Corporate Governance
Cluster 2:	Machine Learning and Data Science Data Science Social Media Mining Machine Learning
Cluster 3:	Network and Internet Mobile Ad Hoc Network IPV6 Networking IOT and Smart City
Cluster 4:	Software Development Visual Programming Agile Software Development Software Quality Assurance
Cluster 5:	Geographic Information System Remote Sensing Geographic Information System Geo Spatial Data Processing and Management

Part III
Micro- Syllabus

Basic Electronics

Program: Bachelor of Information Technology (BIT)
Course Code : ELX111
Year: 1
Semester: I
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of basic principle of Electronics and Com- munication along with its implementation.

Course Details:

1. **Introduction to Circuit Theory** **7hrs**
   Introduction of Voltage, Current and Power, Passive and Active components, Series and parallel circuits, Kirchhoff's Voltage and Current Law, Superposition Theorem, Thevenin's Theorem, Norton's Theorem.
2. **Semiconductor Diode** **10hrs**
   Introduction to Insulator, Semi-conductor and Conductor according to Band Theory, P-Type and N- Type Semiconductor, Semiconductor diode characteristics (Forward and Reverse operation), Modelling the semiconductor Diode, Application of Diode (Half wave and Full Wave Rectifier, Clipper and Clam- per circuit), Types of Diode (Light emitting Diode, Photo Diode, Zener diode) Zener Diode as Voltage Regulator.
3. **Transistor** **7hrs**
   Construction and Working of NPN and PNP transistor, BJT configuration (CB, CE, CC), BJT switch and Logic Circuit, Transistor Biasing (Base Biasing, Emitter biasing, Collector Biasing and Voltage Divider Biasing).
4. **Operational Amplifier** **5hrs**
   Virtual Ground concepts, inverting and non-inverting amplifier, Application of Op-amp (Summer, Subtrator, Integrator, Differentiator).
5. **Introduction to Analog Communication System** **6hrs**
   Analog communication System, Mode of communication (Broadcasting and communication), Wired Vs. Wireless communication, Co-axial and Optical communication, Satellite Communication, Transmission impairment, Needs of Modulation, Types of Modulation.
6. **Introduction to Digital Communication System** **5hrs**

Element of Digital Communication System, Mode of Communications (Line of Sight and Non-Line of sight), Communication channel (Noise and Interference), Shannon Channel Capacity Theorem.

7. **Application of Electronics System** **5hrs**
   Instrumentation System, Transducer, Digital Multimeter, Oscilloscope, Data Logger, Regulated Power Supply, Introduction to Biomedical Equipment's (ECG, Pacemaker, CT scan, Ultra sound Device)

Laboratory Work:

1. Familiarization with basic Electronics Components.
2. Implementation of KCL, KVL, Thevenin's and Norton's Theorem.
3. Characteristics of Semiconductor Diode.
4. Realization of Half and Full Wave Rectifier.
5. Realization of clipper and Clamper Circuit.
6. Implementing Zener Diode as voltage regulator.
7. Transistor configuration (CB, CE)
8. Transistor as Switch
9. Implementing Application of Operational Amplifier (inverting and non-inverting configuration, sum- mer, Subtrator, Integrator, Differentiator)

References:

1. Thomas L Floyd "Electronics Devices" 8th Edition, Pearson Education, Inc.
2. Robert Boylestad and Louis Nashelsky, "Electronic Devices and Circuit Theory" PHI; 4th Edition. 1987
3. Simon Haykin and Michael, "An Introduction to Analog and Digital Communications, 2nd Edition
4. Leslie Cromwell, et Al, "Biomedical Instrumentation and Measurements", Prentice Hall, India.

Mathematics I

Program: Bachelor of Information Technology (BIT)
Course Code : BSM111
Year: 1
Semester: I
Credit Hours : 3
Nature: TH

Course Objectives:

The purpose of the course is to provide fundamental knowledge of basic Mathematical concepts.

Course Details:

1. **Derivatives** **9hrs**
   - 1.1 Limit of a function at a particular point and at infinity
   - 1.2 Properties of limits (without proof)
   - 1.3 Continuity
   - 1.4 Types of discontinuity
   - 1.5 Average rate of change
   - 1.6 Derivative as a slope of tangent to curves
   - 1.7 Method of differentiation (Power rule, Sum rule, Product rule, Quoitent rule and chain rule
   - 1.8 Derivative of implict and parametric functions , Higher order derivatives
2. **Applications of Derivatives** **8hrs**
   - 2.1 Maximum and Minimum of a function of one variable
   - 2.2 Indeterminant forms
   - 2.3 Rolle's theorem
   - 2.4 Mean value theorem
   - 2.5 Taylor's and Maclaurin's Series (without Proof)
3. **Integration and its application** **12hrs**
   - 3.1 Indefinite Integrals and fundamental formula
   - 3.2 Techniques of integration
   - 3.3 Definite integrals
   - 3.4 Improper integrals
   - 3.5 Quadrature

3.6 Rectification
3.7 Volume and Surface integral

4. **Matrices and Determinants** **9hrs**

4.1 Introduction, Types of matrices and Transpose of matrix
4.2 Determinant of a matrices
4.3 Minors and cofactors of a matrix
4.4 Properties of Determinants
4.5 Singular and non –singular matrix
4.6 Adjoint and inverse of matrix
4.7 Solution of system of linear equations having unique solution upto three variables (Cramer's rule, Inverse matrix method and Gaussian elimination method)
4.8 linear transformations
4.9 Rank of matrices

5. **Vectors** **7hrs**

5.1 Introduction
5.2 Types of Vectors
5.3 Vector addittion and scalar multiplication
5.4 Proudct of two vectors
5.5 Scalar proudct of three vectors
5.6 Vector proudct of three vectors
5.7 Collinear and Coplanar vectors
5.8 Vector space
5.9 linear dependence and independence vectors

References:

1. Thomas, G.B. Jr, and Finney, R.L. Calculus and Analytical Geometry. New Delhi : Narosa Publishing House.
2. Berresford, G.C. Calculus with Finite Mathematics. New York : Andrew M. Rockett
3. Dass B.C. and Mukherjee, B.N. (1998) Integral Calculus (48th edition)
4. S.P. Shrestha, H.D. Chaudhary and P.R.Pokharel A Text Book on Engineering Mathematics, Vidyarthi Prakashan (P) Ltd. Kathmandu.

Computer Programming,I (C)

Program: Bachelor of Information Technology (BIT)
Course Code : CIT111
Year: 1
Semester: I
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of computer languages, programming, and software. This course focuses on enhancing the computer programming skills of the students.

Course Details:

1. **Introduction** **3hrs**
   History of programming, hardware, software, Software features & trends, System Software, application Software, Proprietary vs. open source Software, Types of Programming Languages and their generations, Programming Paradigm
2. **Problem Solving** **4hrs**
   Problem analysis, flowchart and algorithm , coding, compiling, execution, debugging, testing, docu- mentation History of C, Structure of C Program.
3. **Elements of C** **5hrs**
   Tokens, Character set, Operators, Keywords, Data types, variables, constants & literals, Identifiers, Escape Sequences, preprocessor directives
4. **Control Statements** **4hrs**
   Looping (for, while, do), branching, decision and conditional statements (if, switch, break, goto, exit, continue)
5. **Input and Output** **2hrs**
   Types, of I/O, I/O operations, Formatted I/O
6. **Functions** **5hrs**
   Library functions and user defined functions, use of functions, recursion Function declaration, defini- tion, arguments, call, return , visibility and scope, Passing by value, Passing by address
7. **Array and Strings** **5hrs**
   Declaration of Single and Multi dimensional array, Sorting arrays, String, string manipulation and string functions
8. **Structure and Unions** **6hrs**

Introduction, structure, union, structure vs. array, structure vs. union, defining and initializing, array of structure, nested structure, function and structure, self referential structure

9. **Pointers** **7hrs**
   Declaration, initialization, pointer to pointer, pointer arithmetic, pointer and array / string / functions/ structure, dynamic memory allocation
10. **File handling** **4hrs**
    Concept and definition of File, file operations, file modes and operation, random file access

Laboratory Work:

1. Programming examples for each chapters (from chapter 3 to 10) A minimum of following labs must be completed:
   a. Introduction to C Programming – basic structure of C program.
   b. To illustrate the use of Input, Output , Variable and Constants,etc.
   c. To illustrate the different operators-Arithmetic, Relational and Logical Operators in C
   d. To illustration of Control Structures (decision making) and processor directives.
   e. To illustration of Control Structures (looping)
   f. To illustrate the function programming- simple and recursive functions
   g. To illustrate the concept of Pointer and String
   h. To illustrate the function programming- Call by value vs. call by reference
   i. To illustrate the concept of Structure
   j. To illustrate the different file operations
2. Use of open source platforms preferable
3. Course Project with a maximum 4 students in a group should submit a group project before the final exam.

References:

1. E. Balagurusami, 2019, “Programming in ANSI C”
2. Dennis Ritchie & Brian W. Kernighan, 1988, “The C Programming Language” , Prentice Hall
3. Byran Gottfried, 2010“Programming with C”
4. Yashavant Kanetkar, 2016, “Let Us C”, BPB
5. Noel Kalicharan, C by Example, Cambridge University Press
6. Zed A. Shaw , Learn C the Hard Way, Pearson, 9789353065720

Basics of IT

Program: Bachelor of Information Technology (BIT)
Course Code : CIT112
Year: 1
Semester: I
Credit Hours : (3)
Nature: TH

Course Objectives:

The main objective of this course is to provide the student with an overall knowledge in the broad arena of IT, its historical background, its growth and development and its role in business and organizational context.

Course Details:

1. **Introduction to Information Technology** **4hrs**
   Information Technology in Business Environment. Elements of Information System and its Archi- tecture. Types of Information System. Role of Information Technology in Organization. Managing Information Technology in Organization
2. **Computer Arithmetic** **6hrs**
   Introduction to number system, Binary system, Measurement of the size of computer memories, Ex- ample use of binary, the hexadecimal system, use of the hexadecimal system, Converting from one number system to another. Computer Codes, Rules and Laws of Boolean algebra, Basic Gates (NOT, AND & OR)
3. **Information Technology Components** **12hrs**
   Hardware Components: CPU. Memory Elements. Types of Computer and Input - Output Com- ponents; Hard Disk Construction, IDE drive standard and features, Troubleshooting, DVD, Blue-Ray disc, Flash Memory, Software Components: System Software. Application Software. Programming Tools (Software) and Enterprise Software.
   Data and Information: Data Organization and Access. File Systems, Database Systems. DBMS. Data Models. Data Warehouse: Introduction to Big Data. Telecommunication and Data Networks: Telecommunication Systems. Data Networks. Network Software. Data Processing Strategies Hill Net- work Environment, Applications of Telecommunication and Computer Network; Network Types, OSI Reference Model, Transmission Control Protocol/Internet protocol Model, Internet

Internets and Intranets: Basics of Internet. its Evolution and its Operations. Services on Internet World Wide Web Future of Internet. Intranets, Introduction to IoT. Internet Security, Uses of Inter- net, Cloud System, Cloud Technologies, Cloud Architecture, Cloud Infrastructure, Cloud Deployment Models

4. **Computer Security** **4hrs**
   Introduction to Computer Security, Basic security threats, Virus, Antivirus, Security best practices, Cookies, Loss of data and data corruption, Firewalls and proxy servers, Security protocols, Introduction to Encryption, Applications, Computer ethics, Free software, freeware and shareware
5. **Applications of Information Technology and E-Commerce** **15hrs**
   Inter-Organizational, Information System, Global Information System, IT Application in Organiza- tional Context, Extranets. Implementation of Inter-Organizational Information System. Role of IT in Business Functions: Transaction Processing. Accounting. Finance. Marketing. Sales, Production, Decision Support. Human Resource Management, Integrated Information System and Enterprise Re- source Planning-. Data, Knowledge and Decision Support Systems. Intelligent Systems in Business: AI and Intelligent Systems. Expert Systems. Intelligent Agents: Fundamentals of E-Commerce. B- to-B and B-to-C Applications, Supporting Elements of E-Commerce, Consumer. Market Research. Customer Support. Infrastructure. Payments and Other Supports
6. **Group Case Study and Report Presentation: Computer Based Information System** **4hrs**
   Computer Based Information System, Students should prepare at least one case study examining any business house and present these to the general class taking into consideration the following outline:

   - Workplace Productivity:
     A survey of techniques for improving the productivity of practices and procedures in the workplace using IT systems.
   - Software and Hardware Concepts:
     A survey of computer systems. Emphasis is on the use of hardware architecture, system software, branded, Antivirus, application software and pricing. Study also covers the conceptual design of hardware and software within that organization.
   - Computer Network and Internet:
     The idea is to make a sketch of the existing computer network model of the organization. The work should also clearly bring out the internet-based model and services used in the organization. (speed, distribution and control mechanisms). IP address schemes and how each device is configured. Study of security measures like firewall and other mechanisms. Produce a report and make and effective presentation for the above.

References:

1. Turban, E., Rainer, R. K. and Potter, Introduction to Information Technology. John Wiley and Sons. Inc., 2000, ISBN: 9971-51-321-8
2. Ramesh Behl, James A. O'Brien, George Marakas, Management Information Systems, 11th Edition, ISBN-10 : 9353164656

Workshop: Problem Solving and Logic

Program: Bachelor of Information Technology (BIT)
Course Code : CIT113
Year: 1
Semester: I
Credit Hours : 1
Nature: PR

Course Objectives:

The main objective of this course is to provide the students with a basic knowledge of understanding, the theory and practice behind computer programming and logic building. The students will be provided with enough examples from real world that would be transformed into computer programs with the help of flowcharts, pseudocode and algorithms. A complete cycle of program design from inception to dry run of the code will be taught.

Course Details:

Problem-solving and design: Introduction to problems in computer system, Pseudocode and flowcharts, Standard flowchart symbols, Assignment, Identification of Variables and constants, Conditional statements, Loop structures, Input and output statements, Standard actions, Examples of algorithms in pseudocode, Algorithms, Test data, Validation and verification, Using trace tables, Identifying and correcting errors, Producing algorithms, Programming concepts (Programming, Declaration and use of variables and constants, Basic data types, How to make your program work)

References:

1. Cambridge IGCSE Computer Science, David Watson and Helen Williams 2014, ISBN-13: 978-1471809309, ISBN-10: 1471809307
2. Programming Logic and Design, 5th Edition, Tony Gaddis, Haywood Community College 2019, Pearson, ISBN-13: 9780133985078

Business Communication Technique

Program: Bachelor of Information Technology (BIT)
Course Code : BCT111
Year: 1
Semester: I
Credit Hours : 2
Nature: TH

Course Objectives:

- To comprehend the lecture or written text and take notes
- To help the students summarize the text
- To help them gain technical knowledge
- To develop communicative skills

Course Details:

1. **Grammar** **3hrs**
   - 1.1 Fundamentals of sentences
   - 1.2 Rank in English: sentence/clause, phrase, word and morpheme
   - 1.3 Components of sentence
   - 1.4 Types of sentences (simple, compound and complex)
2. **Sound System** **3hrs**
   - 2.1 English consonant sounds, monophthongs and diphthongs
   - 2.2 Syllable, stress and intonation
3. **Language Variety** **3hrs**
   - 3.1 Geographical variety: British and American
   - 3.2 Other variety: formal, polite and familiar
4. **Oral communication skills** **2hrs**
   - 4.1 Soft skills
   - 4.2 Etiquettes
5. **Technical Talk: Meaning, procedure, presentation** **3hrs**
   - 5.1 Topics for technical talk: mobile, television, computer, urbanization, banking, promotion of business
6. **Skills in Technical Writing** **8hrs**

6.1 Memo: meaning, types and formats
6.2 Emailing: meaning and usefulness
6.3 Calling a meeting
6.4 Notifications
6.5 Agenda of a meeting: meaning and preparation
6.6 Writing minutes
6.7 Business letters: placing an order, apology, explanation, complaints, appreciation, thank you
6.8 Application for a job
6.9 CV: meaning, types and preparation of CVs

7. **Textual Reading Skills** **10hrs**

7.1 Reading skills for recommended texts
7.2 Reading skills for comprehension passages
7.3 Reading skills for recommended texts
7.4 Recommended texts for reading:
- How much land does a man need (Leo Tolstoy),
- Grief (Anton Chekov),
- Marriage proposal (G.K. Chesterton),
- All the world a stage (William Shakespeare),
- First speech as a prime minister (Winston Churchill),
- Knowledge and wisdom (Bertrand Russell).

8. **Note Taking** **2hrs**

8.1 Note taking from lectures
8.2 Taking notes from written sources
8.3 Summary writing from the text

9. **Tutorial Work**

9.1 Report writing: Meaning and format
9.2 Proposal writing: Meaning and format
9.3 Seminar: Meaning and conduction
9.4 Interview: confidence building and facing an interview

References:

1. Rutherfoord, A.J., Basic Communication Skills for Technology, Pearson Education Asia
2. Khanal Arjun, Communication Skills in English, Sukunda Pustak Bhawan,2010
3. Anne Eisenberg, Effective Technical Communication Mc-Graw Hill 1982.

Discrete Mathematics

Program: Bachelor of Information Technology (BIT)
Course Code : CIT121
Year: 1
Semester: II
Credit Hours : 3
Nature: TH

Course Objectives:

The purpose of the course is to provide fundamental knowledge of discrete structure. This course aims to introduce basic discrete structures, Logic, Reasoning, Algorithms, Recurrence Relation, Graph theory and theory of Automata. Also, it intends to provide fundamental knowledge in discrete Mathematics, Finite state Automata and graph in an Algorithmic approach.

Course Details:

1. **Logic and Induction** **8hrs**
   Proposition and Truth Functions; Propositional logic, Predicates and Quantification; Expression state- ment in the Language of logic; Deduction in Predicate Logic; Introduction to Induction
2. **Mathematical Reasoning** **6hrs**
   Axioms; Rule of Inference and Proofs; Direct proof and Indirect Proofs
3. **Finite state Automata, Grammars and Languages** **8hrs**
   Sequential circuit, Finite State Machines (FSM) and Finite State Automata(FSA); Alphabets, Lan- guage and Grammars; Non-Deterministic Finite State Automata(FSA), Conversion of NonDetermin- istic FSA to Equivalent FSA
4. **Recurrence Relation** **8hrs**
   Introduction, Differencing and summation; Solution of Linear Recurrence Relations; Solution of Non- Linear Recurrence Relations
5. **Graph Theory** **15hrs**
   Definitions, Undirected and Directed Graphs; Walk, Path, Circuit, Components; Introduction to Com- plete Graph, Bipartite Graph, Regular Graph, Hamiltonian Graph, Euler Graph, Planar Graph; Com- puter Representation of Graphs; Shortest path Algorithm, Isomorphic of Graphs; Concept of graph coloring, Application of Graph Theory in Computer Science

References:

1. Richard Johnsonbaugh, Discrete Mathematics, Fifth Edition, Addison Wesley, Pearson Education Asia (LPE), ISBN: 81-780-82799, 2000
2. Kenth Rosen, Discrete Mathematics Structures with Applications to Computer Science, WCB/ McGraw Hill
3. Joe L. Mott, Abrahan Kandel, and Theodora P. Baker, Mathematics Structures for Computer Scientists and Mathematicians, Second Edition, Prentice-Hall of India, ISBN: 81-203-1502-2

Computer Programming II (Java)

Program: Bachelor of Information Technology (BIT)
Course Code : CIT122
Year: 1
Semester: II
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of Object-Oriented Programming. This course focuses on enhancing the computer programming skills of the students following Object Oriented Programming Principles. It also aims the students to understand and develop event driven graphical user interfaces (GUIs) applications, database applications and network programs.

Course Details:

1. **Introduction to Object Oriented Programming** **4hrs**
   Object Oriented Programming concepts and benefits, Difference from Procedure Oriented Program- ming (Java versus C/C++)
2. **Introduction to Java** **3hrs**
   History of Java, Features of Java, Java Development Kit (JDK), bytecode
3. **Java Basics** **5hrs**
   Main Program Execution, Main Method, Class and Objects, Variable, Constants, Data types, Array and String
4. **Operators and Control Flow** **4hrs**
   Operators (arithmetic, logical, relational, bitwise, ternary assignment), control flow (If-else, switch, for, while, break, continue)
5. **Inheritance, Package and Interface** **6hrs**
   Inheritance, Abstract classes, Access Protection, Inheritance, Package and Interface, Interfaces versus Abstract Classes
6. **Exceptions and streams** **5hrs**
   Exceptions, handling error and exception, Exception Classes, Common Exceptions Examples, Excep- tion Handling and Debugging Techniques
7. **Java I/O and streams** **5hrs**
   java.io, Streams Concept, Stream Classes, Readers and Writers, Random Access Files, Serialization
8. **Event Handling** **6hrs**

Event concepts and Classes, Event Hierarchy and Delegation Model, Event Listener; Event-Handling, Adapter Classes, Inner Classes, Anonymous Classes

9. **GUI with AWT & Swing** **7hrs**
   Overview and difference between Swing and AWT, Components and Containers, Separating GUI and Application code, Multicasting, Layout management

Laboratory Work:

1. Programming examples for each chapters (from chapter 2 to 9) . Minimum practical sessions include:
   a. Introduction to Basic Java Program Structure, Simple Programs without classes and methods.
   b. Program to learn the concepts of classes, objects and constructors.
   c. To illustrate Inheritance and Abstract classes
   d. To illustrate interface and Package
   e. To illustrate the java programs using Exception handling (e.g. try catch throw) and Stream (e.g. byte, character, etc.)
   f. To illustrate Multithreading concept in java.
   g. To illustrate GUI components using AWT and Swing
   h. To illustrate Layout management and Event Handling
   i. To illustrate Graphics and multimedia example with text, image and font.
2. Use of open source platforms preferable
3. Course Project with a maximum 4 students in a group should submit a group project before the final exam.

References:

1. Devkota B., 2016, Java Programming Basic,MK Publishers
2. Balagurusamy E., 2019, Programming in Java: Sixth Edition, Tata McGraw Hill
3. Dietel H.M and Dietel P.J., 2017, Java: How to Program, Eleventh Edition, Pearson Education Asia
4. Cay S. Horstmann, 2019, Big Java, John Wiley & Sons
5. Sierra, k. and Bates, B., Head First Java, O'Reilly Media, Incorporated

Mathematics II

Program: Bachelor of Information Technology (BIT)
Course Code : BSM121
Year: 1
Semester: II
Credit Hours : 3
Nature: TH

Course Objectives:

The purpose of the course is to provide fundamental knowledge of permutation, combination, different series, and differential equations concepts.

Course Details:

1. **Permutation and Combination** **6hrs**
   Introduction, principle of counting, factorial notation, permutation, permutation of Objects alike, permutation with restrictions, circular permutation, combination, combination with restriction
2. **Infinite Series** **10hrs**
   Introduction, convergent test of infinite series, Direct comparison test, Limit comparison test, P-series Test , De'Alembert's ratio test , Cauchy root test, Alternative series test, Interval and radius of convergence
3. **Function of Several Variables** **9hrs**
   Introduction, Functions of two or more variables, partial differentiation (First order partial derivatives, Higher order partial derivatives), total derivatives, homogeneous functions, eulers theorem, maxima and minima of functions of several variables
4. **Differential Equations** **8hrs**
   Introduction, Order and degree of differential equations, solution of first order and first degree differ- ential equations (separation of the variables, first order homogeneous differential equation, fist order linear differential equations, exact differential equation), second order liear differential with constant coefficient, initial and boundary value problems
5. **Functions of Complex Variable** **12hrs**
   Basic Definitions, functions of complex variable, algebra of complex numbers, properties of complex numbers, conjugate of a complex number, modulus of a complex number and its properties, arg and diagram, polar representation, square roots of a complex number, De'Moivre's theorem (statement only) and its application to find up to cube root of a complex number, limit and continuity an differ- entiation, cauchy-Riemann

equations, analytical functions, harmonic functions, complex exponential trigonometric hyperbolic functions, statements of Taylor's theorem and Laurent's theorem

Textbooks:

1. Kreyszing , E. Advanced Engineering Mathematics, New Delhi : John wiley and sons Inc.
2. Thomas , G.B. Jr and Finney , R.L. Calculus and Analytical Geometry , New Delhi, Narosa Publishing House

References:

1. Sastry, S.S. , Engineering Mathematics- Volume II, Prentice Hall of India
2. Grewal, B.S., Higher Engineering Mathematics, New Delhi, Khanna Publications

Digital Logic

Program: Bachelor of Information Technology (BIT)
Course Code : ELX121
Year: 1
Semester: II
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of basic principle of digital logic design and its implementation.

Course Details:

1. **Introduction to Digital Logic** **4hrs**
   Analog and Digital signal, Analog system, Digital System, Digital Operation, Application of Digital System, Modern Digital Computer, Clock signal.
2. **Number System** **7hrs**
   Introduction to Number Systems, Binary Number System, Decimal Number system, Octal Number System, Hexa-Decimal Number System and Conversion, Complements and Need of Complements, r's and (r-1)'s Complements, Binary Coded Decimal (BCD) and its Properties, Gray code, ASCII code, Excess-3 code, Negative and Positive Logic.
3. **Boolean Algebra** **9hrs**
   Basic Logic Gates (NOT, AND, OR, EX-OR, EX-NOR), Universal Gates, De-Morgan's Theorem, Boolean laws and Theorems, Sum of Product (SOP), Product of Sum (POS), Boolean Simplification using Boolean Algebra, The Karnaugh Map Method, Don't Care Map Entries, Implementation using Universal gates.
4. **Combinational Circuits** **12hrs**
   Introduction to combinational circuit, Half Adder, Full Adder, Half Subtractor, Full Subtractor, Adder- Subtractor circuit, Multiplexers/De-Multiplexers, Encoder/Decoder, Design of Encoder/Decoder, 7- segment Decoder, Design of Multiplexer/De-Multiplexer, Introduction to ROM and PLA, Design of ROM and PLA using K-map.
5. **Sequential Circuits** **7hrs**
   Introduction to sequential circuit, Synchronous and Asynchronous sequential circuit, Edge Triggered and Level Triggered, Flip-Flop: R-S Flipflop,

D-Flipflop, Clocked R-S flipflop, T-Flipflop, J-K flipflop, Master Slave J-K Flipflop.

6. **Counter and Register** **6hrs**
   Introduction to counter, Concept of up and Down counting Asynchronous counter (Ripple counter), Modulo Counter, Synchronous Counter, Register and their types: SISO, SIPO, PIPO, PISO.

Laboratory Work:

1. Familiarization with basic logic gates.
2. Realization of De-Morgan's Theorem and Universal gates.
3. Adder and Subtractor circuits
4. Multiplexers and De-Multiplexers
5. Encoders and Decoders
6. Seven Segment Display Decoder
7. Flip-flops (R-S Flipflop, D-Flipflop, Clocked R-S flipflop, T-FlipFlop, J-K Flipflop)
8. Asynchronous Counter, Synchronous Counter, Modulo Counter
9. Shift Register

References:

1. Donald P. Leach, Albert Paul Malvino and Goutam Saha, Digital Principles and Applications, 6th edition, Tata McGraw-Hill, 2006
2. William H. Gothmann, Digital Electronics, An introduction to Theory and Practice, 2nd edition, PHI, 2009
3. M. Mano, Digital Logic and Computer Design, Prentice Hall of India, 4th Edition.

Web Technology I

Program: Bachelor of Information Technology (BIT)
Course Code : CIT123
Year: 1
Semester: II
Credit Hours : 3
Nature: TH + PR

Course Objectives:

This course is designed to serve as an ideal beginners guide to writing web pages using HTML, using CSS to make pages more appealing and add interactivity to pages using JavaScript. After completing the course, students will be familiar with the recent technologies such as web technology, client side programming, server side programming, and will be able to develop web based applications.

Course Details:

1. **Basic Concept** **3hrs**
   - 1.1 Introduction to Internet, WWW, Web Browser, Web Servers, URL;
   - 1.2 Overview of different protocols: HTTP, HTTPS, POP, SMTP, FTP, WAP;
   - 1.3 Domain name and hierarchy, domain name registration process, web hosting.
2. **HTML and XHTML** **8hrs**
   - 2.1 Structuring Documents for the Web: Introducing HTML and XHTML, Tags and Elements, Basic Syntax;
   - 2.2 Basic Text Formatting , Presentational Elements, Phrase Elements;
   - 2.3 Lists, Creating Links, Images, Audio, and Video, Using Images asLinks, Adding Flash, Video, and Audio to Your Web Pages;
   - 2.4 Tables : Introducing Tables, Basic Table Elements and Attributes,Spanning Columns Using the colspan Attribute, Spanning Rows Using the rowspan Attribute;
   - 2.5 Forms and Form Controls: Introducing Forms, Creating a Form with the Element, The action Attribute , The method Attribute, The id Attribute, The onsubmit Attribute, The onreset At- tribute, The enctype Attribute, The accept-charset Attribute, The accept Attribute, The

target Attribute, White Space and Text Inputs, Buttons, Checkboxes, Radio Buttons, Select Boxes, File Select Boxes, Hidden Controls
After completion of this section following web-pages have to be developed:
- Web page with list and menus
- Different look and feel web pages using divisions
- Photo gallery
- Page with internal and external links
- User interface with different controls to take user data should be developed

3. **Introducing Cascading Style Sheet** **8hrs**

3.1 Introduction, Where You Can Add CSS Rules, Levels of Style Sheets
3.2 Style Specification Formats, Style classes, Properties and Property Values, color, The :focus and :active Pseudo-Classes , Positioning and Layout with CSS
After completing CSS, Program with different kinds of styles should be developed.

4. **Learning JavaScript** **10hrs**

a. Basic of JavaScript and The Document Object Model
b. Starting to Program with JavaScript: Variables, Assigning a Value to a Variable, Lifetime of a Variable, Operators, Arithmetic Operators, Assignment Operators, Comparison Operators, Log- ical or Boolean Operators, String Operator (Using + with Strings)
c. Functions, Conditional Statement, Looping
d. Event and Event handling
e. DOM Event Model and Element Positioning
f. Changing colors and fonts
g. Form Validation: When to Validate, What You Can Check For, How to Check a Form, Checking Text Fields, Select Box Options, Radio Buttons, Checkboxes
After completion of JavaScript: User interface with form validation capability should be developed, (validation: check for empty fields, specific length string, email validation)

5. **Programming in PHP and MYSQL** **16hrs**

5.1 PHP's Place in the Web World, Basic Rules of PHP Programs, Uses of PHP
5.2 Text, Numbers, Variables in PHP
5.3 Primitives, Operations, Expression and Control Statements
5.4 Array, Functions, Form Handling
5.5 Working with Databases : Organizing Data in a Database, Connecting to a Database Program, Creating a Table, CRUD operation
5.6 Cookies, Session Tracking

After completion of this section, web site with data entry form with form validation functionality, attractive look and feel and basic database handling should be developed.

Laboratory Works:

1. Every topic of the course content should be included for the lab.
2. Individual or group project work to develop a web application should be assigned. This should cover most of the technologies included in the course content. Final evaluation for project work should be done by External assigned by Department.

References:

1. Beginning HTML, XHTML, CSS, and JavaScript, Jon Duckett, Addison-Wiley
2. Learning PHP 5, A Pain-Free Introduction to Building Interactive Web Sites, David Sklar, O'Reilly Media
3. Programming the World Wide Web, Robert W. Sebesta, Addison- Wesley
4. PHP and MySQL Web Development, Luke Welling, Addison-Wesley Professional O'Reilly Media

Project I

Program: Bachelor of Information Technology (BIT)
Course Code : CIT124
Year: 1
Semester: II
Credit Hours : 2
Nature: PR

Course Objectives:

1. To provide the knowledge of the computer-based subjects studied during first year.
2. To provide the knowledge on planning, design, development and implementation of the project.
3. To provide the knowledge to formulate project documentation and oral presentation.

Course Requirements:

1. Project must be chosen in an area related to Computer Science/IT/ICT/Software
2. A proposal including a literature survey must be submitted.
3. A midterm progress report to be submitted.
4. A oral presentation to be given.
5. The final report to be submitted.
6. The oral defense of the final report should be given.

Pre-requisite: Computer Programming I, Web Technology I and Problem Solving and Logic

The project document shall include the following items:

1. Project Team members (At least 2 and maximum of 3)
2. Project supervisor(s)
3. Technical description of the project.
4. Project Analysis, Design, Coding, Testing and Implementation details.

Data Structure and Algorithms

Program: Bachelor of Information Technology (BIT)
Course Code : CIT214
Year: 2
Semester: III
Credit Hours: 3
Nature: TH + PR

Course Objectives:

This course targets to cater ample theoretical and practical knowledge/skills of data structure and algorithms necessary to develop efficient programs. It teaches students about algorithms and data structures, giving the concept of each algorithm and data structure while avoiding complex details.

Course Description:

This is a rigorous course on data structure and their associated algorithms. An overview of data structure concepts, linked list, stack, queues, trees, and graphs is covered. This course also examines algorithms for recursion, sorting, searching and hashing techniques. Moreover, algorithms and their efficiency analysis is explained.

Course Outcomes:

By the end of this course, students will be able to use the data structures and algorithms for problem solving and they are supposed to develop an understanding about different data structures and algorithms - what they are, and how they are implemented. Moreover, students will be able to describe the techniques of algorithm analysis and design efficient algorithms.

Course Details:

1. **Introduction** **5hrs**

    1.1 Overview of Data Structure and Algorithms, Object-Oriented Programming, Software Engineering,

1.2 Abstract Data Types (ADT), Review of Array, Structure, Pointer.
1.3 Algorithm Complexity(Time and Space Complexity), Asymptotic Notations, Time-Space Tradeoff

2. **Linked Lists** **5hrs**

2.1 Linked List as an ADT
2.2 Singly Linked Lists: Insertion, Deletion, Search
2.3 Two Way Lists: Insertion, Deletion, Search
2.4 Circular Linked List

3. **Stacks, Queue and Recursion** **10hrs**

3.1 Stack and Queue as an ADT
3.2 Array and Linked List Implementation of Stack and Queue
3.3 Expression Conversion, Evaluation of Infix, Postfix and Prefix Expressions.
3.4 Circular Queue and Its Implementation
3.5 Application of Stack and Queue
3.6 Recursion, Recursion vs. Iteration. Tail Recursion vs. Nontail Recursion, Indirect Recursion, Towers of Hanoi , Backtracking, Recursive Applications, Efficiency of Recursion.

4. **Trees** **7hrs**

4.1 Introduction to Trees, Basic Concepts and Terminology (Height, Level, Depth, Path, Root, Parent, Child, Leaf, Node, Subtree, Keys, etc.)
4.2 Binary Trees, Tree Traversal , The Efficiency of Binary Trees
4.3 Binary Search Tree (BST), BST Operations (Insertion, Deletion, Search)
4.4 Heaps and Operations
4.5 AVL Tree and Operations
4.6 M-Way Search Trees, B Tree and B+ Tree, B Tree Operations
4.7 Splay Tree
4.8 Application of Tree: Huffman Codes

5. **Sorting** **6hrs**

5.1 Introduction to Sorting, Sorting Types (Internal vs. External)
5.2 Elementary Sorting Algorithms: Insertion Sort, Selection Sort, Bubble Sort
5.3 Divide & Conquer Sorting Algorithms: Heap Sort, Quicksort, Merge Sort
5.4 Efficiency of Sorting Algorithms

6. **Searching and Hashing** **4hrs**

6.1 Searching: Linear Search and Binary Search, Search Efficiency
6.2 Hashing : Hash Function and Hash Tables, Collision Resolution Techniques, Hashing Efficiency

7. **Graph** **6hrs**

   7.1 Graphs as an ADT, Graph Types, Graph Representation,
   7.2 Graphs Traversal ,
   7.3 Minimum Spanning Trees(Kruskal and Prim Algorithm),
   7.4 Single-Source Shortest Paths(Bellman-Ford Algorithm, Dijkstra's Algorithm)

8. **Case Studies In Algorithms** **2hrs**

   8.1 Investigate well known Computer Science problems, such as the Traveling Salesman Problem, the Shortest Path Problem, the Knapsack Problem, etc. and develop solutions to them using approaches such as Brute-Force, Greedy, and Dynamic Programming.

Laboratory Work:

Practical sessions shall be conducted using object oriented programming language like C++ or Java.

1. Implementations of linked lists
2. Implementations of Stack
3. Implementations of Queue
4. Implementation to solve Problems like Tower of Hanoi or Fibonacci Recursion
5. Implementation of tree(at least 1): AVL tree
6. Implementation of elementary Sorting (at least 1): Insertion, Bubble, Selection,
7. Implementation of divide & conquer Sorting (at least 1): Heap Sort, Quicksort, merge Sort
8. Implementations of Search: Linear and Binary Search
9. Implementation of Hashing
10. Implementation of Minimum spanning trees
11. Implementation of Shortest path algorithm

References:

1. S. Lipschutz, Data Structuires, McGraw Hill Education.
2. T. H. Coremen, C.E. Leiserson, R. L. Rivest, C. Stein, Introduction to Algorithms, MIT Press.
3. R. Lafore, Data Structures & algorithms in Java, SAMS.
4. Y. Langsam, MJ, Augenstein and A.M, Tenenbaum Data Structures using C and C++ Prentice Hall India.

5. E. Horowitz, S. Sahni, S. Rajasekarn, Fundamentals of Computer Algorithms, Galgotia Publications.

Database Management System

Program: Bachelor of Information Technology (BIT)
Course Code : CIT213
Year: 2
Semester: III
Credit Hours: 3
Nature: TH

Course Objectives:

The objective of this course is to provide the fundamental concepts and understanding of the database systems, theory and practices in design and implementation of DBMS.

Course Outcomes:

At the end of the course, the students will be able to

- get the broad understanding of database and database management software
- have a high-level understanding of different models and allows to use these models as per user requirements.
- to use different SQL commands in relational DBMS.
- apply different normal forms to design the database.

Course Details:

1. **Introduction** **5hrs**
   Introduction to database and database management system, Characteristics of database management system and its applications, Data Abstraction and Data Independence, Database users and database administrator, Application Architecture, Basics of Database Language(DDL, DML, DCL).
2. **Data Model** **6hrs**
   Different data model concepts; conceptual, physical, logical model, Types of model ; E-R model, Net- work/Hierarchical Model, Relational Model, Object Oriented Model, Entities, Entities Types and Entities Set, Strong Entity Sets and Weak Entity Sets, Relationship and Relationship Set, Attributes and Keys; Primary Key, Candidate Key, Super Key, E-R diagram, Reducing E-R diagram to tables, Generalization and Specialization, Aggregation, Mapping Cardinalities in E-R diagram , Extended E-R Features

3. **Relational Model** **4hrs**
   Introduction to Relational Database, Database Schema and views, Relational Model Constraints, Re- lational Operations, Relational Algebra
4. **Relational Language and Database Constraints** **11hrs**
   Brief Introduction to Structured Query Language, Features, SQL queries and sub queries, Joined and Derived Relations, Different DDL and DML Operations, Views, Modification of database, Joined con- ditions and categories, transaction control language(Commit, Rollback), Integrity Constraints, Domain Constraints, Referential Integrity in SQL, Assertions and Triggering.
5. **Relational Database Design** **7hrs**
   Features of good database design, Concept of functional dependencies, Decomposition using func- tional dependencies, Concept of multi-valued and joined dependencies, Normalization, Different Nor- mal Forms(INF, 2NF, 3NF, 4NF, 5NF, BCNF).
6. **Transaction management and concurrency control** **5hrs**
   Concept of Transaction, ACID properties, Transaction Atomicity and Durability, General Concept of Concurrency Control and Recovery, Serializability Concept, Lock based Protocols, Time Stamp based Protocols
7. **Recovery System** **3hrs**
   Basic concept of Crash Recovery, Classification of failure, Recovery and Atomicity, Log based recovery, Shadow paging
8. **Advanced Database Model** **4hrs**
   Extension of Relational Model, Object -Oriented Model, Distributed Model, NOSQL Systems, XML database

Laboratory Work:

There will be minimum 10 laboratory exercises which covers basic operations using MS-Access or MySQL, Installing and working with database server, practicing DDL and DML commands,. Also, the students are required to perform the group project work.

References:

1. Abraham Silberschatz, Henry F.Korth, S. Sudarshan “Database System Concepts”, Mc Graw Hill, 7th Edition
2. Ramez Elmasri and Shamkant B. Navathe , “Fundamentals of Database Systems”, Pearson Education Asia
3. Raghurama Krishnan, Jonannes Gehrke, “Database Management System”, TATA McGrawHill

4. Pramod Sadalage, Martin Fowler, NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence, Pearson

Microprocessor and Computer Architecture

Program: Bachelor of Information Technology (BIT)
Course Code : ELX211
Year: 2
Semester: III
Credit Hours: 4
Nature: TH + PR

Course Objectives:

The main objective of the course is to provide the basic knowledge of the components of Microprocessor, block diagram and assembly language programming using Intel 8085, Intel 8086, SAPI and SAP II computer architecture, overview of the computer organization, principle of the memory system, Central processing unit, RISC, CISC, Direct memory access, Interrupts.

Course Description:

This course is an introduction to microprocessor and computer architecture which aims at providing fundamental knowledge about computer architecture, instructions cycle, components of Microprocessor, Intel 8085, Intel 8086 and Assembly programming.

Course Outcomes:

After completion of the course, students will be able to

- Know basics of microprocessor and microprocessor-based system
- Know internal architecture of microprocessor
- Write assembly language programs
- Interface various devices such as keyboard, seven segment display
- Handle interrupts.

Course Details:

1. **Fundamental of Microprocessor** **6hrs**
   Introduction and History of Microprocessors, Microprocessor system with Bus organization, Compo- nents of a Microprocessor: Registers, ALU,

Control and Timing, Introduction to SAP I and SAP II with architecture, Applications of Microprocessor.

2. **Intel 8085** **10hrs**
   Functional Block Diagram and Pin Configuration, Features of 8085, Timing and Control Unit, Reg- isters, Data and Address Bus, Intel 8085 Instruction sets, Opcode and Operands, Addressing Modes, Interrupts, Flags, Instruction Sets, Assembly Language Programming with 8085 microprocessors.
3. **Overview of Intel 8086** **5hrs**
   Block Diagram and pin configuration, Features and comparison with 8085, Introduction to 8086 in- structions, Addressing Modes, Introduction to 8086 Assembly Language programming.
4. **Basic Computer Architecture and Microoperations** **5hrs**
   History of Computer Architecture, Overview of Computer Organization, Memory Hierarchy, Microop- erations, Arithmetic Microoperations, Logic Microoperations, Shift Microoperations, Arithmetic Logic Shift Unit
5. **Control Unit and Central Processing Unit** **8hrs**
   CPU organization, Register organization, instruction cycle, Control Unit of Basic Computer, Micro- instruction and Microoperation format, Symbolic Microinstructions, Symbolic Microprogram, Data Transfer and Manipulation, Introduction to RISC and CISC.
6. **Computer Arithmetic** **6hrs**
   Data Representation, Integer Arithmetic, Addition/Subtraction, Multiplication (Unsigned, Booth's Algorithm), Division, Floating point Representation
7. **Input and Output Organization** **5hrs**
   Introduction to Peripheral Devices, I/O Interface, Direct Memory Access, I/O processor, Data Com- munication Processor.
8. **Memory Organization** **5hrs**
   Hierarchy of Memory System, Primary and Secondary memory, Virtual Memory, Cache Memory and Mapping techniques, Memory Management Hardware.
9. **Pipelining** **5hrs**
   Concept of Pipelining, Arithmetic Pipeline, Instruction pipeline, Data Dependency, Handling of Branch Instructions.
10. **Parallel Processing** **5hrs**
    Parallel Processing, Parallelism in Multiprocessor System, Flynn's Classification, Interconnection struc- ture in Multiprocessor System, Vector Arithmetic Unit, Multithreaded Structure.

Laboratory Work:

Assembly language programming using 8085 trainer kit. The programming should include: data transfer, arithmetic operation, logic operation, conditional branching, base conversion.

Text Books:

1. Ramesh S. Gaonkar: Microprocessor Architecture, Programming and Applications with 8085, prentice Hall
2. Morris Mano: Computer System Architecture, Third Edition Prentice Hall

References:

1. Malvino: Digital Computer System Electronics (An Introduction to Microcomputers)
2. Douglas V.Hall : Microprocessor and Interfacing programming and Hardware , McGraw hill

Principles of Organization and Management

Program: Bachelor of Information Technology (BIT)
Course Code : BCT211
Year: 2
Semester: III
Credit Hours: 3
Nature: TH

Course Objectives:

The basic purpose of this course is to provide essential knowledge and exposure of the basic concepts, theories and practices prevalent and evolving in the field of management. Particularly it focuses on the major areas of management process that has broader applications in organizational context. The course also emphasizes to enable students to better understand the functions and opportunities of management that is contributing in successful operations and performance of organizations. Initially, the course comprises of the overview of management and its historical context followed by the concepts and practices of major management functions i.e. planning, organizing, leadership and motivation, and controlling necessary to get an insight in managing the organizational components. The course also aims to provide students a brief insight in contemporary issues with the inclusion of contemporary issues and developments in the management functions.

Course Outcomes:

Upon successful completion of the course, the students should be able to:

- describe fundamental concepts, nature and principles of management;
- understand the broader spectrum of management applications and developments;
- understand the historical context of management and relate it with the contemporary management practices;
- explain the role and responsibilities of managers and adapt to the various styles of management across organizations;
- develop analytical abilities to better manage the business and organizations by implementing essential managerial functions;
- apply various tools to facilitate the decision-making process in organizations for better management.

Course Details:

1. **Introduction to Management** **4hrs**

   1.1 Concept, characteristics and levels of management; Kinds of managers; Basic management func- tions. 2hrs
   1.2 Managerial roles and skills; Management as science and art; Concept of CSR and Corporate Governance; Skill's for today's managers.2hrs

2. **Evolution of Management and the Environmental Context of Management** **6hrs**

   2.1 Theories on evolution of management: Classical management perspective, Behavioral management perspective, Quantitative management perspective, Systems perspective, Contingency perspective. 3hrs
   2.2 Business environment: Concept, characteristics and components; Organization and environment interface; Emerging management issues and challenges. 2hrs
   2.3 Organizational change: concept and drivers; Employee resistance to change and overcoming tech- niques; Concept of organizational development and OD intervention. 1hr

3. **The Planning and Decision Making** **6hrs**

   3.1 Organizational planning: Concept, features, process, types, and methods; Strategic planning: concept and process. 3hrs
   3.2 Managerial decision making: Concept, features, process, types of managerial decision making; Decision making under uncertainty.3hrs

4. **Organizing** **8hrs**

   4.1 Organizing: Concept and elements; Types of organizational design. . 2hrs
   4.2 Job Design: Concept and techniques; Elements of job specification and description. 2hrs
   4.3 Departmentalization: Concept and types; Concept of authority, power and responsibility. 2hrs
   4.4 Delegation and Decentralization of Authority: Concept, advantages and disadvantages; Concept and techniques of coordinating various units in organization. 2hrs

5. **Motivation and Leadership** **6hrs**

   5.1 Nature of motivation; Content perspectives on motivation: Needs-hierarchy theory, two-factor theory and their implications. 2hrs
   5.2 Process perspective on motivation: Expectancy theory, equity theory, goal-setting theory and their implications; Reinforcement perspectives on motivation. 2hrs

5.3 Leadership: Nature of leadership; Approaches to leadership: Generic, Situational approaches to leadership, Related approaches to leadership, Emerging approaches to leadership. 2hrs

6. **Communication and Managing Workforce** **6hrs**

6.1 Organizational communication: Concept and types; Barriers to communication; Methods of enhancing effective communication. 3hrs

6.2 Groups and teams in organization: Characteristics of groups and teams; Interpersonal and inter- group conflict, Managing conflict. . 3hrs

7. **Controlling** **3hrs**

7.1 Concept, nature and purpose of control; Control process; Levels of control: strategic, structural, operations and financial control. 2hrs

7.2 Role of information technology on operational and behavioral control in organization. 1hr

8. **Functional areas of Management** **6hrs**

8.1 Marketing management: Concept and relevance of marketing management; Components of mar- keting mix. 2hrs

8.2 Human resource management: Concept and functions; Concept of Succession Planning; Overview of knowledge management and talent management. 2hrs

8.3 Research and development: Concept and process of business research; Overview of organizational agility and learning organization; Concept and strategies for promoting corporate entrepreneur- ship; Overview on other fundamental functional areas of management. 2hrs

Addendum: At least one case will be administered at the end of each chapter. The students will also complete a project work and a few other assignments as specified by the faculty member.

Textbook:

1. Robbins, S. P., Coulter, M., Viswanathan, R., & Randel, A. Management. Pearson Education.

References:

1. Griffin, R. W. Fundamentals of Management. Cengage Learning Asia.
2. Tripathi, P. C., Reddy, P. N., & Bajpai, A. Principles of Management. McGraw Hill.

Software Engineering

Program: Bachelor of Information Technology (BIT)
Course Code : CIT212
Year: 2
Semester: III
Credit Hours: 3
Nature: TH + PR

Course Objectives:

This course aims to introduce a systematic approach of planning, designing, implementation and maintaining a software product. This course expects the learners have basic background of computer systems and object-oriented programming. It covers the fundamental of software engineering, agile development, requirement engineering, system modeling, architectural design, design reuse, testing, quality management and configuration management. The laboratory exercise will enable learners to know the software engineering procedure practically.

Course Details:

1. **Software Engineering Fundamentals** **4hrs**
   - 1.1 Professional software products
   - 1.2 Software engineering diversity
   - 1.3 Unified Process
   - 1.4 Software engineering ethics
2. **Agile Software Development** **5hrs**
   - 2.1 Agile methods
   - 2.2 Plan-driven and agile development
   - 2.3 Extreme programming
   - 2.4 Scrum methodology
   - 2.5 Agile project management
3. **Requirement Engineering** **5hrs**
   - 3.1 Functional and non-functional requirements
   - 3.2 Software requirement specification documents
   - 3.3 Requirement engineering process
   - 3.4 Requirements management
   - 3.5 Business/Domain Modeling

3.6 Business Process Reengineering(BPR)

4. **System Modeling** **6hrs**
   - 4.1 Context models
   - 4.2 Interaction models
   - 4.3 Structural models
   - 4.4 Behavioral models
   - 4.5 Package Structure

5. **Architectural design** **5hrs**
   - 5.1 Architectural design decisions
   - 5.2 Model-View-Controller architectural pattern
   - 5.3 Layered architectural pattern
   - 5.4 Client-server architecture

6. **Design, reuse and implementation** **5hrs**
   - 6.1 Use case Realizations
   - 6.2 Mapping Designs to Code
   - 6.3 Design patterns
   - 6.4 Implementation issues
   - 6.5 Open source development

7. **Software testing and cost estimation** **5hrs**
   - 7.1 System testing
   - 7.2 Component testing
   - 7.3 Test case design
   - 7.4 Test automation
   - 7.5 Estimation techniques
   - 7.6 Project duration and staffing

8. **Quality Management** **6hrs**
   - 8.1 Quality concepts
   - 8.2 Software Quality Assurance(SQA)
   - 8.3 Software reliability
   - 8.4 Software maintenance
   - 8.5 ISO standards
   - 8.6 CMMI
   - 8.7 SQA plan
   - 8.8 Software certification

9. **Configuration Management** **4hrs**
   - 9.1 Configuration management planning
   - 9.2 Change management
   - 9.3 Version and release management
   - 9.4 Case tools for configuration management

Laboratory Work:

1. Development of user requirement and corresponding system requirements.
2. Introduction to standard Integrated Development Environment(IDE) like eclipse and accessing its features.
3. UML diagrams for particular project/case. Use of open source/free tools like https://sequencediagram.org/
4. Understanding program flow and code structure of MVC architecture through any open source framework.
5. Implementation of different Design patterns like singleton, factory, etc.
6. Implementation of Source code management software like git.

Textbooks:

1. Sommerville, Software engineering, tenth edition, Pearson, 2016

References:

1. R. Pressman, Software Engineering: A Practioner's approach, McGraw Hill, 2005
2. C. Ghezzi, M. Jazayeri and D. Mandrioli, Fundamentals of Software Engineering, Prentice Hall, 2003
3. C. Larman, Applying UML and Patterns, Third Edition, Prentice Hall, 2005.

Web Technology II

Program: Bachelor of Information Technology (BIT)
Course Code : CIT211
Year: 2
Semester: III
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The course will enhance the students' knowledge about the advanced web application development along with the database for web application, and web security.

Course Outcomes:

After completing the course, students will be familiar with the recent technologies on frontend and backend web development and also integrating both. Students will be familiar with various cyber-attacks and securing the web.

Course Details:

1. **Introduction** **2hrs**
   - 1.1 Introduction to Web 1.0, 2.0, 3.0,
   - 1.2 Introduction to Web Technologies: Introduction to Web servers like Apache and IIS, XAMPP (Bundle Server), WAMP (Bundle Server), Handling HTTP Request and Response, installations of above servers,
   - 1.3 Recall on HTML 5.0, XHTML, CSS 3 and their role on web development.
2. **Styling more with CSS** **5hrs**
   - 2.1 CSS Selectors, CSS Properties and Values, Browser Compatibility,
   - 2.2 Dynamic vs Static Web Page.
   - 2.3 The page layout techniques (Normal flow, The display property, Flexbox, Grid, Floats, Position- ing, Table layout, Multiple-column layout),
   - 2.4 Designing responsive web pages, Media queries,
   - 2.5 introduction to Bootstrap and its advantages

3. **AJAX and XML** **8hrs**
   - 3.1 Basics of AJAX and asynchronous programming.
   - 3.2 Introduction to XML, Advantages and its Application of XML, Syntax and Rules for creating XML document, XML elements, XML attributes, XML Tree, XML Namespace, XML schema languages: Document Type Definition(DTD).
   - 3.3 XML Schema Definition(XSD), Extensible Style Sheet Language/ Transformations(XSL)/ (XSLT), XQuery, XPath, Simple API for XML(SAX), XML DOM.

4. **Advanced JavaScript** **12hrs**
   - 4.1 Introduction to JSON, Introduction to jQuery
   - 4.2 Variables, scope, let, const and var keywords
   - 4.3 for-of loops [ES6], for-await-of loops [ES2018], for-in loops [ES1], for-each loops for iterating arrays
   - 4.4 Object Literals, Constructors and Object Oriented JavaScript
   - 4.5 Classes and Inheritance in JavaScript
   - 4.6 callback functions in JavaScript
   - 4.7 Promises Basics, promise.then() & promise.catch()
   - 4.8 Arrow functions in JavaScript, getter and setter
   - 4.9 Feth API in JavaScript, Async/Await in JavaScript

5. **Web Security** **5hrs**
   Cryptography, Authentication, Encryption/Decryption, Digital Certificates, Digital signature, Secure Socket Layer, Attacks, Cross-site scripting attacks, SQL injection.
6. **Combining Together** **5hrs**
   Creating the Database, Establishing the Site, Coding the Non-JavaScript Version, Creating the Ajax Resources, Adding the JavaScript, Securing the site.
7. **Web Application Development Frameworks** **8hrs**
   Introduction to CMS, MVC Architecture, Unit Testing, Introduction to Frameworks, Introduction to Popular Case Study on popular frameworks

Laboratory Work:

1. Every topic of the course content should be included for the lab.
2. Individual or group project work to develop a web application should be assigned. This should cover most of the technologies included in the course content. Final evaluation for project work should be done by External assigned by Department.

References:

1. Jeffrey C. Jackson. Web Technologies: A Computer Science Perspective, Pearson
2. P. J. Deitel and H. M. Deitel, Internet and World Wide Web: How to Program, 5th Edition, Pearson
3. Dafydd Stuttard, Marcus Pinto, The Web Application Hacker's Handbook, 2nd Edition, Wiley Publi- cation

References:

1. Larry Ullman "Modern JavaScript Develop and DeSign"
2. Thomas A. Powell, "HTML & CSS: The Complete Reference", 5th Edition, Mc Graw Hill
3. Max Bramer, "Web Programming with PHP and MySQL", Springer
4. Robin Nixon, "Learning PHP, MySQL & JavaScript With jQuery, CSS & HTML5", 4 th Edition, O'Reilly
5. Eric Newcomer, "Understanding Web Services: XML, WSDL, SOAP, and UDDI", AddisonWesley

Computer Graphics Technology

Program: Bachelor of Information Technology (BIT)
Course Code : CIT224
Year: 2
Semester: IV
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The objective of this course is to understand the theoretical foundation as well as the practical applications of computer graphics, 2D and 3D algorithms used in graphics.

Course Description:

The course covers concepts of graphics hardware, software, and applications, data structures for representing 2D and 3D geometric objects, drawing algorithms for graphical objects, techniques for representing and manipulating geometric objects, illumination and lighting models, and concept of virtual reality.

Course Outcomes:

On completion of the course the student should have the following learning outcomes:

- Knows fundamental computer graphics algorithms and data structures.
- Understand the techniques used to approximate the physical process of image generation.
- Understand the mathematics behind geometric transformations and techniques for modelling objects.

Course Details:

1. **Introduction of Computer Graphics** **6hrs**

   1.1 A Brief Overview of Computer Graphics, Areas of Applications.

- 1.2 Graphics Hardware: Display Technology, Architecture of Raster Scan Display, Vector Display, Output Devices, Input Devices.
- 1.3 Graphics Software: Software standards, Need of machine independent graphics language, Introduction to OpenGL
- 1.4 Color Models: RGB, HSV

2. **Scan Conversion Algorithms** **9hrs**

- 2.1 Line Drawing Algorithms
  - i. DDA
  - ii. Bresenham's Algorithm with derivation for positive and negative slope
- 2.2 Circle Generation Algorithm: Midpoint Algorithm with derivation
- 2.3 Area Filling - Scan Line Algorithm
- 2.4 Boundary Fill Techniques
- 2.5 Flood Fill Techniques

3. **Two Dimensional Geometric Transformations and Viewing** **9hrs**

- 3.1 Basic Transformations
  - i. Translation
  - ii. Rotation
  - iii. Scaling
- 3.2 Other Transformations
  - i. Reflection
  - ii. Shearing
- 3.3 Homogeneous Co-ordinate systems
- 3.4 Composite Transformations
- 3.5 Two Dimensional Viewing
  - i. Viewing pipeline
  - ii. Window to viewport coordinate transformation
- 3.6 Clipping
  - i. Introduction and application of clipping
  - ii. Line Clipping
    - A. Cohen Sutherland line clipping
  - iii. Polygon clipping
    - A. Sutherland Hodgeman polygon clipping

4. **Three Dimensional Graph** **8hrs**

- 4.1 3D object representation
  - i. Polygon surface
  - ii. Polygon tables
  - iii. Plane equations
  - iv. Polygon meshes
- 4.2 Projections
  - i. Parallel Projections: Isometric and Oblique
  - ii. Perspective Projection

4.3 3D Transformation: Transformation, Rotation, Scaling, Reflection, Shear

5. **Visible Surface Detection** **5hrs**

   5.1 Hidden Surfaces and their Removal Techniques
      i. Back-Face Detection
      ii. Depth Buffer Method
      iii. A- buffer method
      iv. Scan Line Method
      v. Depth Sorting Method(Painter's Algorithm)

6. **Illumination and Shading** **5hrs**

   6.1 Illumination Theory
   6.2 Ambient Light
   6.3 Reflections: Diffuse, Specular
   6.4 Surface Shading methods
      i. Constant Shading
      ii. Gouraud Shading
      iii. Phong Shading

7. **Introduction to Virtual Reality** **3hrs**

   7.1 Concept of Virtual Reality
   7.2 Components of Virtual Reality System
   7.3 Types of VR System
   7.4 Application of VR

Laboratory Work:

Implementation of various 2D and 3D graphics algorithms covered in the course using C / Java or any other language or tool.

Textbooks:

1. Donald Hearn and M. Pauline Baker: Computer Graphics, Prentice-Hall.

References:

1. James D. Foley, Andries van Dam, Steven K. Feiner, John F. Hughes, Computer Graphics: Principles and Practice in C, Addison-Wesley.

2. J.D. Foley, S.K. Feiner and J.F. Hughes, “Computer Graphics – Principles and Practises” (Second Edition in C)
3. R.K. Maurya, “Computer Graphics with Virtual Reality”, Wiley India
4. F.S. Hill, Stephen M.Kelley, “Computer Graphics using Open GL” Prentice Hall
5. Fundamentals of Computer Graphics, Peter Shirley, Michael Ashikhmin, Steve Marschner, Third edition, A K Peters/CRC Press
6. Mason Woo, Jackie Neider, Tom Davis, Dave Shreiner, Open GL Programming Guide Third Edition, The Official Guide to Learning OpenGL, Version 1.2, OpenGL Architecture Review Board, LPE Pearson Edition Asia.

Data Communication and Computer Networks

Program: Bachelor of Information Technology (BIT)
Course Code : CIT223
Year: 2
Semester: IV
Credit Hours : 4
Nature: TH + PR

Course Objectives:

The main objectives of the course are:

1. To learn the basic principles behind the interconnection between different hosts across private, organizational and public networks.
2. To learn to develop programs for data communication on various network layers and understand the security concepts on these layers.

Course Description:

The course is designed to teach the principles of networking by using protocol layering of the internet and the TCP/IP protocol suite. The topics flow in bottom-up approach. The learner shall understand how bits move in physical layer before learning how programs exchange messages at application layer. The learner shall complete 10 quizzes, 4 assignments, 1 semester Project, regular examinations and topic-based problem sets. Moreover, the learner will work through Lab Assignments, Programming Assignments and Simulation Experiments throughout the course.

Course Outcomes:

The expected outcome of the course are:

1. The learner shall learn to design the network of IPv4 and IPv6 for interconnecting various network components at different layers.
2. The learner shall learn to configure various data communication strategies for maintaining Quality of Service while securing the network at different layers.
3. The learner shall learn different tools and strategies for securing the network environment.

Course Details:

1. **Introduction to Computer Network and Data Communication** **3hrs**
   - 1.1 Internet and Intranet
   - 1.2 History of Computer Networking and Internet
   - 1.3 Introduction to Packet Switched Network and Circuit Switched Network
2. **Physical Layer** **7hrs**
   - 2.1 Analog and Digital Transmission
   - 2.2 Bandwidth Utilization: Multiplexing and Spectrum Spreading
   - 2.3 Transmission Media
   - 2.4 Switching
3. **Data-link Layer** **10hrs**
   - 3.1 Introduction
   - 3.2 Error Detection and Correction
   - 3.3 DLC (Data Link Control) and MAC (Media Access Control)
   - 3.4 Wired and Wireless Networks.
     - Coding Schemes: CDMA, 4G LTE Cellular networks, 5G Cellular Networks
     - Spread spectrum signals
4. **Network Layer** **10hrs**
   - 4.1 Network Protocols
   - 4.2 Routing Protocols: Unicast and Multicast
   - 4.3 Routing Algorithm
   - 4.4 Software Defined Network (SDN): Principles, Protocols, Implications
   - 4.5 Virtual LANs, NAT, DHCP, ARP, BGP/OSPF
   - 4.6 NetFlow
   - 4.7 MPLS (Multi-protocol label switching) and SD-WAN (Software-defined Wide Area Network)
5. **Transport Layer and Application Layer** **6hrs**
   - 5.1 Transport Layer Protocols: TCP/UDP
   - 5.2 Standard Client-Server Protocols
   - 5.3 Network management & Multimedia Networking
6. **Network Essentials** **6hrs**
   - 6.1 Quality of Service
   - 6.2 Information Theory
   - 6.3 The Network Management Framework - SNMP, NETCONF/YANG, ICMP, BGP, OSPF.
7. **Networking as a Career** **6hrs**

7.1 Cisco Certifications on Network Design
7.2 IOT, VoIP
7.3 SCADA, Realtime systems.

8. **Network Security** **9hrs**

8.1 Introduction
8.2 Principles of Cryptography
- Symmetric Key Cryptography
- Public Key Encryption
- Cryptographic Hash Functions, Message Authentication Code and Digital Signatures

8.3 Securing Applications
- End-point Authentication
- Email
- SSL
- IP Sec and VPNs
- Securing Wireless LAN
- Operational Security with Firewall and Intrusion Detection Systems

9. **Network SCAMS** **3hrs**

9.1 DMZ, Honeypots, Honey Nets, Proxy Servers

Laboratory Work:

1. Study of different types of Network cables and practically implement the cross-wired cable and straight through cable using clamping tool.
2. Study of Network Devices (Repeater, Hub, Switch, Bridge, Router, Gateway) in Detail.
3. Connect the computers in LAN.
4. Study of basic network command and Network configuration commands.
5. Performing an Initial Switch Configuration
6. Performing an Initial Router Configuration
7. Configuring and Troubleshooting a Switched Network
8. Examining WAN Connections
9. Interpreting Ping and Traceroute Output
10. Demonstrating Distribution Layer Functions
11. Placing ACLs
12. Implementing an IP Addressing Scheme
13. Planning Network-based Firewalls
14. Configuring DHCP, VLANs, IOT and SIP devices.

Programming Assignments(s):

1. Network Socket Programming
2. Network Packet Analysis - Deep Packet Inspection
3. Network Packet Analysis - Log File Analysis

Textbooks:

1. Computer Networking, A Top-Down Approach. J.F. Kurose and K.W. Ross. Pearson. Eighth Edition.
2. Data Communications and Networking. Forouzan. McGraw-Hill
3. Networking Essentials. J.S. Beasley, P. Nilkaew. Pearson

References:

1. TCP/IP Protocol Suits
2. Cryptography and Network Security
3. Computer Networks. A.S. Tanenbaum, N. Feamster, D.J. Wetherall

Management Information System

Program: Bachelor of Information Technology (BIT)
Course Code : CIT222
Year: 2
Semester: IV
Credit Hours : 3
Nature: TH + PR

Course Objectives:

This course aims to introduce students to comprehensive knowledge and understanding on information systems and its importance in business and management. This course also highlights the use and advantages of information systems in organizations with a focus on managerial aspects of MIS to promote an awareness of the economic, social and ethical implications of such systems on society and IT professionals.

Course Description:

This course provides a brief introduction to foundations of information system (IS) in business, components of IS, different types of ISs for business and management. It is designed to familiarize students with organizational and managerial foundations of systems, the role of information systems in enhancing business processes and management decision making across the enterprise, and the process of building and managing systems in organizations along with security and ethical challenges.

Course Outcomes:

Upon completion of this course, students will be able to:

- Understand the basic concepts and technologies used in the field of information systems;
- Have the knowledge of the different types of information systems;
- Understand the role of information systems in organizations, the strategic management processes, and the implications for the management;
- Understand the concepts of database and decision support in business
- Have the knowledge of basic enterprise applications
- Understand the processes of developing and implementing information systems;

- Develop an understanding of how various security, privacy and ethical issues while using IS along with security measurement.

Course Details:

1. **Foundations of Information Systems in Business** **6hrs**
   Data, Information, Information System (IS), Resources of IS, Characteristics of IS, Roles and Potential Risks of IS in business, Managerial challenges of IS, Types of IS: TPS, MIS, DSS, ESS; Objectives, Characteristics and Applications of MIS
2. **Competing with Information Technology** **5hrs**
   Competitive Strategic Concepts-Competitive forces and Strategies, Value chain and strategic IS, Strategic uses of Information Technology-Reengineering business processes, Becoming an Agile company, Creating a virtual company, Building a knowledge-creating company
3. **Data Resource Management** **5hrs**
   Fundamental Data Concepts, Database structures, Types of Databases (Operational, Distributed, External and Hypermedia), Data warehouses and Data Mining
4. **Decision Support System** **5hrs**
   Decision support in business-information, decision and management, Decision support system (DSS)-concepts and components, Online Analytical Processing, Using Decision Support Systems
5. **Enterprise Business System** **5hrs**
   Cross-functional enterprise applications-Enterprise application architecture, Enterprise application Integration, Customer Relationship Management (CRM), Phases of CRM, Benefits and Challenges of CRM, Trends in CRM; Enterprise Resource Planning (ERP), Benefits and Challenges of ERP, Trends in ERP; Supply Chain Management SCM, Roles of SCM, Benefits and Challenges of SCM, Trends in SCM
6. **Developing Business Systems** **7hrs**
   The system development life cycle-System Investigation, Analysis, Design, Development, Testing, Implementation and Support and Maintenance; Feasibility Studies, System conversion strategies, Prototyping, End-User Development, Outsourcing, Rapid Application Development, Agile Development and Extreme Programming, Scrum
7. **Electronic Commerce Systems** **5hrs**
   Introduction to E-Commerce, E-Commerce vs E-Business, Benefits and Limitations of E- Commerce, Essential E-Commerce Processes, Introduction to Electronic Payment System
8. **Security and Ethical Challenges** **7hrs**
   IT Security Challenges-Hacking cracking, Cyber theft, DoS attack, unauthorized use at work, software piracy, theft of intellectual property, computer viruses and worms, adware and spyware and more; Privacy Issues,

Health Issues; Introduction to various tools for security management; Ethical and Social Issues, Moral dimensions of Information age

Laboratory Work:

1. Database Design and Implementation
    - Create a database using a database management system (e.g., MySQL, Oracle, MS Access) to store and retrieve data related to a specific business scenario.
    - Design and implement appropriate tables, relationships, and constraints within the database.
    - Perform queries to retrieve specific information from the database, such as generating reports or extracting meaningful insights.
2. Data Visualization and Reporting
    - Use data visualization tools (e.g., Tableau, Power BI) to create interactive dashboards and reports based on real-world data.
    - Analyze the data and present it visually in a way that facilitates decision-making and provides meaningful insights to stakeholders.
3. Systems Analysis and Design
    - Conduct a feasibility study for a business process or system improvement.
    - Identify the requirements of the system, including user needs, system functionality, and constraints.
    - Develop a system design, including creating use cases, activity diagrams, and entity-relationship diagrams.
4. E-commerce and Online Business:
    - Develop a prototype for an e-commerce website using platforms like Figma, Adobe XD
    - Implement online payment integration and order processing functionality.
    - Perform testing and debugging to ensure the website functions properly.
5. Enterprise Resource Planning(ERP) Implementation
    - Choose an ERP system (e.g., SAP, Oracle ERP ,Odoo) and simulate the implementation process for a fictional organization.
    - Map business processes to the ERP system's functionalities and configure the system accordingly.
    - Test the system, analyze the results, and make necessary adjustments.
6. Cybersecurity and Risk Management

- Conduct a vulnerability assessment for a computer network or system.
- Identify potential risks and threats and develop strategies to mitigate them.
- Implement security measures, such as firewalls, encryption, and access controls, to protect the system from unauthorized access or data breaches.

7. Decision Support Systems

    - Develop a decision support system (DSS) for a specific problem or decision- making process.
    - Utilize tools like Excel or specialized DSS software to create models, perform calculations, and generate recommendations based on different scenarios.

8. Business Process Improvement

    - Analyze an existing business process and identify areas for improvement.
    - Redesign the process using process modeling techniques (e.g., BPMN) to streamline operations, reduce costs, or enhance efficiency.
    - Implement and test the improved process to measure its impact on business performance.

Textbooks:

1. O'Brien, J. A, Marakas, G. M. Management Information Systems, McGraw Hill
2. Kenneth, C. L. Jane P. L Management Information Systems, Pearson Education

References:

1. Jawadekar W.S, Management Information Systems, McGraw Hill
2. Davis, G Olson M Management Information System: Conceptual Foundations, Structure and Development, Tata McGraw Hill

Operating Systems

Program: Bachelor of Information Technology (BIT)
Course Code : CIT221
Year: 2
Semester: IV
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objectives of the course are to

- Describe need and role of operating system.
- Understand OS components such a scheduler, memory manager, file System handlers and I/O device managers.
- Analyze and criticize techniques used in OS
- Demonstrate and simulate algorithms used in OS

Course Description:

This course includes the basic concepts of operating system components. It consists of process management, deadlocks and process synchronization, memory management techniques, File system implementation, and I/O device management principles used in providing the functions. It also includes case study on some type of operating system.

Course Outcomes:

By the end of the course students should be able to

- describe the general architecture of computer, operating system structures and components
- understand and analyse theory and implementation issues related in designing operating systems
- familiarize with the functions of operating system and their implementation approach and strategies.

Course Details:

1. **Operating System Overview** **4hrs**
   - 1.1 History and Definition of OS Functions of OS, OS as Resource Manager and Extended Machine, Evolution of operating system, Types of OS.
   - 1.2 Operating System Structure: Monolithic, Layered, Microkernel, Virtual Machine.
   - 1.3 System Calls, System Programs, Kernel, Shell, Shell Commands, Shell Programing.
   - 1.4 Open-Source Operating Systems.
2. **Process Management** **10hrs**
   - 2.1 Introduction to process, Concept of Uniprogramming, Multiprogramming and Multiprocessing.
   - 2.2 Process Model, Process States and Transitions, Process Control Block.
   - 2.3 Threads, Thread Vs. Process, User and Kernel Space Threads.
   - 2.4 Process Synchronization and Inter Process Communication, Race Condition in Process Synchronization and Critical Section Problem.
   - 2.5 Implementation Mutual Exclusion: Mutual Exclusion with Busy Waiting (Disabling Interrupts, Lock Variables, Strict Alteration, Peterson's Solution, Test and Set Lock), Sleep and Wakeup, Semaphore, Monitors, Synchronous and Asynchronous Message Passing
   - 2.6 Classical IPC problems: Producers Consumer Problem using Semaphore, Sleeping Barber Problem. Dining Philosopher Problem, Reader's Writers Problem
   - 2.7 Process Scheduling: Goals, Types, Scheduling in Batch and Interactive System: First-Come First-Serve, Shortest Job First, Shortest Remaining Time Next, Round- Robin, Priority, Multiple Queues, Highest Rank Ratio Next (HRN/HRRN), Real Time Scheduling Concept.
3. **Deadlock** **6hrs**
   - 3.1 Definition of Deadlock, Resource Type: Preemptable and Non-preemptable Resources, Conditions for Deadlock
   - 3.2 Handling Deadlocks: Deadlock Prevention, Deadlock Avoidance, Deadlock Detection and Resolution
   - 3.3 Deadlock Prevention: Attacking Condition for Deadlock
   - 3.4 Deadlock Avoidance: Safe and Unsafe States, Resource Allocation Graph (For Single and Multiple Resource Instance-Bankers Algorithm)
   - 3.5 Recovery from Deadlock
   - 3.6 Other Issues: Communication Deadlock, Live lock, Starvation.
4. **Memory Management** **8hrs**

4.1 Introduction To Memory Address, Logical Vs. Physical Address Space, Swapping, Multiprogramming with fixed and variable partitions.
4.2 Memory Management (Bitmaps & Linked-List), Memory Allocation Strategies: Contiguous and Non-contiguous.
4.3 Virtual Memory: Need, Concept of Paging, Page Table, Table Structure, Page Faults, Demand Paging, Thrashing.
4.4 Page Replacement Algorithms: FIFO, LRU, Optimal, Second Chance, Clock Page.
4.5 Segmentation: Need of Segmentation, Drawbacks, Segmentation with Paging, Segmentation Vs. Paging

5. **File Management** **6hrs**

5.1 File Overview: File Naming, File Structure, File Types, File Access, File Attributes and File Operations
5.2 Directory Systems: Single Level, Two-level and Hierarchical, File System Layout, Directory Operations, Path names
5.3 File Access Mechanism: Sequential, Direct and Indexed
5.4 Implementing Files: Contiguous Allocation, Linked List Allocation, Linked List Allocation using Table, I-nodes
5.5 Free Space Management: Bitmaps, Linked List

6. **I/O Management** **5hrs**

6.1 Principle of I/O Hardware and Software, Controllers, Memory Mapped I/O, DMA Operation, Interrupts
6.2 Goals of I/O Software, Handling I/O (Programmed I/O, Interrupt Driven I/O, I/O using DMA)
6.3 I/O Software Layers (Interrupt Handlers, Device Drivers)
6.4 Disk Structure, Disk Scheduling: FCFS, SSTF, SCAN, CSCAN, LOOK, CLOOK

7. **Security** **3hrs**

7.1 Security Breaches, Types of Attacks, Authentication and Authorization
7.2 Basics of Cryptography and its Types
7.3 Protection Mechanism, Access Control Matrix and Access Control List

8. **Case Study** **3hrs**
UNIX, LINUX, Windows

Laboratory Work:

The laboratory work includes solving problems in operating system discussed in course contents. The lab work should include;

1. Demonstration of basic Shell Commands (In Linux OS)
2. Implementation of some system calls (of Linux OS)
3. Simulation process Scheduling algorithms
4. Implementation of Producer and Consumer Problem
5. Simulation of page replacement algorithms
6. Simulate free space management techniques
7. Simulation of disk scheduling algorithms

Textbooks:

1. Andrew S. Tanenbaum and Herbert BOS, ”Modern Operating System”, PHI

References:

1. Stalling William, ”Operating Systems”, Pearson Education
2. Silbcrschatz A., Galvin P., Gagne G., ”Operating System Concepts”, John Wiley and Sons Maurice Bach, ”Design of the UNIX Operating Systems”
3. Crowley, C. ”Operating Systems: A Design-Oriented Approach”, Tata McGraw Hill

Fundamentals of Probability and Statistics

Program: Bachelor of Information Technology (BIT)
Course Code : BSM221
Year: 2
Semester: IV
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The main objective of this course is to familiarize students about the application of statistics in information technology. However, the specific objectives are to enhance the fundamental knowledge on data presentation, use of descriptive and inferential statistics in information technology.

Course Description:

This course aims to develop the statistical skill for solving information technology related problems. The primary focus of the course is to provide the basic knowledge on data management, classification and presentation of data, use of descriptive statistics and inferential statistics in information technology. Initially, it provides the concept of data, data collection methods, data management and presentation. The following units provide skills on measures of central tendency and dispersion, probability and probability distribution and inferential statistics (estimation, sampling, hypothesis testing). The final chapter provides the skill for finding relationship and cause and effect relationship among the variables.

Course Outcomes:

After the completion of the course, the students will be able to;

- Manage the raw data in the form of frequency distribution, diagrams and charts.
- Select the suitable techniques of measure of central tendency and dispersion
- Choose an appropriate methods of sampling
- Test the hypothesis
- Assess the relationship and cause and effect relationship between the variables

Course Details:

1. **Presentation of Data** **4hrs**
   Definition of Statistics; use of statistics in information technology (IT); data management; level of measurement; preparation of frequency distribution; diagrammatic and graphical presentation; graphical location of mode and median.
2. **Measures of Location and Dispersion** **10hrs**
   Measures of central tendency (mean, median and mode); calculation of the different measures of central tendency; partition values; measures of dispersion (range, quartile deviation, mean deviation, standard deviation, coefficient of variation); measure of symmetry and normality of data (excluding method of moments); exploratory data analysis (EDA): stem and leaf, Box and Whisker plot, detection of extreme values in the dataset.
3. **Probability and Probability Distribution** **10hrs**
   Basic terminologies; competing approaches in probability; additive and multiplicative laws of probability; conditional probability; Bayes theorem; mathematical expectation (calculation of mean and standard deviation); application of probability distribution; discrete (binomial and Poisson distribution); assumptions of binomial and Poisson distribution; fittings of binomial and Poisson distribution; continuous (Normal) distribution, assumptions of normal distribution; application of normal distribution.
4. **Sampling and Distribution of Sampling Statistics** **3hrs**
   Concept of universe and sample; parameter and statistic; sampling methods; advantages of probability and non-probability sampling; sampling and non-sampling errors; central limit theorem; sampling distributions of mean and proportion.
5. **Parameter Estimation** **3hrs**
   Concept of estimation; characteristics of good estimator; point and interval estimates; construction of confidence interval; level of significance; margin of errors; construction of confidence interval for mean and proportion; determination of sample size.
6. **Hypothesis Testing** **6hrs**
   Meaning of hypothesis; hypothesis testing; types of errors; one tail and two tails; critical values; processes of hypothesis testing; hypothesis testing for mean and proportion for large sample (Z- test)
7. **Simple Linear Correlation and Regression** **9hrs**
   Concept of correlation; linearity and non-linearity; different approaches of measuring the relationship: scatter plot and statistical methods; Pearson's correlation coefficient; Spearman's rank correlation; coefficient of determination; test of significance of correlation coefficient using probable error; concept of regression; linear models; assumptions of linear models; simple linear regression model; computation of fitted regression models using least square method; interpretation of regression coefficients.

Laboratory Work:

1. Presentation of data (table, diagram and graph)
2. Computation of measures of central tendency and partition Values (ungrouped and grouped data)
3. Computation of measures of dispersion (ungrouped and grouped data)
4. Computation of Measures of skewness and kurtosis, Box and whisker plot
5. Computation of correlation coefficient.
6. Fitting of lines of regression.
7. Sampling distribution, random number generation, and computation of sample size
8. Computation of probability, descriptive statistics of probability distributions
9. Solving the numerical problem of probability distribution (Binomial, Poisson and Normal)
10. Fitting probability distributions in real data (Binomial, Poisson)
11. Computation of confidence interval estimation of Population mean
12. Problems solving of hypothesis testing (Z-test/ t-test).

Textbooks:

1. Sheldon, M. Ross. Introduction to probability and Statistics for Engineers and Scientists. Academic Press.
2. David, M & Levine et al. Statistics for Managers using MS Excel. Pearson Publication.

Project II

Program: Bachelor of Information Technology (BIT)
Course Code : CIT225
Year: 2
Semester: IV
Credit Hours : 2
Nature: PR

Course Objective:

- To provide the knowledge of the computer-based subjects studied in first and second year.
- To provide the knowledge on planning, design, development and implementation of the project.
- To teach students to work and solve problem in a team environment.
- To provide the knowledge to formulate project documentation and oral presentation.

Pre-requisite:

Computer Programming I, Web Technology I&II, Computer Programming II, Database Management System, Computer Networking, Computer Graphics.

Course Requirements:

The project must be started at the beginning of the semester and finished by the end of the very semester. Project must be chosen in an area related to Computer Science/IT/ICT/Software.

Project Work Phase:

The entire project work shall be divided into three phases and evaluation shall be done accordingly.

First Phase: The students are required to form a team of 2-3 members and come up with a conceptual framework for their project and must be documented in the form of a proposal as per the format provided by the department and presented in front of the examiner.

Second Phase: The students are required to show the progress of the work done. They must have finished the design phase including the overall system/architectural design and validation scheme. Progress Report must be presented in front of the internal evaluator committee.

Third Phase: All students must have finished all the phases before the final presentation. Students must come up with their output of the product they have developed and should give oral defense in the presence of external examiner

The project document shall include the following items:

- Project Team members (At least 2 and maximum of 3)
- Project supervisor(s)
- Technical description of the project.
- Project Analysis, Design, Coding, Testing and Implementation details.

Economics

Program: Bachelor of Information Technology (BIT)
Course Code : BCT311
Year: 3
Semester: V
Credit Hours: 3
Nature: TH

Course Objectives:

The aim of this course is to enhance understanding of economic theories and its application to develop student's skills in personal and professional decision making in the areas of management, IT, and business.

Course Description:

This course of Economics consists of the introduction to economic theories and application. It consists of theory of demand and supply, theory of consumer behavior, theory of production, cost and revenue curves, theory of product pricing and contemporary macroeconomics like national income accounting, money banking and international trades with reference to Nepal.

Course Outcomes:

After successful completion of the course the students shall be able to:

- understand the basic concept of microeconomics and macroeconomics;
- analyze the behavior of consumer, producer, market and firms and their decision-making process in practical sense;
- acquaint with concept of national income and its various concepts and their determinants;
- apprehend the role and importance of money, banking and international trade.

Course Details:

1. **Introduction** **3hrs**
    1.1 Introduction to economics with various definition and ten basic principles of economics
    1.2 Concept of microeconomics, macroeconomics; distinction between microeconomics and macroeconomics and their mutual interdependence
2. **Demand Supply and Elasticities** **8hrs**
    2.1 Concept of demand, demand schedule, demand curve, demand function and its types and law of demand
    2.2 Concept of supply, supply schedule, supply curve, supply function and its types and law of supply
    2.3 Concept and measurement of price, income and cross elasticity of demand
    2.4 Concept and measurement of elasticity of supply
3. **Theory of Consumer Behavior** **9hrs**
    3.1 Concept of utility and types of utility analysis
    3.2 Cardinal utility analysis: assumptions, consumer's equilibrium and criticisms
    3.3 Ordinal utility analysis: concept, indifference curve, marginal rate of substitution, budget line and consumers equilibrium, Price Effect: derivation of PCC, Income effect: derivation of ICC, substitution effect, decomposition of price effect into income and substitution effect (Hicksian approach)
4. **Cost and Revenue Curves** **7hrs**
    4.1 Concept to costs: opportunity cost, implicit cost, explicit cost, accounting cost, economic cost, fixed cost, variable cost
    4.2 Derivation of various short run cost curves (total, average and marginal) and long run cost curves
    4.3 Relationship between short run and long run AC and MC curves
    4.4 Concept of revenue: total revenue, average revenue and marginal revenue and revenue curves under perfect and imperfect competition
    4.5 Numerical exercises
5. **Market Structure** **7hrs**
    5.1 Perfect Competition: meaning and characteristics of perfect competition, short run and long run equilibrium of the firm and industry. Derivation of short run and long run supply curve of a firm and industry.

5.2 Monopoly: Meaning and characteristics of monopoly, short run and long run equilibrium of monopoly firm; price discrimination and degrees of price discrimination
5.3 Monopolistic competition: meaning and characteristics of monopolistic competition, short run and long run equilibrium of the firm and industry under monopolistic competition
5.4 Oligopoly: meaning and characteristics of oligopoly
5.5 Numerical exercises

6. **National Income Accounting** **3hrs**

6.1 Meaning and different concept of national income: GDP, NDP, GNP, NNP, national income (NI), personal income (PI), disposable income (DI), Per capita income (PCI) Real GDP, nominal GDP, GDP deflator

7. **Forms of Business Organization** **2hrs**

7.1 Sole trading concern, Partnership, Joint Stock Companies, State owned Enterprises, Multinational Companies

8. **Money, Inflation, Banking, and International Trade** **6hrs**

8.1 Concept and functions of money, value of money, concept of money supply (M1, M2 etc.)
8.2 Concept of inflation, its types, causes and effects of inflation
8.3 Banking and its types, role and functions of commercial bank, role and functions of central banks with reference to Nepal Rastra Bank
8.4 International Trade: concept and importance of international trade, concept of balance of payment (BOP), concept of balance of trade (BOT)

Practcal Works:

Excel and other relevant statistical software should be used to compute numerical exercise.

Instructional Methods

The general instructional pedagoy includes class lectures, presentations, group works, case studies, guest lectures, research works, project works, assignments. The teaching faculty will determine the choice of teaching pedagogy and statistical tools as per the requirements of topics.

Textbooks:

1. Dominick Salvatore, International Economics: (8th Ed.). Wiley India.
2. Dwibedi, D.N. (2001). Macroeconomic Theory and Policy. Tata McGraw-Hill Publishing Compnay Limited, New Delhi
3. Koutsoyianis, A. (1991). Modern Microeconomics, Hongkong: ELBS
4. Mankiew, N.G. Microeconomics, Newyork: Latest Edition
5. Shapiro, Edward. (2004). Macroeconomic Analysis. New Delhi: Galgotia Publication (P) Ltd

Mobile Application Development

Program: Bachelor of Information Technology (BIT)
Course Code : CIT311
Year: 3
Semester: V
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objective of the course are:

- Introduce students to mobile application development and its tools and technologies.
- Provide an understanding of the development environment for mobile platforms.
- Teach the basics of Android and iOS application development.
- Provide an understanding of mobile application architecture and its importance.
- Teach the techniques and tools for testing and debugging mobile applications.
- Provide an understanding of the process of deployment.

Course Description:

This course provides an introduction to mobile application development, covering the basics of Android and iOS application development, mobile application architecture, testing, debugging, and deployment.

Course Outcomes:

Expected outcome of the course are:

- Develop basic Android and iOS applications.
- Design user interfaces using UI components and native elements.
- Implement Model-View-Controller (MVC) architecture pattern in mobile apps
- Ensure data privacy and security in mobile applications.
- Test and debug mobile applications using different techniques and tools.

- Deploy mobile applications by bundling, signing and complying with app store guidelines and policies.

Course Details:

1. **Introduction to Mobile Application Development** **2hrs**
   - 1.1 Overview of Mobile Application Development
   - 1.2 History of Mobile Application Development
   - 1.3 Mobile application development Vs Web application development
   - 1.4 Types of Mobile Applications (Native, Hybrid, Web)
2. **Overview of Mobile Platforms** **4hrs**
   - 2.1 Mobile Application Platforms (iOS, Android, Cross-Platform)
   - 2.2 Choosing the right platform
   - 2.3 Understanding the development environment for mobile platforms
   - 2.4 Introduction to mobile application development lifecycle
3. **Mobile Application Architecture & Security** **12hrs**
   - 3.1 Understanding the basic architecture of mobile applications
   - 3.2 Introduction to Model-View-Controller (MVC) architecture pattern
   - 3.3 Introduction to data storage options in mobile applications
   - 3.4 Understanding application programming interfaces (APIs) and how to use them
   - 3.5 Understanding the importance of data privacy and security in mobile applications
4. **Introduction to Android Development** **12hrs**
   - 4.1 Introduction to Android Studio, Android SDK and emulator
   - 4.2 Basic Android application development using Java and XML
   - 4.3 Understanding of Android UI components and layouts
   - 4.4 Various sensors
   - 4.5 Local Storage SQLite, Realm etc.
   - 4.6 Firebase, Push Notifications
5. **Testing and Debugging Mobile Applications** **8hrs**
   - 5.1 Techniques and tools for testing and debugging mobile applications
   - 5.2 Introduction to unit testing in mobile application development
   - 5.3 Alpha testing and Beta testing in Mobile Application
6. **Deployment of Mobile Applications** **5hrs**
   - 6.1 Understanding bundling of mobile applications
   - 6.2 Introduction to signing of mobile application
   - 6.3 Understanding the importance of app store guidelines and policies
   - 6.4 Understanding the process of deployment

7. **Introduction to iOS Development** **2hrs**
    7.1 Introduction to Xcode, Swift and iOS simulator
    7.2 Basic iOS application development using Swift
    7.3 Basic features of Xcode Integrated Development Environment
    7.4 Understanding iOS Native UI elements

Laboratory Work:

The laboratory should contain all the features mentioned in a course, which should include

- Overview of Java programming language and its object-oriented concepts.
- Introduction to Android application architecture and terminology.
- Understanding the concept of resource folders in Android development.
- Understanding different layouts in Android, including linear layout, relative layout, toolbar, list view, recycler view, grid view, scroll view, view pager, and tab layout.
- Creating forms in Android.
- Understanding different types of dialogs in Android, including alert dialogs, toasts, and popups.
- Overview of shared preferences in Android.
- Understanding menus in Android, including option menu and context menu.
- Introduction to activity and fragment in Android, along with working with intents.
- Understanding themes and styles in Android.
- Overview of databases in Android, including simple queries and SQLite.
- Implementation of APIs in Android using libraries like Volley and Retrofit.
- Advanced concepts in Android, including threading, JSON parsing, and working with Google Play Services such as maps and GPS.

Practical questions:

1. Create a simple Android application that displays ”Hello World” using a text view.
2. Bundling Android application to APK for distribution & Installing Android application using AVD emulator or USB debugging.
3. Create a linear layout with two text views and a button. On button click, the text views should display the current date and time.
4. Implement a list view that displays a list of items with images and text.
5. Implement a form with several text input fields and a button.
6. Implement a pop-up window that appears when a button is clicked.

7. Implement a simple menu with two options that display a toast message when clicked.
8. Create an activity that uses an intent to navigate to another activity.
9. Implement a simple SQLite database with one table and display the table data in a list view.
10. Implement an API call using the Retrofit library to fetch data from a web service and display it in a list view.
11. Create a thread that displays a message after a specified amount of time.
12. Implement JSON parsing to fetch data from a web service and display it in a list view.
13. Use Google Maps API to display a map and mark a location.
14. Implement a toolbar with a search view that filters a list view based on the search query.
15. Create a tab layout with multiple tabs that display different data in a list view.

Textbooks:

1. Bill Phillips and Brian Hardy ”Android Programming” The Big Nerd Ranch Guide
2. Christian Keur and Aaron Hillegass ”iOS Programming” The Big Nerd Ranch Guide
3. Neil Smyth ”Android Studio Development Essentials”, Android 7 Edition
4. Bart Jacobs ”Beginning iOS 12 Programming with Swift”', Learn how to build apps from the ground up.

References:

1. F. Ahmed and M. E. Hoque, ”An Overview of Mobile Application Development: Challenges, Opportunities, and Future Trends,” IEEE Access, vol. 6, pp. 11231-11253, 2018.
2. C. Wu, H. Wu and Y. Tsai, ”A Comparative Study of Android and iOS App Development,” IEEE Access, vol. 6, pp. 34623-34633, 2018.
3. M. Abd-El-Barr, M. G. Sobh and A. E. Abdel-Maksoud, ”Mobile Application Development: Issues, Considerations and Guidelines,” International Journal of Advanced Computer Science and Applications, vol. 9, no. 5, pp. 298-306, 2018.
4. R. Mahendiran and S. P. Singh, ”Mobile Application Development for Android and iOS Platforms: A Comparative Analysis,” Journal of Information Systems Engineering & Management, vol. 2, no. 2, p. 13, 2017.
5. Google Android Developer Documentation: https://developer.android.com/docs

6. Apple iOS Developer Documentation: https://developer.apple.com/ documentation/
7. ”Android Studio Tips and Tricks” blog post by Google: https://android-developers.googleblog.com/2017/10/android-studio-tips-and-tricks.html
8. ”10 Best Practices for Mobile App Testing” blog post by Testlio: https://testlio.com/blog/10-best-practices-for-mobile-app-testing/
9. ”How to Publish an App on the App Store: A Step-by-Step Guide” blog post by Appinstitute: https://appinstitute.com/how-to-publish-an-app-on-the-app-store/

Numerical Methods

Program: Bachelor of Information Technology (BIT)
Course Code : BSM311
Year: 3
Semester: V
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of this course is to develop the techniques of solving complicated mathematical problems numerically with certain error and approximation, to provide sound knowledge of Numerical methods.

Course Description:

This course covers the different numerical computing techniques for non-linear equations, system of linear equations. interpolation, extrapolation using the concept differentiation and integration and numerical solution of differential equations with error analysis of each method.

Course Objectives:

At the end of the course students will be able to choose the appropriate numerical techniques and develop the algorithms to apply it, interpret the results, and assess accuracy of the used techniques to the different mathematical problems.

Note: Instructor may choose the suitable computer language (python or C) or software (Mathematica, Matlab, or Octave) to write the programs prescribed in the syllabus.

Course Details:

1. **Concepts of Numerical Computing, Approximations and Errors4hrs**
   1.1 Introduction, Process and Characteristic of Numerical Computing
   1.2 Computer Codes and Arithmetic: Representation of Numbers

1.3 Significant Digits, Errors: Inherent, Numerical, Modeling, Blunders, Absolute and Relative Errors
1.4 Stability, Convergence of Iterative Process, Error Estimation

2. **Numerical Method for Solving Nonlinear Equations** **7hrs**

2.1 Review: Algebraic, Polynomial, Transcendental Equations
2.2 Iterative Methods, Starting and Stopping Criteria
2.3 Bisection Method, Rate of convergence
2.4 Newton-Raphson Method, Rate of convergence.
2.5 Fixed Point Method, Rate of Convergence
2.6 System of Nonlinear Equations: Fixed point method and Newton-Raphson methods

3. **Solution of the System Linear Equations** **8hrs**

3.1 Review: Existence of Solution,
3.2 Direct Solution by Gaussian Elimination Method
3.3 Direct Solution by Gauss-Jordan Method
3.4 Jacobi Iteration Method
3.5 Gauss-Seidel Iteration Method
3.6 Convergence of Iterative Methods

4. **Curve Fitting: Interpolation, Regression** **8hrs**

4.1 Lagrange Interpolation Polynomial
4.2 Newton Interpolation Polynomial
4.3 Interpolation with Equidistant Points
4.4 Chebyshev Interpolation Polynomial
4.5 Error in Polynomial Interpolation
4.6 Fitting Linear Equations: least squares Regression
4.7 Fitting Transcendental Equations

5. **Numerical Differentiation and Integration** **9hrs**

5.1 Differentiating Continuous Functions: Forward, Central difference quotient
5.2 Differentiating Tabulated Functions, Error analysis, higher order derivatives
5.3 Richardson Extrapolation
5.4 Lower and Upper Sum
5.5 Newton-Cotes Methods
5.6 Trapezoidal Rule: Error Analysis, Composite trapezoidal rule
5.7 Simpson's 1/3 Rule, 3/8 Rule, Error analysis
5.8 Romberg Integration

6. **Numerical Solutions of Differential Equations** **9hrs**

6.1 Review: Order, Degree of Differential Equations, Linear and non-linear Differential Equation General and Particular Solutions, Initial value problems

6.2 Taylor Series Method, Euler's Method
6.3 Runge-Kutta Method of Order 2 and Order 4
6.4 Review: Partial Differential Equations
6.5 Parabolic Equations: Heat Equation
6.6 Elliptic Equations: Laplace's Equation, Poisson's Equation
6.7 Hyperbolic Equations: Wave Equation

Textbooks:

1. W. Cheney and D Kincaid "Numerical Mathematics and Computing", 7th Edition, Brooks/Cole Publishing Co. 2012
2. Cleve Moler, "Numerical Computing with MATLAB", SIAM, 2004

References:

1. W.Y. Yang,W. Cao, T.S. Chung, J. Morris "Applied Numerical Methods Using MATLAB", John Wiley & Sons, Inc, 2005
2. G.R. D.C. and T.B. Budhathoki "A Textbook of Numerical Methods", Heritage Publishers & Distributors Pvt.Ltd., 2078

Object Oriented Analysis and Design using UML

Program: Bachelor of Information Technology (BIT)
Course Code : CIT312
Year: 3
Semester: V
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objective of the course are

1. To study the main features of the software development process in an object-oriented framework.
2. To provide exposure to Visual Object-Oriented Modeling languages, specifically UML (Unified Modelling Language).
3. To develop a specification and implementation using UML from a given system requirements description.

Course Description:

This course contains introduction to object-oriented analysis and design, elaboration, System sequence diagrams, domain model, GRASP, class diagrams.

Course Outcomes:

After successful completion of this course, student will be able to

- understand the fundamental concepts and principles of UML, including the purpose and benefits of using UML in software development.
- model and visualize complex software systems or processes using UML.

Course Details:

1. **Object Oriented Fundamentals** **10hrs**

    1.1 Introduction
    1.2 Object Oriented Analysis and Design
    1.3 Iterative development and unified process

1.4 Case Study
1.5 Understanding Requirements
1.6 Use Case modeling, Relating Use cases — include, extend and generalization
1.7 Overview of the Unified Modeling Language: UML Fundamentals and Notations

2. **Object Oriented Analysis** **10hrs**

2.1 Representation of System Behavior
   i. System Sequence Diagram
   ii. System sequence diagram and use case
2.2 Building Conceptual/object Model
   i. Identifying object
   ii. Adding Associations
   iii. Adding attributes

3. **Object Oriented Design** **13hrs**

3.1 Writing contracts: Describing and Elaborating Use Cases
3.2 Collaboration Diagram and sequence diagram
3.3 Objects and Patterns (information expert, creator, low coupling, high cohesion, controller)
3.4 Determining Visibility
3.5 Class Diagram

4. **Implementation** **12hrs**

4.1 Programming and Development Process
4.2 Mapping Design to Code
4.3 Creating Class Definitions from Design Class Diagrams
4.4 Creating Methods from Collaboration Diagram
4.5 Updating Class Definitions
4.6 Classes in Code
4.7 Exception and Error handling

Laboratory Work:

Laboratory Exercise will include handling a object oriented design and modeling activity in a ACSE Environment. UML pattern design and modeling will be taken up with the help of UML Software.

Textbooks:

1. Larman, C., Applying UML and Patterns, Pearson Education Asia, 2008

References:

1. Martin Fowler UML Distilled: A Brief Guide to Standard UML
2. Booch, G., Jacobson, I., Rumbaugh, J., The Unified Software Development Process, Addison-Wesely, 2009.
3. Jacobson I., Object-Oriented Software Engineering – A Use Case Driven Approach, Addison-Wesely, 2009.

Research Methodology

Program: Bachelor of Information Technology (BIT)
Course Code : BCT312
Year: 3
Semester: V
Credit Hours : 3
Nature: TH

Course Objectives:

The objective of this course is to understand the principle of conducting business research in diverse setting. This course aims to deliver the skills to the students on data collection, analysis and interpretation of business-related issues. This course describes the basic concepts in research and business research and delivers the knowledge in the processes and steps to be followed in scientific research. It explains the process of searching relevant literatures through various sources.

Course Description:

It also introduces the basic research design used in business research, variable and their measurement and sampling techniques. It will also provide the basic information on data collection (qualitative and quantitative) and analysis and interpretation. It discusses the idea of research proposal preparation, report writing and interpreting the results.

Course Outcomes:

After completion of this course, the students should be able to:

- know the types, processes and value of research;
- describe the different types of research methods used in business research;
- Search the relevant literatures from different sources and develop the skill on writing literature review;
- choose an appropriate research design;
- choose the suitable variables and constructs for specific research problem and sampling methods;
- prepare the research instruments, test their reliability and validity;
- analyze the data using appropriate statistical tools;
- prepare the research proposal and report writing.

Course Details:

1. **Introduction to the Research** **7hrs**
   - 1.1 Concept of research; Nature and types of research.
   - 1.2 Scientific research and its process; Characteristics of scientific research; Application of scientific research in social science and management research.
   - 1.3 Research paradigms and contemporary issues; Approaches to research.
   - 1.4 Management research: Concept, Types and value of research in decision making processes.
   - 1.5 Ethical issues in business research.
   - 1.6 Status of business research in Nepal.
2. **Literature Review** **5hrs**
   - 2.1 Concept, Importance and function of literature review, Steps in literature review.
   - 2.2 Selection of appropriate literature; Searching literature through internet and looking for business issues.
   - 2.3 Practices of writing literature review.
3. **Research Problem, Research Question, Theoretical and Conceptual Framework** **5hrs**
   - 3.1 Concepts of research problem; Guiding principles of formulating research problem; Formulation of research questions and objectives.
   - 3.2 Theory: Theoretical framework, Conceptual framework, Role of theory in research; Deduction and induction.
   - 3.3 Concept of hypothesis: Types, Hypothesis formulation and good criteria of hypothesis formulation.
4. **Research Design** **5hrs**
   - 4.1 Concept, importance and components of research design; Research question and research design.
   - 4.2 Types of research design: Exploratory, Descriptive, Case study, Comparative, experimental, Cross sectional.
   - 4.3 Research designs commonly used in business research.
5. **Measurement, Scaling and Sampling** **9hrs**
   - 5.1 Measurement: Concept, Nature Types-nominal, ordinal, interval and ratio scales.
   - 5.2 Characteristics of measurements: Validity and reliability: Measurement of validity and reliability of research instruments; Relationship between reliability and validity; Sources of measurement problems.

5.3 Scale construction; Importance of scaling in research; Issues of scaling; Attitude measurement; Common scaling techniques used in business research with their advantages and disadvantages.
5.4 Sampling, types: Probability and non-probability with their major advantages and disadvantages; Choices of appropriate sampling methods; Key issues in sampling techniques.
5.5 Sampling and non-sampling errors; Minimization of sampling and non-sampling errors.

6. **Data Collection and Analysis** **9hrs**

6.1 Nature and sources of data; Methods of obtaining primary and secondary data.
6.2 Preparation of research instruments and questionnaire design: Principles, components and types.
6.3 Sources and method of obtaining qualitative data.
6.4 Data entry; Editing and processing; Preparation of frequency distribution; Diagrams and charts.
6.5 Data analyses: Descriptive statistics- Choices of appropriate measures of central tendency, Dispersion; Correlation and regression.
6.6 Inferential statistics - Construction of confidence interval for mean and proportion, Hypothesis testing for the measures of association, Relationship and differences in means.
6.7 Analysis of qualitative data.

7. **Research Proposal and Scientific Writing** **5hrs**

7.1 Selection of research title; Research proposal: Goal, Types and structure; Components of research proposal; Characteristics of good proposal.
7.2 Research reports-method of writing research report; Components of research report; Typing and layout of research report.
7.3 Citations and references.

Textbooks:

1. Zikmund, W. G. Business research methods. New Delhi: Thompson Publishing

References:

1. Cooper, D. R., Schindler, P. S., & Sharma, J. K. Business research methods. New Delhi: Tata McGraw Hill Ltd.

2. Saunders, M., Lewis, P., & Thornhill, A. Research methods for business students. New York: Pearson Education Ltd.
3. Bryman, A., & Bell, E. Business research methods. New Delhi. Oxford University Press.
4. Sekaran, U. Research methods for business: A skill building approach. New York: John Wiley and Sons.

Technical Proposal Writing

Program: Bachelor of Information Technology (BIT)
Course Code : BCT313
Year: 3
Semester: V
Credit Hours : 1
Nature: PR

Course Objectives:

The objectives of the course will be to enable students to:

1. differentiate different forms and types of proposal
2. write the structures and components of proposal and report
3. produce technical writing such as technical proposals and reports
4. cite the sources and keep the reference in MLA and APA format

Course Description:

The course of technical proposal writing course offers students the opportunity to develop an in-depth understanding of technical proposal and report writing. The course comprises five units which incorporates the essential aspects of proposal writing. General concept of technical proposal with types, components of proposal, grant proposal and contract documents, classification and structure of report, the approaches of research and citation and referencing are covered by splitting in different units.

Course Details:

1. **Basic Concept of Technical Proposal** **3hrs**
   Introduction of Proposal; Getting started on proposal; Types of proposal: solicited/unsolicited, informal/formal and internal/external; Persuasion in proposals – Logos, ethos, and pathos
2. **Structuring the Technical Proposal** **3hrs**
   Components of formal proposal: Title page, executive summary, statement of problem, rationale, objectives, literature review, significance of the study, methodology, limitation, cost estimate or budget, time schedule, feasibility, expected outcomes, references; common mistakes in proposal writing; models of technical proposal, Literature Review, Research questions

3. **Grant Proposal and Contract Documents** **3hrs**
   Defining grant proposal, grant writing steps: planning, identifying Grant makers and their requirements, research and drafting, writing, formatting and revising, making the submission; introduction of contract; types of contract documents, preparing sample contract documents.
4. **Report Writing** **3hrs**
   Introduction; Classification of reports, long report structure, short reports, model of technical report
5. **Research Project Work and Technical Writing** **3hrs**
   Research Approaches: qualitative, quantitative, and integrative; designing/creating the research; research document elements; In-text citation and referencing; APA, IEEE.

References:

1. Adhikari, Dharma, and Phanindra Upadhyaya (2013). Technical Communication II. Kathmandu.
2. Khanal, Arjun (2010), Communication Skills in English, Sukunda Pustak Bhandar, Bhotahity, Kathmandu
3. Sharma R.C. et al. (2009)," Business Correspondence and Report Writing": A Practical Approach to Business and Technical communication. Tata Mc Graw Hill.

Artificial Intelligence

Program: Bachelor of Information Technology (BIT)
Course Code : CIT323
Year: 3
Semester: VI
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objective of the course are:

- To provide the students with basic foundation on concepts of searching and knowledge representation in AI systems.
- To make students more pragmatic in knowledge of AI.

Course Description:

The course consists of Introduction, Agents and Environments, Informed and Uninformed Search, Knowledge Representation, Learning, Applications of AI, Production systems, and Uncertainty in AI.

Course Outcomes:

This course allows students to develop the AI skills, ROBOTIC process automation, and cognitive computing.

Course Details:

1. **Introduction** **4hrs**
   1.1 Artificial Intelligence (AI), AI Perspectives: acting and thinking humanly, acting and thinking rationally
   1.2 Turing test approach: Chinese room argument
   1.3 Foundations of AI
   1.4 Applications of AI
2. **Intelligent Agents** **4hrs**

- 2.1 Introduction of agents, Structure of Intelligent agent, Properties of Intelligent Agents
- 2.2 Relationship between agents and environments
- 2.3 Types of Agents: Simple Reflexive, Model Based, Goal Based, Utility Based.
- 2.4 Performance evaluation of agents: PEAs Description

3. **Problem Solving by Searching** **9hrs**

- 3.1 Definition, State Space Search: Water Jug problem, puzzle problem
- 3.2 Solving Problems by Searching, Search Strategies, Performance evaluation of search Techniques
- 3.3 Uninformed Search: Depth First Search, Breadth First Search, Depth Limited Search, Bidirectional Search
- 3.4 Informed Search: Greedy Best first search, A* search, Hill Climbing, Simulated Annealing
- 3.5 Constraint Satisfaction Problems, Constraint Satisfaction search

4. **Knowledge Representation** **10hrs**

- 4.1 Definition and importance of Knowledge, Issues in Knowledge Representation, Knowledge Representation Systems, Properties of Knowledge Representation Systems.
- 4.2 Types of Knowledge Representation Systems: Semantic Nets, Frames, Rule Based Systems,
- 4.3 Propositional Logic (PL): Syntax, Semantics and properties, Backward Chaining and Forward Chaining
- 4.4 Predicate Logic: FOPL, Syntax, Semantics, Quantification, Clausal Normal form Resolution

5. **Artificial Neural Networks** **9hrs**

- 5.1 Concepts of Learning, Supervised, Unsupervised and Reinforcement Learning
- 5.2 Learning with Neural Networks: Introduction, Biological Neural Networks Vs. Artificial Neural Networks (ANN), Mathematical Model of ANN, Types of ANN: Feed-forward, Recurrent, Single Layered, Multi-Layered,
- 5.3 Activation Function and Gradient Descent
- 5.4 Perceptron, Multilayer perceptron: Back propagation Algorithm.

6. **Applications of AI** **9hrs**

- 6.1 Expert Systems, Development of Expert Systems, Knowledge Engineering, Case Based Reasoning
- 6.2 Natural Language Processing, Fundamentals, Associated problems and steps in NLP
- 6.3 Genetic Algorithms
- 6.4 Fuzzy Logic, Fuzzy Inference system
- 6.5 Bayes Theorem

Laboratory Work:

Students are required to carry out at least 7 lab tasks on predicate calculus, searching and neural networks using LISP or Prolog. Some of the lab tasks may be on:

- Relationship programs (e.g., mother, father, brother etc)
- Recursive programs: Factorial, Fibonacci series etc
- Ancestor programs
- Tower of Hanoi (TOH) program
- Program for implementing Neural Networks for realization of AND, OR gates. Use library tools

References:

1. Russel S. and Norvig P., Artificial Intelligence: A modern Approach, Prentice Hall Ritch and Knight, Artificial Intelligence, Prentice Hall
2. Artificial Intelligence in the 21st century, Stephen Lucci, Danny Kopec, Mercury Learning and Information
3. Artificial Intelligence: Foundations of computational Agents, David L. Poole, Alan K. Mackworth, first edtion, Cambridge University Press

Digital Forensic Security Technologies

Program: Bachelor of Information Technology (BIT)
Course Code : CIT322
Year: 3
Semester: VI
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objectives of the course are

- Getting acquainted with basic concepts and computer forensics fundamentals
- Describe privacy and security concepts related to computer forensics.
- Introducing cybercrime and contrasting cybercrime from conventional crime.
- Introducing common forensics tools and best practices.

Course Description:

The Digital Forensic Security Technology course will be basically targeted towards the concept of computer forensics, cybercrime and the spectrum of available computer forensics tools. It also covers basic memory forensic investigation with introduction to different file systems, and the minimal knowledge of prevailing laws related to the subject matter. It also includes case study on some interesting cybercrime incidents.

Course Outcomes:

Upon completion of this course, students will be able to

- Describe computer forensics fundamentals and security concepts.
- Explain how cybercrime is different from conventional crime.
- Understand computer forensics technology and best practices for computer forensics.
- Simulate a memory-level OS forensic analysis.

Course Details:

1. **Introduction and basic concepts** **6hrs**
   - 1.1 Basic concepts
     - i. Networking Refresher (The OSI Layer, The TCP/IP Model, Subnetting)
     - ii. Encryption and Cryptography basics (E2EE, Hashing, Hashing Techniques, the CIA triad, concepts on steganography and cryptocurrency)
   - 1.2 Data Backup concepts
     - i. Storage Fundamentals (Understanding File Systems: FAT16, VFAT, FAT32, NTFS, EXT4)
     - ii. Basics of Data Backup and Restore, the 3-2-1 backup rule, RAID, Disaster Recovery and High Availability
   - 1.3 Proprietary and Open-source Systems
   - 1.4 Introduction to Digital Forensics
2. **Basic Security Concepts** **8hrs**
   - 2.1 Basic Security Concepts
     - i. Basics of Information Security, Network Security, Firewall, PAM, ACLs
     - ii. Computer and Network threats (including but not limited to Malware, Trojans, Adware, D/DoS, Ransomware, Honeypot, Social Engineering)
     - iii. Security, Privacy and Trust
   - 2.2 Introduction to Penetration Testing
     - i. Basic Concepts of Information Gathering, Vulnerability Analysis, Forensics Tools, Sniffing, Spoofing, Password Attacks and Reverse Engineering
   - 2.3 Threat, Vulnerability and Risk
   - 2.4 Data Confidentiality (Privacy, Confidentiality, and Secrecy)
3. **Cybersecurity and Cybercrime** **10hrs**
   - 3.1 Conventional Crime vs. Cybercrime
   - 3.2 Cyber Crimes
     - i. Phishing, Spamming, Web spoofing, Email spoofing, cyber stalking, pornography, commercial espionage, piracy, illegal betting, financial fraud, identity theft, and other trending cybercrimes
     - ii. Cybercrime prevention measures and cybersecurity
   - 3.3 Case Study in Cybercrime (LOVE-LETTER-FOR-YOU.TXT.vbs, Stuxnet, WannaCry, e.tc.)
4. **Computer Forensics Technology** **10hrs**
   - 4.1 Overview of Computer Forensics and Forensic Tools
     - i. Introducing Kali Linux

   ii. Nmap, Wireshark, Metasploit, FTK Imager, Autopsy, e.tc.
   iii. Data Recovery Tools (Photorec, AccessData, TSK, e.tc.)

4.2 Rules of Evidence (Admissible, Authentic, Complete, Reliable, Believable)

4.3 Good Forensic Practices (Identification, Securing the evidence, Collecting, Packaging, Preserving the evidence, Documenting the evidence, Documenting all changes)
   i. Getting Familiar with command line tools (SSH, Telnet)
   ii. Password-protected and Passwordless SSH
   iii. Network Monitoring tools, Live Threat Maps, Intrusion Detection System (IDS), Intrusion Prevension System (IPS), Packet Sniffing

5. **Operating System Investigation** **8hrs**

   5.1 Windows File Systems, Registry Hives and Event Logs
   5.2 Linux File System basics
   5.3 Android File System basics
   5.4 Memory acquisition and analysis in Windows

6. **Prevailing National Laws and Security Concepts** **3hrs**

   6.1 Prevailing Laws
      i. Electronic Transaction Act (Chapter 9)
      ii. The Privacy Act, 2075 (Chapter 6, 9, 10)
      iii. Cybersecurity Policy 2021 (Draft)
   6.2 Other Concepts
      i. Cyber Strategy of Government of Nepal
      ii. The CERT Team
      iii. Laws related to computers:- CFAA, DMCA, etc.
      iv. Open-Source Intelligence (OSINT)
      v. The OWASP Top ten

Laboratory Work:

1. Email header analysis
2. Network Forensics
3. Windows Forensics Analysis
4. Hardening your OS (Windows, Linux)

Textbooks:

1. Oettinger William, “Learn Computer Forensics”, Packt Publishing
2. Årnes André, “Digital Forensics”, Wiley & Sons Inc.

References:

1. Nelson Bill, “Guide To Computer Forensics and Investigations - Standalone Book”, Cengage Learning
2. Sikorski Michael, “Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software”, No Starch Press
3. Ligh Michael, “The Art of Memory Forensics: Detecting Malware and Threats in Windows, Linux, and Mac Memory”, Wiley
4. Casey Eoghan, “Handbook of Digital Forensics and Investigation”, Academic Press
5. Petzold Charles, “Code: The Hidden Language of Computer Hardware and Software”, Microsoft Press

Financial Accounting

Program: Bachelor of Information Technology (BIT)
Course Code : BCT322
Year: 3
Semester: VI
Credit Hours : 3
Nature: TH

Course Objectives:

The objective of this course is to acquaint the students with generally accepted accounting principles of financial accounting and their applications in business organizations. Its primary goal is to impart the basic knowledge and skills of accounting so that students will understand and work out how financial statements are prepared and analyzed in business firms.

Course Description:

This course gives students a strong foundation in financial accounting to prepare them for future business courses and the real world. It motivates students to learn by showing how accounting is important to a business. The major topics covered in this course are: nature and types of business, introduction to accounting, recording business transactions, journalizing, posting to accounts, preparing the trial balance, the adjusting process in book-keeping, preparing financial statements, accounting cycle, accounting systems, accounting standard, internal control & cash and analysis of financial statements.

Course Outcomes:

After the completion of the course the students should be able to:

- understand the nature of business and the role of accounting and ethics in business;
- understand the generally accepted accounting principles – major accounting rules, standards, and ways of reporting financial information;
- develop skills of journalizing transactions using the double-entry accounting system, posting of transactions to accounts, preparing trial balance and explaining how it can be used to discover errors;
- prepare and analyze financial statements;
- illustrate the accounting cycle for one period.

Course Details:

1. **Introduction to Accounting and Business** **4hrs**
   - 1.1 Nature of business; Types of businesses: Service businesses, Merchandising businesses, Manufacturing businesses; Forms of business entities.
   - 1.2 Meaning of accounting; Bases of accounting: Cash and accrual bases of accounting; Users of accounting information; Generally accepted accounting principles; Basic accounting concepts; Business transactions and the accounting equation.
   - 1.3 Financial statements; Ethics in accounting.
2. **Recording Business Transactions** **10hrs**
   - 2.1 Nature of business transactions: The basis for recording transactions; The role of source documents in recording transactions.
   - 2.2 Using accounts to record transactions; Double-entry accounting system; Use of accounting equation.
   - 2.3 Journalizing and posting to accounts.
   - 2.4 Sub-division of journal; Preparing the trial balance.
3. **The Adjusting Process** **5hrs**
   - 3.1 Nature of the adjusting process; Adjusting entries for accruals; Adjusting entries for deferrals; Adjusting entries for depreciation.
   - 3.2 Adjusted trial balance; Preparing worksheets.
4. **Financial Statements and Accounting Cycle** **10hrs**
   - 4.1 Brief overview of annual report of company and its components; Report of independent registered public accounting/auditing firm; Concept and types of financial statements.
   - 4.2 Preparation of Balance sheet and Income statement.
   - 4.3 Preparation of statement of owner's equity and statement of cash flows.
   - 4.4 Closing entries - Journalizing and posting closing entries; Accounting cycle.
5. **Accounting Systems and Standard** **5hrs**
   - 5.1 Basic accounting systems; Manual accounting systems; Computerized accounting systems.
   - 5.2 Accounting standards: Meaning, Nature, Need and significance of accounting standards; NAS, IAS, NFRS, IFRS.
6. **Receivable and Current Liabilities** **7hrs**
   - 6.1 Accounting for receivables: Account receivables and note receivables, Recognizing account receivables, Valuation of account receivables, Accounting for provision for doubtful debts and bad debts.

6.2 Accounting for current liabilities: Account payable, Tax payable, Contingent liabilities; Balance sheet presentation.

7. **Financial Misrepresentation, Internal Control and Cash** **4hrs**

7.1 Financial statement misrepresentation; Basic concept of forensic accounting.
7.2 Internal control; Cash controls over receipts and payments.
7.3 Bank accounts; Bank reconciliation statement; Petty cash funds.

References:

1. Warren, C. S., Reeve, J. M., & Duchac, J. E., Financial accounting. Delhi: Cengage Learning India Private Limited.
2. Norton, C. L., & Porter, G. A. Introduction to financial accounting. New Delhi: Cengage Learning India Private Limited.
3. Narayanswamy, R. Financial accounting: A managerial perspective. Delhi: PHI Learning Pvt. Ltd.
4. Accounting Standards Board. Nepal financial reporting standards (NFRS) 2018. Kathmandu: ASB

Human Computer Interface and UI Design

Program: Bachelor of Information Technology (BIT)
Course Code : CIT321
Year: 3
Semester: VI
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The general objectives of the course are

- To understand the key issues in the field of human-computer interaction.
- To know the approaches to designing and evaluating systems that emphasize usability, that are effective and that can be cost-justified.
- To be able to design and evaluate interfaces on the basis of relevant knowledge and appreciation of human capabilities applied to the tasks we increasingly depend on computer-based technology to perform.

Course Description:

This course is an introduction to the principles and methods of Human-Computer Interaction (HCI) and User Interface (UI) Design. Students will learn about the theories and concepts that underpin HCI, the methods used to evaluate and design interfaces, and the practical skills needed to develop effective UIs. Through a mix of lectures, readings, and hands-on exercises, students will develop a deep understanding of user-centered design and gain experience with a range of tools and techniques used in HCI.

Course Outcomes:

By the end of this course, students will be able to:

- Understand the fundamental concepts and theories that underpin HCI and UI design.
- Understand Characteristics of Graphical and Web User Interfaces
- Know the approaches to design and evaluate systems that emphasize usability
- Appreciate a range of human population capabilities and differences that impinge on interface design and use

- Identify needs and requirements for UI design
- Design and develop effective UIs using tools such as wireframing and prototyping software.
- Use user-centered approaches to interaction design
- Describe the evaluation paradigms and techniques used in interaction design.

Course Details:

1. **Introduction and Overview** **3hrs**
   - 1.1 Defining the User Interface
   - 1.2 The Importance of Good Design
   - 1.3 Introduction and history of Human Computer Interaction
2. **Characteristics of Graphical and Web User Interfaces** **6hrs**
   - 2.1 Interaction styles.
   - 2.2 The Graphical User Interface
     - i. The Popularity of Graphics
     - ii. The concept of direct manipulation.
   - 2.3 The concept of direct manipulation.
   - 2.4 The characteristics of graphical interfaces.
   - 2.5 The characteristics of Web interfaces.
   - 2.6 Web page versus Web application design.
   - 2.7 The general principles of user interface design
3. **Usability Engineering** **6hrs**
   - 3.1 Usability and its attributes
   - 3.2 Usability Trade-offs
   - 3.3 Categories of Users and Individual User Differences
4. **The Usability Engineering Lifecycle 4 User Profiling** **4hrs**
   - 4.1 The Human Action Cycle
   - 4.2 Important Human Characteristics in Design (perception, memory, visual acuity, foveal and peripheral vision, sensory storage, information processing, learning, skill, and individual differences.)
   - 4.3 Human Considerations in the Design
   - 4.4 Performance versus Preference
5. **Identifying needs and establishing requirements** **4hrs**
   - 5.1 Introduction to requirements
   - 5.2 Types of requirements
   - 5.3 Data Gathering Techniques
   - 5.4 Task Description and Analysis

6. **Prototyping and Construction** **6hrs**

   6.1 Introduction
   6.2 Prototyping and construction
      i. What is a prototype?
      ii. Why prototype?
      iii. Low-fidelity prototyping
      iv. High-fidelity prototyping.
      v. Compromises in prototyping
      vi. Construction: from design to implementation

7. **Conceptual and Physical Design** **6hrs**

   7.1 Conceptual design: moving from requirements to first design
      i. perspectives for developing a conceptual mode
      ii. Expanding the conceptual model
      iii. Using scenarios in conceptual design
      iv. Using prototypes in conceptual design
   7.2 Physical Design: getting concrete
      i. Guidelines for physical design
      ii. Different kinds of widget
   7.3 Tools Support

8. **User-centered approaches to interaction design** **4hrs**

   8.1 Advantages of involving users in development
   8.2 Principles of a user-centered design
   8.3 Understanding users' work: applying ethnography in design
      i. Coherence
      ii. Contextual Design
   8.4 Participatory Design
      i. PICTIVE
      ii. CARD

9. **Evaluation** **6hrs**

   9.1 Introduction (What, When and why)
   9.2 Formative vs Summative Evaluation
   9.3 Evaluation Paradigms
   9.4 Evaluation Techniques
   9.5 Relationship between evaluation paradigms and techniques
   9.6 DECIDE framework

Laboratory Work:

Students will develop and present a UI design for a real-world application or website where they will apply the principles and techniques learned throughout the course.

Textbooks:

1. The Essential Guide to User Interface Design: An Introduction to GUI Design Principles and Techniques, 3rd Edition, Wilbert O. Galitz
2. Interaction Design: Beyond Human-Computer Interaction 5th Edition by Helen Sharp, Jennifer Preece, Yvonne Rogers
3. Usability Engineering by JAKOB NIELSEN

References:

1. Human-Computer Interaction Preece J, Rogers Y, Sharp H, Benyon D, Holland S and Carey T, Addison-Wesley 1994
2. The Design of Everyday Things by Donald A. Norman
3. About Face The Essentials of Interaction Design Fourth Edition by Alan Cooper Robert Reimann David Cronin Christopher Noessel with Jason Csizmadi and Doug LeMoine
4. Designing the User Interface-4th Edition by Ben Shneiderman
5. Handbook of Human Factors and Ergonomics, Fourth Edition by Gavriel Salvendy

IT Project Management

Program: Bachelor of Information Technology (BIT)
Course Code : BCT321
Year: 3
Semester: VI
Credit Hours : 3
Nature: TH

Course Objectives:

The objective of this course is to provide a comprehensive knowledge to the students with the fundamental concepts of IT project management principles and integrate those principles with the managerial activities that they have to perform in future.This course teaches students how to develop approaches and management strategies for the software projects.

Course Description:

This course covers the introduction of project management that covers the concepts, roles and responsibilities of an IT manager. It allows the students to plan, analyze and perform project and managerial activities in all the fields including IT sector. This knowledge can be used in a variety of places, making the course useful in pursuing careers in IT management and related fields.The fundamental tasks of an IT manager has been incorporated in the course which seems a valuable opportunity for students to develop essential project management skills and knowledge.

Course Outcomes:

By the end of this course, students will be able to:

- Develop an understanding of the concept of various project management related tools and techniques.
- Acquiring the skills for using and managing the different IT Projects.
- Communicating, presenting and working as a role model for managing different projects.
- Getting the ideas of different trends in the IT sector for effective project management.

Course Details:

1. **Introduction** **8hrs**
   Basics of IT project and IT project management, Classification of IT projects, Components of project management, concepts of conventional IT project management and modern IT project management, Scope of project Management, project management cycle, project management framework
2. **Project Analysis and Time Management** **7hrs**
   Project planning and scheduling, Estimation of Activity resources and duration, Work Breakdown Structures, Planning model: Critical path Method(CPM) and Program Evaluation and Review Technique(PERT). Phases of IT project life cycle, line-of business organizations, project organizations
3. **Project Cost Management Process** **6hrs**
   Introduction, Types of cost estimates, cost budgeting, cost control methods and process, Earned Value Management and Project Portfolio Management
4. **Project Quality Management** **4hrs**
   IT Project quality, Quality requirements, Quality management planning, Quality Assurance, Different Quality tools
5. **Project Communication Management** **4hrs**
   Communication Planning requirement and planning procedures, Effective communication and Reporting
6. **Project Risk Management** **5hrs**
   Basics of risks, IT project risks, positive and negative risks, Risk management plan, types of risk analysis process, risk modeling techniques, risk examination and control
7. **Project Procurement Management** **4hrs**
   Procurement concepts, IT procurement management, planning for purchase and acquisition, contract management, Evaluation criteria, procurement legalities
8. **Custom Process in IT projects** **5hrs**
   Creating IT project management methodology, certification process flow, code of ethics, certification process flow
9. **Case Study** **2hrs**
   It should cover all the concepts for effective IT project management.

References:

1. Kathy Schwalbe, Information Technology Project Management, Seventh Edition, Cengage,(2013)
2. The Project Manager's Guide to Software Engineering's Best Practices, M.C. Christensen and R.H. Thayer, 2001,IEEE computer Society

3. Clifford F. Gray, Erik W. Larson, Project Management: The Management Process, MCGraw Hill
4. Software Project Management, A Unified Framework, Walker Royce, Pearson

Project III

Program: Bachelor of Information Technology (BIT)
Course Code : CIT324
Year: 3
Semester: VI
Credit Hours : 2
Nature: PR

Course Objectives:

The objectives of this course are to provide project management skills (developing, implementing, managing collaboration) and to learn working as a team. The student will also learn about formulating project documentation.

Course Description:

To develop small scale project based on the application development platforms and tools (JAVA, PHP, Computer Graphics, Python) or platform of any current trend. This course provides practical skill-based knowledge.

Project Work Phases:

The entire project work shall be divided into three phases and evaluation shall be done accordingly.

First Phase: The students are required to form a team of 2-3 members and come up with a conceptual framework for their project and must be documented in the form of a proposal as per the format provided by the department and presented in front of the examiner.

Second Phase: The students are required to show the progress of the work done. Project group must submit their progress report to their respective supervisor weekly. They must have finished the design phase including the overall system/architectural design and validation scheme. Progress Report must be presented in front of the internal evaluator committee.

Third Phase: All students must have finished all the phases before the final presentation. Students must submit and defend the project work before final board examination. All the project group must come up with their output of the product they have developed and should give oral defense in presence of external examiner

The project document shall include the following items:

1. Project Team members (At least 2 and maximum of 3)
2. Project supervisor(s)
3. Technical description of the project.
4. Project Analysis, Design, Coding, Testing and Implementation details.

Cloud Computing

Program: Bachelor of Information Technology (BIT)
Course Code : CIT413
Year: 4
Semester: VII
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The main objectives of this course are

- To provide students a sound foundation of the Cloud Computing so that they are able to start using and adopting Cloud Computing services and tools in their real-life scenarios.
- To provide theoretical as well as practical knowledge of cloud computing including designing, implementing and managing the cloud computing.

Course Description:

This course covers different concepts of cloud computing including introduction, architectures, cloud virtualization, programming models, security, and platforms and applications of cloud computing.

Course Outcomes:

After successful completion of this course, student will be able to

- Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing.
- Analyze various cloud programming models and apply them to solve problems on the cloud.
- Identify resource management fundamentals, i.e. resource abstraction, sharing and sandboxing and outline their role in managing infrastructure in cloud computing.

Course Details:

1. **Introduction to Cloud Computing** **5hrs**
   - 1.1 Evolution of Cloud Computing,
   - 1.2 Characteristics of Cloud Computing,
   - 1.3 Benefits and challenges of cloud computing,
   - 1.4 Cloud Storage, Cloud services requirements, Cloud adoption
2. **Cloud Virtualization technology** **10hrs**
   - 2.1 Cloud computing service models
     - i. Platform as service,
     - ii. Software as a service,
     - iii. Infrastructure as service,
   - 2.2 Deployment models
     - i. Public clouds
     - ii. Private clouds
       - A. Community cloud
     - iii. Hybrid clouds
3. **Cloud Virtualization Technology** **6hrs**
   - 3.1 Introduction to Virtualization
   - 3.2 Level of Virtualization (Desktop, Hardware,Storage, Application and Operating system)
   - 3.3 Benefits of virtualization
   - 3.4 Introduction to virtualization software
   - 3.5 Hypervisor and types
   - 3.6 Overview of Kubernetes
4. **Service Oriented Computing** **10hrs**
   - 4.1 Monolithic
   - 4.2 Service SoA, Sjab Services, Microservices
5. **Storage in Cloud Computing** **6hrs**
   - 5.1 Database as service, Storage services
   - 5.2 File Stoarage
   - 5.3 Block storage
   - 5.4 Archive Stoarge
   - 5.5 Object Storage
   - 5.6 Database: Relational and Non-relational
6. **Resource management in Cloud Computing** **7hrs**
   - 6.1 Sealing and its types, Way of scales
   - 6.2 Load Balancing, Redundancy and High availability
   - 6.3 Cloud Monitoring
     - i. SLA

ii. Billing
iii. Security

Laboratory Work:

The lab should cover at least the concepts given in each chapter.

References:

1. Raj Kumar Buyya, Christian Vecchiola, S. Thamarai Selvi, Mastering Cloud Computing
2. Arshdeep Bahga and Vijay Madiseti Cloud Computing: A hand on Approach

Data Mining and Warehousing

Program: Bachelor of Information Technology (BIT)
Course Code : CIT411
Year: 4
Semester: VII
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The course provides a broad overview of data mining and warehousing, including their application in business intelligence, customer relationship management, marketing, healthcare, and other fields. The course emphasizes practical hands-on experience with data mining and warehousing tools and techniques, enabling students to analyze data and solve real-world problems.

Course Description:

Data Mining and Warehousing are essential components of modern data-driven organizations. They help extract valuable insights from large datasets, make informed decisions, and gain a competitive edge in the market. The course aims to equip students with the necessary skills and knowledge to understand, develop, and apply data mining and warehousing techniques in real-world scenarios.

Course Outcomes:

Upon completing the course on Data Mining and Warehousing, students should be able to:

- Understand the key concepts, principles, and techniques of data mining and warehousing.
- Apply data preprocessing techniques to prepare data for mining and warehousing.
- Apply data mining algorithms, such as association rule mining, classification, and clustering, to extract meaningful insights from data.
- Identify and handle outliers in data.

Course Details:

1. **Introduction to Data Mining** **5hrs**
    1.1 Introduction to Data Mining
    1.2 The Origins of Data Mining
    1.3 Data Mining Tasks
    1.4 Data and Patterns used in Data Mining
    1.5 Technologies Used in Data Mining
    1.6 Major Issues in Data Mining
2. **Data Preprocessing** **7hrs**
    2.1 Data Objects and Attribute Types
    2.2 Basic Statistical Descriptions of Data
    2.3 Data Preprocessing
    2.4 Data Cleaning
    2.5 Data Integration
    2.6 Data Reduction: Principle Component Analysis
    2.7 Data Transformation and Data Discretization
    2.8 Measures of Similarity and Dissimilarity
3. **Data Warehousing and Online Analytical Processing** **7hrs**
    3.1 Data Warehouse: Basic Concepts and features
    3.2 Characteristics of Data Warehouse
    3.3 Extract, Transform and Load
    3.4 Multitiered Architecture
    3.5 Conceptual Modeling of Data Warehouse
    3.6 Data Warehouse Modeling: Data Cube, Data Mart and OLAP
4. **Classification** **10hrs**
    4.1 Basic Concepts: General Approach to Classification
    4.2 Decision Tree
    4.3 Rule-Based Classifier
    4.4 Nearest-Neighbor Classifiers
    4.5 Naive Bayes Classifier
    4.6 Ensemble Method: Random Forest
    4.7 Model Evaluation: Confusion Matrix- Accuracy, Recall, Precision and F1-score.
5. **Association Analysis** **5hrs**
    5.1 Basic Concepts
    5.2 Association Rules and Analysis
    5.3 Apriori Algorithm
    5.4 FP-Growth Algorithm
6. **Cluster Analysis** **6hrs**

6.1 Introduction: Basic Clustering Methods
6.2 K-means Clustering
6.3 Hierarchical Clustering
6.4 DBSCAN Clustering

7. **Anomaly Detection** **5hrs**

7.1 Introduction to Anomalies
7.2 Issues of Anomaly Detection
7.3 Approaches to Anomaly Detection
7.4 Statistical Approaches: Normal distribution and Box plot

Laboratory Work:

Using either MATLAB or Python or any other Data Mining tools (such as WEKA), students should practice enough on real-world data intensive problems like IRIS or Wiki dataset.

References:

1. Jiawei Han, Micheline Kamber, Jian Pei - Data mining - concepts and techniques-Elsevier, Morgan Kaufmann
2. P. Tan, M. Steinbach, V. Kumar - Introduction To Data Mining
3. Mark W. Humphries, Michael W. Hawkins, Michelle C. Dy - Data warehousing_ architecture and implementation-Prentice Hall PTR
4. Ponniah P., Reddy P. - Data Warehousing Fundamentals. Volume 1

Software Development and Operations (DevOps)

Program: Bachelor of Information Technology (BIT)
Course Code : CIT412
Year: 4
Semester: VII
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The key objectives of learning Software Development and Operations (DevOps)are:

- Learn and understand how various technologies work together in DevOps. Get a firm understanding in DevOps Processess, Tools and Technologies.
- Learn the basics of working in DevOps environments like Linux, AWS, Bash & Python Scripting, Jenkins, Ansible, Docker, Kubernetes and more

Course Description:

This course provides an introduction to the concepts, practices, and tools used in DevOps. Stu- dents will learn about the fundamental principles of DevOps and how they can be used to improve the software development and deployment process. The course will cover a range of topics including automation, continuous integration and delivery, testing, and monitoring. This course starts from very basics of command line, hands on demonstrations of many tools & technologies. Also most importantly it will show the readers how various technologies in DevOps work together by setting up their own projects.

Course Outcomes:

Upon completing the course on Data Mining and Warehousing, students should be able to:

- understand the fundamental principles and benefits of DevOps and how it can improve software development and deployment processes.
- apply continuous integration and delivery practices to automate the software build, test, and deployment process.

Course Details:

1. **Introduction** **3hrs**
   - 1.1 Introduction to DevOps: Definition, Scope, Application.
   - 1.2 Introduction to Continuous Integration(CI) and Continuous Delivery(CD).
   - 1.3 Prerequisites Info and Setup: Chocolatey (Windows), Homebrew(MacOS), AWS Setup, Linux Setup. Server Management in Linux.
   - 1.4 Basics of Networking
2. **DevOps Basics with Virtualization** **5hrs**
   - 2.1 Basics of Virtualization: Definition, Types and Heirarchy of Virtualizations.
   - 2.2 Learn Virtualization: Manual and Automatic. OS Environments: Linux, MacOS, and Win- dows
   - 2.3 Bash Scripting. Basics of Scripting.
   - 2.4 Variables, Conditons, Loops etc.
   - 2.5 Automating day to day Admin Tasks.
3. **DevOps on Cloud** **6hrs**
   - 3.1 Essentials of Cloud Computing
   - 3.2 Cloud and Virutalization Architecture
   - 3.3 Cloud Deployment Architecture
   - 3.4 Cloud Providers - An Overview
   - 3.5 Introduction to DevOps on Cloud Infrastructure like Amazon Web Service(AWS)
4. **GIT- A Version Controlling Tool** **6hrs**
   - 4.1 Introduction to Version Control
   - 4.2 Git - A CLI, Essentials of GIT in industry.
   - 4.3 Practice with GIT: Tracking, Monitoring, Ignoring, Staging, Merging etc.
   - 4.4 Working with Remotes
   - 4.5 GIT Workflows
5. **Jenkins** **5hrs**
   - 5.1 Essentials of Continuos Integration.
   - 5.2 Architecture of Jenkins.
   - 5.3 Jenkins Tool Management System.
   - 5.4 User Management in Jenkins.
   - 5.5 Jenkins Workshop: Authentication, Authorization.
   - 5.6 Maven Overview.
   - 5.7 Creating Jobs and Automatic Build Settings
   - 5.8 Building Delivery Pipeline. Plugin Management and Notification System.

6. **Docker** **8hrs**
   - 6.1 Introduction to Real World Transportation Challenges. Docker Introduction.
   - 6.2 Architecture of Docker
   - 6.3 Docker Containers: Creation, Share and Publish.
   - 6.4 Docker Networking: Network Types, Docker Compose.
   - 6.5 Docker Swarm: Introduction, Creating and Scaling docker bundles.
7. **Kubernetes** **4hrs**
   - 7.1 Introduction to Kubernetes. Kubernetes Cluster Architecture - Brief Overview.
   - 7.2 Concepts: Pods, Replica Sets, Deployments and Namespaces.
   - 7.3 Kubernetes Services and Networking
   - 7.4 Persistent volumes and persistent volume claims - An Overview.
   - 7.5 Design of PODS.
   - 7.6 Understanding labels, selectors, jobs and schedulers.
8. **Ansible** **8hrs**
   - 8.1 Introduction to Ansible: Basics and Architecture of Ansible. Control Machine and Managed Node Requirements
   - 8.2 Inventory: Hosts, Groups.
   - 8.3 Ansible Modules
   - 8.4 Adhoc Commands
   - 8.5 YAML script
   - 8.6 Playbook: Handlers and Variables, Tasks, Privilege Escalation.
   - 8.7 Roles: Directory Structure, Duplication and Execution, Default Variables, Dependencies, Search Path, Ansible Galaxy
   - 8.8 Including and Importing: Playbooks, Task Files and Roles.

Laboratory Work/Case Studies:

1. Study on different search engines like Google, Bing etc and key business strategies incor- porated by those engines.
2. Study of different web development frameworks for SEO.
3. A case study of SEO and digital marketing in a business(choosen by student) domain: Global and Local Perspective.

Textbooks:

1. Gene Kim, Patrick Debois, John Willis, Jez Humble and John Allspaw (2021). The DevOps Handbook: How to Create World-Class Agility, Reliability, and Security in Technology Organizations

2. Gene Kim, Kevin Behr, George Spafford(2018). The Phoenix Project: A Novel about IT, DevOps, and Helping Your Business Win.

References:

1. Jennifer Davis and Ryn Daniels(2016). Effective DevOps: Building a Culture of Collabora- tion, Affinity, and Tooling at Scale.
2. Jez Humble and David Farley(2010). Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation.

Technology Entrepreneurship

Program: Bachelor of Information Technology (BIT)
Course Code : BCT411
Year: 4
Semester: VII
Credit Hours: 3
Nature: TH

Course Objectives:

This study provides students with requisite concepts, skills of entrepreneurship, and factors affecting entrepreneurship so that students are able to apply them while launching and operating technology entrepreneurial ventures.

Course Description:

The course focuses on the fundamentals of technology entrepreneurship which has guided through the changes in the society. This course empowers the students for understanding the key aspects of entrepreneurship and the formation of new technology companies. Different topics allow the students to learn so that they can identify and evaluate innovation opportunities, and access the industry.

Course Outcomes:

After the end of the course, the students will be able to:

- Understand the key concepts of technology entrepreneurship.
- Acquire creativity for finding and evaluating opportunities in technology enterprises.
- Figure out the opportunities that fit students and gather the resources and manage the ventures.

Course Details:

1. **Introduction to Technology Entrepreneurship** **4hrs**

Definition of entrepreneur,Types of Entrepreneur, Fundamentals of entrepreneurship and technology entrepreneurship, Characteristics of Technology entrepreneurship, Technology based firms

2. **Technology Entrepreneur Development** **4hrs**
   Technological Entrepreneurship and Economic Development, Role of enterprises for economic development
3. **Opportunity identification and Analysis** **4hrs**
   Identification of the opportunities, conduction of market research, emerging idea for the probable business concept
4. **Development of IT Business Plan and Modeling** **3hrs**
   Understanding different business models, Creation of sustainable IT business model
5. **Learn StartUp in Software Development** **5hrs**
   Concept of lean startup methodology including minimum viable product, rapid prototyping and customer development in technology.
6. **IT Product Development, Marketing and Financial Management8hrs**
   Using ideas for Product development including design, development, testing and validation of the product, Marketing Strategies and tactics, building brand effective selling products and services, basics of financial statements, budgets
7. **Technology Intellectual Property, Legal and Regulatory** **7hrs**
   Issues in Technology, Protection of Intellectual Property including trademark, copyrights, Ideas regarding legal and regulatory issues that impact on startup including corporateformation, contracts, agreements
8. **Pitching and fundraising** **4hrs**
   Build pitch deck, raising capital, Knowledge of fundraising process.
9. **Scaling and growth** **3hrs**
   Approaches for scaling of business including hiring, partnerships.
10. **Technology Entrepreneurial Mindset** **3hrs**
    Developing the mindsets and skills needed to be a successful entrepreneur.

Laboratory Work/Case Study:

All the students are required to perform a case study.

References:

1. Barringer, B. R. and Ireland, R. D. Entrepreneurship: Successfully Launching New Ventures. Pearson Education, New Delhi.
2. Blundell, R, and Lockett, N. Exploring Entrepreneurship: Practices and Perspectives. Oxford University Press, New Delhi.

3. Kuratko, D. F. and Rao, T. V. Entrepreneurship: A South-Asian Perspective. Cengage Learning, New Delhi.
4. Technology Ventures, Richard Dorf and Thomas Byers, 1st Edition The Monk and the Riddle, Randy Komisar

Wireless Communication Systems

Program: Bachelor of Information Technology (BIT)
Course Code : CIT414
Year: 4
Semester: VII
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The main objectives of the course are:

- To provide the basic understanding of Wireless Technology.
- To be familiar with GSM and CDMA System.
- To understand Wireless standard and its future Trend.

Course Description:

The course is intended to prepare the students to understand the underlying theories related to wireless technology. The course includes history/fundamental concept of wireless communication and its future trend of transformation of the technology. The course is focused on architecture, services features and operation of GSM and CDMA technology and the basic fundamentals on the wireless network design.

Course Outcomes:

After completion of the course , students will be able to

- Know basics of Wireless Technology.
- Know Architecture of GSM and CDMA Technology.
- Know the facts underlying in the design of wireless communication system.

Course Details:

1. **Introduction** **6hrs**

Definition and History of wireless communication, Mode of Communication, Communication: TDD, FDD. Wireless Radio Frequency Spectrum, Evolution of Mobile Communication.

2. **Fundamental of Cellular mobile Communication** **6hrs**
   Concept of Cellular mobile system, Hexagonal Cell Geometry, Frequency Reuse and channel assignment strategy, Handoff, Coverage and capacity enhancement in cellular network: cell splitting, sectoring, Interference: Co-channel & Adjacent channel.
3. **Modulation/Multiple Access Technique** **10hrs**
   Fundamental concept of modulation and its type: A.M, F.M, P.M. Detail study on Amplitude Shift Keying, Frequency Shift Keying, Phase Shift Keying, Pulse Code Modulation, Differential Phase shift Keying (DPSK), Quadri phase Shift Keying (QPSK), Minimum Shift Keying: Gaussian Minimum Shift Keying(GMSK). Multiple Access: Frequency Division Multiple Access (FDMA), Time Division Multiple Access (TDMA), Code Division Multiple Access (CDMA), Spatial Division Multiple Access (SDMA).
4. **Radio wave Propagation/ Propagation Loss in Mobile Network 10hrs**
   Overview of Radio Wave Propagation. Review of Propagation mechanism: large scale path loss, Reflection, Diffraction, Scattering. Practical link budget design using path loss models: Indoor Propagation model (Partition loss, Long distance model), Outdoor Propagation Models(Okumura, Hata model, Longley-Rice model), Parameters of mobile multipath channel (time dispersion, coherence bandwidth, Doppler spread and coherence time), Fading: Small Scale Fading: flat, frequency selective, fast, slow, Rayleigh and Ricean fading distribution.
5. **Global System for Mobile Communication (GSM)** **5hrs**
   Introduction to Global System for Mobile Communication (GSM): System Architecture, radio sub-system, Services and features, channel types (traffic and control), frame structure, Frequency Hopping, Speech Channel, MSC, Call Handover, Transmitters, Authentication, Encryption and User Identification, Different Services.
6. **Recent Trends in Wireless Communication** **8hrs**
   LTE, Evolution beyond 5G, MIMO, NOMA

Laboratory Work:

- Study of AM/FM Modulation System.
- Study of AM/FM Demodulation System.
- Study of PCM System
- Study of ASK, FSK, PSK
- Study of Eye Diagram.
- Field Visit and Report

References:

1. TS Rappoport: Wireless Communications: Principles & Practice. Prentice Hall-1996.
2. S.Haykin, Digital Communication Systems, Latest Edition.
3. B.P Lathi, Analog and Digital Communication Systems, Latest Edition.

Big Data Technologies

Program: Bachelor of Information Technology (BIT)
Course Code : CIT421
Year: 4
Semester: VIII
Credit Hours: 3
Nature: TH + PR

Course Objectives:

The general objective of the course are to

- Understand the Big Data Platform and its Use cases
- Provide an overview of GFS and Map Reduce principles
- Provide an overview of Apache Hadoop
- Provide HDFS Concepts and Interfacing with HDFS
- Understand Map Reduce Jobs in Hadoop
- Provide hands on Hadoop Ecosystem
- Apply analytics on Structured, Unstructured Data.
- Provide an overview of Indexing and Searching

Course Description:

Big Data is the hot new buzzword in IT circles. The proliferation of digital technologies with digital storage and recording media has created massive amounts of diverse data, which can be used for marketing and many other purposes. The concept of Big Data refers to massive and often unstructured data, on which the processing capabilities of traditional data management tools result to be inadequate.

Course Outcomes:

After completion of the course , students will be able to

- Identify Big Data and its Business Implications.
- List the components of Hadoop and Hadoop Ecosystem
- Access and Process Data on Distributed File System
- Manage Job Execution in Hadoop Environment
- Develop Big Data Solutions using Hadoop Ecosystem

Course Details:

1. **Introduction To Big Data And Hadoop** **8hrs**
   Types of Digital Data, Introduction to Big Data, Big Data Analytics, Role of Distributed System in Big Data, Current trends in Big Data Analytics. Structure and Unstructured Data, Introduction to Nosql databases.
2. **Google File System And Map Reduce Framework** **10hrs**
   GFS: Architecture of GFS, read, write, append algorithms in GFS, Fault tolerance in GFS. Map Reduce: Anatomy of a Map Reduce Job Run, Architecture, Fault tolerance in Map Reduce, Job Scheduling, Shuffle and Sort, Task Execution, Parallel efficiency of MapReduce, Map Reduce Features.
3. **Apache Hadoop And HDFS(Hadoop Distributed File System)** **10hrs**
   History of Hadoop, Apache Hadoop, Analyzing Data with Unix tools, Analyzing Data with Hadoop, Hadoop Streaming, Hadoop Ecosystem. The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization Avro and File-Based Data structures.
4. **Hadoop EcoSystem** **10hrs**
   Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators.
   Hive : Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions.
   Hbase : Basics, Concepts, Clients, Example, Hbase Versus RDBMS.
   Spark : Introduction and Basic concepts
5. **Searching And Indexing** **7hrs**
   Full Text Indexing and Searching, Indexing with Lucene, Introduction to Apache Solr, Distributed search with Elasticsearch. Data store, list, update index and delete with elasticsearch.

Laboratory Work:

1. Installation of Apache Hadoop and familiar with HDFS.
   Setup HDFS in a single node to multi node cluster, perform basic file system operation on it using commands.
2. Map-Reduce on Apache Hadoop.
   Write various MR programs dealing with different aspects of it.
3. Hbase.
   Setup of Hbase in single node and distributed mode, write a program to write in Hbase and query it.

4. ElasticSearch.
   Setup elasticsearch in single and distributed mode, define template, write data in it and finally query it.
5. Apache Solr.
   Setup Apache Solr in single, define template, write data in it and finally query it.

Textbooks:

1. Tom White " Hadoop: The Definitive Guide" Third Edit on, O'reily Media.
2. Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley.

References:

1. Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)
2. Tom Plunkett, Mark Hornick, "Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R Enterprise and Oracle R Connector for Hadoop", McGraw-Hill/Osborne Media (2013), Oracle press.
3. Pete Warden, "Big Data Glossary", O'Reily, 2011.
4. Michael Mineli, Michele Chambers, Ambiga Dhiraj, "Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Wiley Publications, 2013.
5. Paul Zikopoulos, Dirk DeRoos, Krishnan Parasuraman, Thomas Deutsch, James Giles, David Corigan, "Harness the Power of Big Data The IBM Big Data Platform ", Tata McGraw Hill Publications, 2012.
6. https://research.google.com/archive/mapreduce-osdi04.pdf

Society, IT and Law

Program: Bachelor of Information Technology (BIT)
Course Code : BCT421
Year: 4
Semester: VIII
Credit Hours: 3
Nature: TH

Course Objectives:

The objective of this course is to provide students with an understanding of the relationship between society, IT, and law. Students will learn about the impact of IT on society, the legal framework for IT, and the ethical and social issues surrounding IT. The course will focus on the Nepalese legal framework and provide students with practical skills for navigating the digital landscape.

Course Description:

This course will cover the following topics: the role of IT in society, the legal framework for IT, cybercrime and cybersecurity, privacy and data protection, intellectual property rights and digital piracy, E-governance and cyber law in Nepal, and the impact of social media on society.

Course Outcomes:

Expected outcome of the course are:

- Develop an understanding of the impact of IT on society
- Understand the legal framework for IT and its application in Nepal
- Identify ethical and social issues surrounding IT and the digital landscape
- Develop practical skills for navigating the digital landscape in Nepal

Course Details:

1. **Introduction to Society, IT, and Law** **3hrs**
   - 1.1 Course objective and description 1hr
   - 1.2 Definition and scope of Society, IT, and Law 1hr

1.3 Interactions and relationships among Society, IT, and Law 1hr

2. **Ethical and Social Implications of IT** **6hrs**

2.1 Ethical considerations in IT and their impact on society 2hrs
2.2 Social implications of IT, including privacy and security concerns 2hrs
2.3 Legal frameworks and guidelines for addressing ethical and social issues in IT 2hrs

3. **Intellectual Property Law** **9hrs**

3.1 Overview of intellectual property law 3hrs
3.2 Copyright, trademark, and patent law 3hrs
3.3 Protection of intellectual property in the digital age 3hrs

Case Studies:

a. Apple vs. Samsung patent dispute: This is a well-known legal battle between two tech giants over alleged infringement of patents related to smartphones and tablets.
b. Nepal Music Copyright Society (NMC): In 2019, the NMC faced allegations of misusing royalty funds collected from music creators. This case highlights the importance of protecting intellectual property rights and the challenges faced by copyright societies in Nepal.

4. **Cybercrime and Cybersecurity Law** **9hrs**

4.1 Overview of cybercrime and cybersecurity law 3hrs
4.2 Types of cybercrime, including hacking and identity theft 3hrs
4.3 Legal and technological measures for preventing and responding to cybercrime 3hrs

Case Studies:
Nepal Rastra Bank data breach: In 2020, the Nepal Rastra Bank suffered a data breach that exposed sensitive information of its employees. This case underscores the need for stronger cybersecurity measures in Nepal and the potential risks of cybercrime.

5. **E-Commerce and Consumer Protection Law** **9hrs**

5.1 Overview of e-commerce and consumer protection law 3hrs
5.2 Legal frameworks and guidelines for online transactions 3hrs
5.3 Protection of consumer rights in e-commerce 3hrs

Case Studies:
Daraz Nepal's refund policy: In 2021, Daraz Nepal faced criticism over its refund policy after customers reported difficulties in obtaining refunds for products. This case raises questions about the adequacy of consumer protection laws in Nepal and the responsibilities of e-commerce platforms in protecting consumer rights.

6. **International and Comparative Law** **6hrs**

6.1 Overview of international and comparative law 2hrs
6.2 Differences and similarities among legal systems around the world 2hrs
6.3 The role of international law in governing IT and society 2hrs

7. **Emerging Issues in Society, IT, and Law** **3hrs**

7.1 Review of current and emerging issues in the field 2hrs
7.2 Discussion of potential future developments and their implications 1hr

References:

1. Information Technology Law: The Law and Society by Andrew Murray
2. Intellectual Property Law by Lionel Bently and Brad Sherman
3. E-Commerce Law: National and Transnational Topics and Perspectives by Julia Hörnle and Christian Twigg-Flesner
4. Cyberlaw: Text and Cases by Nandan Kamath

Internship

Program: Bachelor of Information Technology (BIT)
Course Code : CIT422
Year: 4
Semester: VIII
Credit Hours: 2
Nature: PR

Course Objectives:

Some of the key objectives of Internship are:

1. To engage in team-building activities and company events.
2. To adhere to professional ethics and workplace policies.
3. To learn to maintain a reflective journal to document experiences and insights.

Course Description:

The internship program provides students with an opportunity to gain practical experience in their chosen field of study. Through hands-on training and exposure to real-world projects, students will develop essential skills and knowledge that complement their academic learning. The program aims to bridge the gap between theory and practice, fostering professional growth and preparing students for their future careers.

Course Outcomes:

This course will enable students to

- gain hands-on experience in various technical areas, such as programming languages, database management, networking, and cybersecurity.
- develop a deeper understanding of the IT industry in Nepal, including its trends, challenges, and opportunities.
- develop critical thinking and problem-solving skills by working on real-world IT projects and addressing technical issues faced by the organization.
- enhance collaboration and project management skills by working closely with IT professionals and participating in team projects, fostering effective communication and teamwork.

- gain practical experience and skills that enhance employability in the IT industry, preparing for future career opportunities in Nepal's IT sector

Course Details:

1. **Introduction to Internship** **3hrs**
   - 1.1 Program orientation and introduction to internship guidelines
   - 1.2 Familiarization with the organization's IT infrastructure and systems
   - 1.3 Overview of the IT department's roles and responsibilities
   - 1.4 Defining internship goals and learning objectives
2. **Understanding the organization** **6hrs**
   - 2.1 Shadowing IT professionals in different areas (e.g., networking, software development, cy- bersecurity)
   - 2.2 Assisting with routine IT tasks (e.g., system maintenance, troubleshooting, software instal- lations)
   - 2.3 Introduction to IT project management methodologies
   - 2.4 Familiarization with IT documentation and reporting practices
3. **Collaborating in a project** **9hrs**
   The project could be related to:
   - 3.1 Collaborating with the IT team on ongoing projects
   - 3.2 Hands-on experience with programming languages and frameworks
   - 3.3 Networking configuration and administration
   - 3.4 Database management and optimization
   - 3.5 Exposure to cybersecurity practices and risk management
4. **Undertaking own Project** **9hrs**
   - 4.1 Undertaking independent projects aligned with personal interests and goals
   - 4.2 Implementing software solutions or developing web applications
   - 4.3 Conducting research on emerging technologies and trends
   - 4.4 Participating in code reviews and quality assurance processes
   - 4.5 Gaining exposure to IT project planning and execution
5. **Reporting** **3hrs**
   - 5.1 Finalizing and presenting the internship project
   - 5.2 Conducting a comprehensive analysis of the project's outcomes
   - 5.3 Reflecting on personal and professional growth during the internship
   - 5.4 Participating in an internship review and evaluation process
   - 5.5 Receiving a certificate of completion and recommendations for future career development

Internship Notes:

The specific activities and projects within the IT internship program can be tailored to align with the student's area of interest and the hosting organization's needs. The intership program requires its participants to maintain a log report signed by the as- signed supervisor. After the internship program, the participant will create a video/text article to reflect the changes before and after the program.

Project IV

Program: Bachelor of Information Technology (BIT)
Course Code : CIT423
Year: 4
Semester: VIII
Credit Hours: 6
Nature: PR

Course Objectives:

The objectives of the course are

- To provide the knowledge of the computer-based subjects studied in up to third year.
- To develop theoretical and practical skills needed to develop real world software applications using different software development tools and techniques.

Course Description:

This course covers theoretical and practical concepts needed to develop a real-world software system. The course focuses on enabling students with the skills related to software development.

Nature of Project:

The project work should include development of an application/system software. Students are highly recommended to implement relevant algorithms, flowcharts and the theories they have learned. The project can be done in group with at most three members in each group. The students can choose any programming languages for the implementation of the project. Also, students should be able to write their own program modules rather than relying on predefined APIs.

Phases of Project:

The following are the phases of project work:

1. **Proposal Submission and Presentation:** Students must submit and present project proposal in a prescribed format.
2. **Mid-Term:** Students must submit progress report and should defend midterm progress by the mid of the semester.
3. **Final Submission:** Students must submit and defend the project work before final board examination. The final defense will be evaluated by an evaluation committee in presence of external examiner. The report should be submitted in standard format as prescribed. The hard/soft copy of report should be submitted before one week of the final presentation.

Report Contents:

1. Prescribed content flow for the project proposal
   a. Introduction
   b. Problem Statement
   c. Objectives
   d. Methodology
   e. Schedule
   f. Expected Outcome
   g. References
2. Prescribed Content for the project report
   a. Cover & Title Page
   b. Certificate Page
   c. Acknowledgement
   d. Abstract Page
   e. Table of Contents
   f. List of Abbreviations, List of Figures, List of Tables
   g. Main Report
   h. References
   i. Bibliography (if any)
   j. Appendices

Report Format Standards:

A. Page Number
   The pages from certificate page to the list of tables/ figures/ abbreviations/ approvals should be numbered in roman starting from i. The pages from chapter 1 onwards should be numbered in numeric starting from 1. The page number should be inserted at bottom, aligned center.
B. Page Size and Margin

The paper size must be a page size corresponding to A4. The margins must be set as

- Top = 1”
- Bottom = 1”
- Left = 1.25”
- Right = 1”

C. Paragraph Style
All paragraphs must be justified and line spacing of 1.5.

D. Text Font of Document

- The contents in the document should be in Times New Roman font
- The font size in the paragraphs of document should be 12

E. Section Headings
Font size for the headings should be 16 for chapter headings, 14 for section headings, 12 for sub-section headings. All the headings should be bold faced.

F. Figures and Tables
Position of figures and tables should be aligned center. The figure caption should be centered below the figure and table captions should be centered above the table. All the captions should be of bold face with 12 font size.

E-Governance

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The course aims to familiarize students with the knowledge of E-Governance, different E-Governance models, necessary infrastructure development, implications of emerging technologies, and various implementation issues.

Course Description:

This course builds a foundation for understanding the concepts of E-Governance. It covers the importance, need, and opportunities of E-Governance; infrastructure for implementation; existing models; challenges; and the impact of technologies such as data mining, big data analytics, mobile and cloud computing, and social media. It includes success scenarios and relevant case studies.

Course Outcomes:

By the end of the course, students should be able to:

- Demonstrate sufficient theoretical and conceptual grounding on E-Governance, including its importance and role in public service delivery.
- Understand and analyze the theoretical and practical issues involved in the development of ICT infrastructure for E-services.
- Familiarize themselves with emerging technologies and their integration into E-Governance and related services.

Course Contents:

1. **Introduction** **5 hrs**
   ICTs and Development, Role of ICT in Public Services, Evolution, Scope

and Content of E-Governance, Difference between E-Government and E-Governance, Benefits and Challenges, Global Trends in E-Governance.

2. **Models of E-Governance** **10 hrs**
   Digital Delivery of Public Services, G2C, G2B, G2G, G2E Models, Models of Digital Governance: Broadcasting, Critical Flow, Comparative Analysis, Mobilization and Lobbying, Interactive Service Models, Evolution and Maturity Models of E-Governance.
3. **E-Government Infrastructure Development** **9 hrs**
   E-Readiness and Assessments, Legal, Institutional, Human, and Technological Preparedness, Network and Computing Infrastructure, Data Centers, E-Government Architecture, iCloud Governance, Data Systems, ICT Penetration Issues.
4. **Security for E-Governance** **6 hrs**
   Security Issues and Challenges, Security Frameworks, Security Management Models, E-Government Security Architecture, Security Standards.
5. **Applications of Emerging Technologies on E-Governance** **5 hrs**
   National Data Warehouses (Census, Commodities), Use of Data Mining in Agriculture, Health, Education, Planning, Commerce, Entertainment, and Other Sectors. Social Media, Mobile, and Cloud Computing Implications.
6. **Case Studies** **10 hrs**
   E-Governance in Nepal (Success Stories), Cyber Laws, IT Policy, Local Level E-Governance (Smart Nagarpalika), Applications: Land Reform, HRM Software, NICNET, CARD, Reservoir Monitoring, Andhra Pradesh Initiatives, Ekal Seva Kendra, Sachivalaya Vahini, Bhoomi, IT in Judiciary, E-Khazana, E-Seva, E-Panchayat, NIC Services, SAARC Region E-Governance.

Laboratory Work:

Students are required to submit a case study on E-Governance.

Text Book:

1. C.S.R. Prabhu (2012): *E-Governance: Concepts and Case Studies*, Second Edition, PHI Learning.

Reference Books:

1. P.K. Suri and Sushil (2019): *Strategic Planning and Implementation of E-Governance*, Springer.

2. Backus, Michiel (2001): *E-Governance in Developing Countries*, IICD Research Brief No. 1.
3. Richard Heeks (2006): *Implementing and Managing E-Government*.

IOT and Smart Technologies

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives

Some of the key objectives of learning IoT and Smart City are:

1. Understand the basic concepts of IoT
2. Plan an IoT implementation
3. Manage security, privacy, and safety risks on IoT projects
4. Manage an IoT prototyping and development project throughout the development lifecycle

Course Description

The Internet of Things (IoT) refers to the increasing number of devices that are not traditional computers but are connected to the internet to transmit data and receive instructions. This includes smart appliances, virtual assistants, and internet-enabled sensors used across various industries.

IoT extends the power of the internet to physical objects, enabling interaction with minimal human input. A key application of IoT is in smart cities, where urban infrastructure is managed using smart technologies. This course covers the foundational knowledge of IoT and explores related smart city technologies including Building Information Modeling (BIM), Geographic Information Systems (GIS), and Artificial Intelligence (AI).

Course Outcomes

1. Understand fundamental concepts, technologies, and challenges related to IoT and smart cities.
2. Develop creative designs to transform a municipality into a smart city.

Course Content

1. **Overview of Internet of Things (IoT)** (4 Hrs)
   1.1 Definition of IoT
   1.2 Trends of IoT: History and industry growth
   1.3 Fundamental components of an IoT system
   1.4 Applications of IoT
2. **Sensors and Devices used in IoT** (6 Hrs)
   2.1 Overview of sensors and devices
   2.2 IoT device hardware
   2.3 Scaling, manufacturing, and shipping
   2.4 Gateways
3. **Connectivity** **4 Hrs**
   3.1 Introduction to connectivity
   3.2 Cellular connectivity
   3.3 Satellite connectivity
   3.4 WiFi/Bluetooth/LPWAN connectivity
4. **Data Processing** **4 Hrs**
   4.1 Introduction to the cloud
   4.2 Introduction to Internet of Things (IoT) platforms
      4.2.1 Organization Strategies to choose an IoT platform
      4.2.2 Types of IoT platforms
      4.2.3 Needs to migrate to an IoT Platform.
   4.3 EDGE and FOG computing
   4.4 APIs to integrate IoT to custom application.
5. **IoT Protocols and Machine Learning** **9 Hrs**
   5.1 Overview of IoT protocols
   5.2 Internet of Things (IoT) Protocals: Data protocols:
      5.2.1 MQTT
      5.2.2 CoAP
      5.2.3 AMQP
      5.2.4 M2M Protocol
      5.2.5 XMPP
   5.3 Internet of Things (IoT) Protocals: Network protocols:
      5.3.1 HTTP
      5.3.2 LoRaWAN
      5.3.3 Bluetooth
      5.3.4 ZigBee
   5.4 Machine learning for IoT
6. **User Interface and User Experience in IoT** **2 Hrs**

6.1 Introduction to UI/UX
6.2 Key considerations
6.3 Recent trends in UI/UX design for IoT

7. **Security and Privacy in IoT** **4 Hrs**

7.1 IoT Security challenges and threats
7.2 Authentication and access control in IoT systems
7.3 Data privacy and compliance in smart cities
7.4 Security best practices for IoT deployments

8. **IoT for Smart Cities** **3 Hrs**

8.1 Introduction to smart cities
8.2 Needs for smart city transformation
8.3 Role of IoT in smart cities.
8.4 Case studies of smart city.

9. **Other Technologies in Smart Cities** **3 Hrs**

9.1 Building Informatin Modeling (BIM) and Geographical Information System (GIS) value for smart cities:
9.1.1 BIM dimensions
9.1.2 Sustainability and facility management in BIM
9.1.3 BIM lifecycle
9.1.4 Integrating BIM and GIS
9.2 Big Data
9.3 Case Study: Applications of Smart Cities

10. **Smart City Design** **6 Hrs**

10.1 Smart energy management and grid systems
10.2 Smart transportation and traffic management
10.3 Environmental monitoring and sustainability
10.4 Smart buildings and infrastructure
10.5 Citizen participation in smart city initiatives

Project-Based Learning

1. Students work on a hands-on project related to IoT and smart cities.
2. Guided development of smart city concept for the student's municipality.
3. Regular progress reviews and feedback sessions with stakeholders.
4. Final presentation and demonstration.

Visual Programming

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

- To gain a solid understanding of Visual Programming concepts and their applications in C#.
- To gain practical skills in database connectivity, and advanced topics such as multithreading and asynchronous programming to create robust and interactive applications.

Course Contents:

1. **Introduction to Visual Programming and C# Basics** **(8 hours)**

   1.1 Introduction to Visual Programming and its benefits
   1.2 Visual Studio Integrated Development Environment (IDE)
   1.3 Introduction to .NET framework
   1.4 Creating, Compiling, and executing .NET application
   1.5 C# Program Structure
   1.6 Basic C# Language Constructs: Variables, data types, operators, statement and Namespace in C#
   1.7 Control flow statements (decision making and looping), foreach loop
   1.8 Array and String
   1.9 Introduction to System Collection
   1.10 Functional Programming: LINQ and Querying Collections (array, list etc.) using LINQ

2. **Object-Oriented Concept** **(7 hours)**

   2.1 Object-oriented programming (OOP): Principles, Advantage, and Disadvantage
   2.2 Classes and Objects in C#
   2.3 Properties: Get Accessors and Set Accessors
   2.4 Encapsulation, inheritance, and polymorphism
   2.5 Constructors and Destructors
   2.6 Inheritance and its types (single, multiple, multilevel)

2.7 Method overriding and abstract classes
2.8 Interfaces and their implementation in C#
2.9 Exception handling in C#

3. **Graphical User Interface (GUI) Development** **(15 hours)**

3.1 Graphical System: Advantage, Disadvantage, and Characteristics
3.2 Introduction to GUI development with Windows Forms and WPF
3.3 Common GUI controls: buttons, labels, textboxes, checkbox, combo box, radio button, list box and list view
3.4 Containers: Panel, Group box and Tab Control
3.5 Menus and Toolbar: Context menu Strip, menu strip, status strip and tool strip
3.6 Dataset and Data Grid View
3.7 Control Properties and Layout
3.8 Dialog: Color Dialog, Folder Browser Dialog, Open File Dialog, Save File Dialog, Font Dialog and Message Dialog
3.9 Delegate and Event Handling Mechanism: Mouse Event Handling, Keyboard Event Handling, Window Event Handling and handling events associated with common GUI controls
3.10 Data Binding and Validation in GUI applications
3.11 Introduction to Windows Presentation Foundation (WPF)
3.12 Layout management and organizing controls in WPF
3.13 Building responsive and interactive WPF applications
3.14 Using advanced controls and styling in WPF
3.15 Multiple Document Interface (MDI) Windows

4. **Database Connectivity** **(10 hours)**

4.1 Introduction to Databases and SQL
4.2 ADO.NET basics: Connection, Command, Reader, and Adapter classes
4.3 Connection string and connecting to a database in C#
4.4 Querying and Saving data to database: Create, read, update, and delete records
4.5 Entity Framework (EF) and Object-Relational Mapper (ORM)
4.6 Adding Entity Framework to Winform application: Database Providers, LINQ to Entities, Entity Classes, DbContext and Entity Data Model
4.7 Querying a database using LINQ

5. **Asynchronous Programming** **(5 hours)**

5.1 Multithreading and asynchronous programming: Principle of Asynchrony, Async/Await patterns in C#
5.2 Executing an Asynchronous Task from a GUI Application
5.3 Working with timers and background workers

Laboratory Work:

- The laboratory work includes creating GUI applications covering most of the concepts of above units using C# and .NET Framework.
- Students should submit a final project work that uses the concept of Visual Programming learned in this course.
- The evaluation of the practical work should be based on project work.

Textbook:

1. Dietel P. and Dietel H., 2018, *Visual C#: How to Program*, Sixth Edition, Pearson Education Asia

Reference Book:

1. Albahari J., Albahari B., 2021, *C# 9.0 in a Nutshell: The Definitive Reference*, First Edition, O'Reilly Media

E-Business and E-Commerce

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

- Provide students with an understanding of e-business and e-commerce.
- Teach technologies and strategies necessary to conduct business online.
- Emphasize the importance of e-business and e-commerce in today's digital economy.
- Introduce key concepts, tools, and techniques essential for success in the field.
- Prepare students for the rapidly evolving nature of e-business and e-commerce.

Course Description:

This course introduces students to e-business and e-commerce through lectures, discussions, and practical exercises. It covers technologies and strategies used in online business, including web design and development, online marketing, social media, mobile apps, e-payment systems, security and privacy, and customer relationship management. Students will explore business models, challenges, and opportunities inherent to the online marketplace.

Course Outcomes:

Upon completion of this course, students will be able to:

- Understand fundamental concepts and principles of e-business and e-commerce.
- Analyze and evaluate various e-business and e-commerce models.
- Design and develop a basic e-commerce website using appropriate tools.
- Apply effective online marketing techniques.
- Address key security and privacy concerns in e-commerce.
- Evaluate the impact of social media on e-commerce businesses.
- Implement effective customer relationship management strategies.

Course Contents:

1. **Introduction to E-Business and E-Commerce** **6 hrs**
   1.1 What is E-Business and E-Commerce (2 hrs)
   1.2 Importance (1 hr)
   1.3 Types of E-Commerce (1 hr)
   1.4 E-Commerce vs. Traditional Commerce (2 hrs)
2. **Building an E-Commerce Website** **9 hrs**
   2.1 Website Design and Development (3 hrs)
   2.2 Content Management Systems (CMS) (2 hrs)
   2.3 E-Commerce Platforms (2 hrs)
   2.4 Payment Systems (2 hrs)
3. **Online Marketing for E-Commerce** **9 hrs**
   3.1 Digital Marketing Channels (2 hrs)
   3.2 Search Engine Optimization (SEO) (3 hrs)
   3.3 Pay-Per-Click (PPC) Advertising (2 hrs)
   3.4 Social Media Marketing (2 hrs)
4. **Mobile E-Commerce** **6 hrs**
   4.1 Mobile Commerce Platforms (2 hrs)
   4.2 Mobile Payment Systems (2 hrs)
   4.3 Mobile App Development (2 hrs)
5. **Security and Privacy in E-Commerce** **6 hrs**
   5.1 Online Security Threats (2 hrs)
   5.2 Security Solutions (2 hrs)
   5.3 Privacy Concerns (2 hrs)
6. **Social Media and E-Commerce** **6 hrs**
   6.1 Role of Social Media (2 hrs)
   6.2 Social Media Strategies (2 hrs)
   6.3 Analytics and Metrics (2 hrs)
7. **Customer Relationship Management in E-Commerce** **3 hrs**
   7.1 CRM Strategies (1 hr)
   7.2 Retention and Loyalty Programs (1 hr)
   7.3 Personalization (1 hr)

Laboratory Work:

- Design and develop an e-commerce website using CMS and e-commerce platforms.
- Implement online payment systems.
- Create and manage digital marketing strategies (SEO, PPC, social media).
- Analyze and optimize social media marketing campaigns.
- Develop a mobile e-commerce app with mobile payment functionality.
- Conduct security audits and implement mitigation strategies.
- Formulate CRM strategies with loyalty and personalization features.

- Analyze website and social media data to drive decision-making.
- Collaborate on a comprehensive e-commerce business plan (website, marketing, financials).

Textbook:

- Kenneth C. Laudon and Carol Guercio Traver, *E-Commerce 2021: Business, Technology, Society*.

References:

1. Alan Davidson, *E-Commerce: A Guide to the Law of Electronic Business*.
2. Kenneth C. Laudon and Carol Guercio Traver, *E-Commerce Essentials*.
3. Ryan Deiss, *Social Media Marketing for E-Commerce*.
4. Mark Grossman, *E-Commerce and Internet Law: Treatise with Forms*.
5. Wen-Chen Hu, *Mobile Commerce: Opportunities, Applications, and Technologies of Wireless Business*.

Mobile Ad Hoc Network

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

1. Study of ad-hoc/sensor networks
2. Study of End to End Delivery and Security.
3. Study of Media Access Control (MAC) Protocols.
4. Study of Routing Protocols.
5. Study of Networking Sensors and Applications.

Course Description:

Wireless Ad-hoc Networks will provide a technical overview and introduction to the topic of wireless ad-hoc networks. Wireless ad-hoc networks will be defined. Major requirements and challenges of wireless ad-hoc networks will be covered. The solution space and related technologies at different layers will be discussed.

Course Outcomes:

On successful completion of this course, students will be able to:

1. Describe the unique issues in ad-hoc/sensor networks. This will be assessed through assignments and seminars.
2. Describe current technology trends for the implementation and deployment of wireless ad-hoc/sensor networks. This will be assessed through assignments and classroom interaction.
3. Discuss the challenges in designing MAC, routing and transport protocols for wireless ad-hoc/sensor networks. This will be assessed through assignments and classroom interaction.

Course Details:

1. **Introduction to Ad-hoc Networks** **5hrs**
   1.1 Introduction: Definition, Characteristics, Features, Applications
   1.2 Ad-hoc Mobility Models: Indoor Model, Outdoor Model
   1.3 Wireless Channel: Characteristics
2. **Medium Access Protocol (MAC)** **10hrs**
   2.1 Issues in designing a MAC protocol for wireless ad-hoc networks, goals and classification
   2.2 Classification of MAC protocols, scheduling algorithms, protocols using directional antennas
   2.3 IEEE standards: 802.11a, 802.11b, 802.11g, 802.15, HIPERLAN
   2.4 Real Time MAC protocols, multichannel protocols, Power Aware MAC
3. **End to End Delivery and Security** **8hrs**
   3.1 Transport Layer: Issues in designing transport layer classification, ad-hoc transport protocol
   3.2 Security issues in ad-hoc networks: issues and challenges, network security attacks, secure routing protocols
   3.3 Ad-hoc wireless networks: LAN, WAN, MAN, PAN architecture and applications
4. **Routing Protocols** **8hrs**
   4.1 Routing Protocols: Design issues, goals and classifications
   4.2 Proactive vs Reactive Routing, Unicast Routing Algorithms
   4.3 Multicast Routing Algorithms, Hybrid Routing Algorithm, Energy Aware Routing, Hierarchical Routing, QoS Aware Routing
5. **Cross Layer Design** **8hrs**
   5.1 Need for cross layer design
   5.2 Cross layer optimization, parameter optimization techniques, cross layer cautionary perspective
   5.3 Integration of ad-hoc with mobile IP networks
6. **Networking Sensors and Applications** **8hrs**
   6.1 Networking Sensors: Unique features
   6.2 Deployment of ad-hoc/sensor networks
   6.3 Sensor network programming challenges
   6.4 Applications of ad-hoc/sensor networks and future directions, wireless fidelity system

Laboratory Work/Case Studies:

1. Create a sample wireless topology using a simulation tool.
2. Create a mobile ad-hoc network using a simulation tool.
3. Implement an Ad-hoc On-demand Distance Vector (AODV) protocol using a simulation tool.
4. Implement a Transmission Control Protocol (TCP) using a simulation tool.
5. Implement a User Datagram Protocol (UDP) using a simulation tool.
6. Implement a Low Energy Adaptive Clustering Hierarchy (LEACH) protocol using a simulation tool.

Textbook(s):

1. Toh C.K., *Ad-Hoc Mobile Wireless Networks – Protocols and Systems*, Prentice Hall.
2. Siva-RAM-Murthy, *Ad-Hoc Wireless Networks - Architectures and Protocols*, Addison-Wesley.

References:

1. Stefano Basagni, Marco Conti, Silvia Giordano, and Ivan Stojmenovic, *Mobile Ad-hoc Networking*, Wiley-IEEE Press, 2004.
2. Mohammad Ilyas, *The Handbook of Ad-hoc Wireless Networks*, CRC Press, 2002.
3. T. Camp, J. Boleng, and V. Davies, "A Survey of Mobility Models for Ad-hoc Networks."
4. Wireless Communications and Mobile Computing, *Special Issue on Mobile Ad-hoc Networking Research, Trends and Applications*, Vol. 2, No. 5, 2002, pp. 483–502.
5. Fekri M. Abduljalil and Shrikant K. Bodhe, *A Survey of Integrating IP Mobility Protocols and Mobile Ad-hoc Networks*, IEEE Communications Surveys and Tutorials, 2007.

Software Quality Assurance

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

- Provide a broad understanding of software testing and quality factors.
- Familiarize yourself with the software quality management components.
- Understand how SQA components can be incorporated into the project life-cycle.

Course Description:

This course focuses on techniques aimed at maintaining the quality of software. Quality assurance is considered an activity that runs through the whole development process from understanding the needs of clients to analyzing, documenting the requirements and verifying & validating the solutions through various testing approaches. The course also looks at quality assurance from a technical standpoint. Various aspects of software quality are covered, including tools, strategies and standards.

Course Outcomes:

- Utilize the concepts in software development life cycle with regard to testing.
- Plan and apply the concepts in preparing the quality plan & documents.
- Demonstrate the capability to adopt quality standards.
- Integrate different aspects of quality into the development of information systems.
- Consider established models and methods for the assessment of IT systems and software engineering processes.

Course Contents:

1. **Software Testing & Software Quality** **5 hours**
   - 1.1 Software errors, faults and failures
   - 1.2 Software testing
   - 1.3 Testing principles
   - 1.4 Software quality and Software quality assurance
   - 1.5 Psychology of testing
   - 1.6 Ethics in software testing
   - 1.7 Software testing as a career
2. **SQA Components in Project Life-cycle** **3 hours**
   - 2.1 Software Development Life-cycle models
   - 2.2 Levels of testing
   - 2.3 Types of testing
3. **Software Verification** **5 hours**
   - 3.1 Walkthroughs
   - 3.2 Inspections
   - 3.3 Audits
   - 3.4 Verification process
4. **Software Validation** **5 hours**
   - 4.1 Validation of software design
   - 4.2 Validation of product specifications
   - 4.3 Validation of software product
   - 4.4 Testing different types of software products
5. **Software Product Quality** **3 hours**
   - 5.1 Software disasters - causes and consequences
   - 5.2 Software reliability and improvement
6. **Software Process Quality** **3 hours**
   - 6.1 Process, Process quality
   - 6.2 Software development process and its components
   - 6.3 Process improvement and stabilization
7. **Test Techniques** **6 hours**
   - 7.1 Static testing and Dynamic testing
   - 7.2 Black-box – equivalence partitioning, boundary value analysis, decision table testing, state transition testing, use case testing
   - 7.3 White-box testing – statement testing and coverage, decision testing and coverage
   - 7.4 Experience-based testing – error guessing, exploratory testing, checklist-based testing

8. **Test Management** **7 hours**
   - 8.1 Test organization
   - 8.2 Test planning, estimating and measuring software quality
   - 8.3 Test monitoring, test control and documentation control
   - 8.4 Configuration management
   - 8.5 Risk management
   - 8.6 Defect management

9. **Testing Tools** **6 hours**
   - 9.1 Test tool considerations
   - 9.2 Test automation – benefits and risks
   - 9.3 Test execution and management tools
   - 9.4 Principles for tool selection
   - 9.5 Success factors for testing tools

10. **Quality Management Standards** **2 hours**
    - 10.1 Scope of quality management standards
    - 10.2 Total Quality Management (TQM), ISO, CMMI, Six Sigma, IEEE standards

Laboratory:

Laboratory exercises should be conducted based on the curriculum contents to evaluate and guide students to know about various tools, technologies and methods of software testing.

Textbooks:

- *Software Quality Assurance: From theory to implementation* by Daniel Galin [Pearson Education, 2004]
- *Mastering Software Quality Assurance: Best Practices, Tools and Techniques for Software Developers* by Murali Chemuturi [J.Ross Publishing, 2010]

References:

- *The Economics of Software Quality* by Capers Jones, Olivier Bonsignour [Pearson Education, 2012]
- *Software Testing and Quality Assurance: Theory and Practice* by Kshirasagar Naik, Priyadarshi Tripathy [Wiley, 2008]
- *Software Testing* by Ron Patton [Sams Publishing, 2001]

Remote Sensing

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of Remote Sensing to undergraduate students who wish to explore earth surface imaging remotely from space or airborne systems. Students will understand background concepts and principles of remote sensing. They will be able to collect, use, and interpret satellite images, and apply remote sensing concepts to real-world applications.

Course Description:

This course covers the fundamental concepts of remote sensing and is divided into six chapters: Introduction, Science of Remote Sensing, Sensors, Earth Observation Space Programs, Digital Image Processing, Interpretation and Analysis, and Remote Sensing Applications.

Course Outcomes:

- Evaluate and compare different earth observation systems and their data for desired purposes.
- Perform satellite image processing and accuracy assessment.
- Accomplish serious investigations for research as well as professional assignments.

Course Contents:

1. **Introduction** **5 hrs**
   Remote Sensing, Geographic Information Systems, Historical Overview,

Image Characteristics, GPS, Characteristics of Real Remote Sensing Systems, Applications of Remote Sensing.

2. **Science of Remote Sensing** **5 hrs**
   Radiation and Energy Sources, Electromagnetic Spectrum, Energy Interactions with Atmosphere and Earth Surface, Passive and Active Sensing, Optical Remote Sensing, Microwave Remote Sensing, Hyperspectral Remote Sensing.
3. **Sensors** **5 hrs**
   Sensors, Sensor Types and Characteristics, Scanning (Across Track vs. Along Track), Optical Mechanical Scanners (MSS, TM, LISS, WiFS, PAN), Resolutions (Spatial, Spectral, Temporal, Radiometric).
4. **Earth Observation Space Programs** **10 hrs**
   Satellite Constellations and Parameters, Orbits (GEO, MEO, LEO, SSO, GTO), Satellite Functions (Communication, Earth Observation, Navigation, Astronomical), Recent Satellite Missions and Constellations (Landsat, Sentinel, MODIS, WorldView) and Attributes (Sensor, Subsystem, Resolutions, FOV, Swath, Launch Years, etc.).
5. **Digital Image Processing, Interpretation and Analysis** **10 hrs**
   Image Interpretation, Interpretation Devices and Types, Interpretation Elements and Visual Keys, Digital Image Processing, Image Classification (Supervised vs. Unsupervised), Quantitative Analysis.
6. **Remote Sensing Applications** **10 hrs**
   Applications in Disaster Management (Earthquake, Flood, etc.), Urbanization and Settlement Mapping, Agriculture and Forestry, Land Use and Land Cover Monitoring.

Laboratory Work:

1. Earth Explorer and Map Composition
2. Introduction to Remote Sensing Images and Color Composition
3. Familiarization with Low Resolution Images
4. Familiarization with High Resolution Images
5. Introduction to Image Enhancement
6. Mathematical Operations: DN to Radiance Conversion
7. Indices Calculations: NDVI, NDBI, NDWI
8. Unsupervised Classification
9. Supervised Classification
10. Accuracy Assessment

Text Books and References:

1. Lillesand, T.M., and Kieffer, R.M. (2015): *Remote Sensing and Image Interpretation*, John Wiley.
2. Liu, J.-G., & Mason, P.J. (2016): *Image Processing and GIS for Remote Sensing: Techniques and Applications*, Wiley-Blackwell.
3. Campbell, J.B. (2002): *Introduction to Remote Sensing*, Taylor Publications.
4. Joseph, George (2003): *Fundamentals of Remote Sensing*, Universities Press.
5. Sabbins, F.F. (1985): *Remote Sensing: Principles and Interpretation*, W.H. Freeman and Company.

Journals:

1. *International Journal of Remote Sensing*, Taylor & Francis
2. *Remote Sensing of Environment*, Elsevier
3. *Remote Sensing*, MDPI
4. *IEEE Transactions on Geoscience and Remote Sensing*, IEEE

Data Science

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives

The course provides a solid foundation in the principles and techniques used in data science, along with hands-on experience using popular tools. Students will learn to clean and prepare data, perform analysis using statistical and machine learning methods, and visualize and communicate insights effectively.

Course Description

Data Science is an interdisciplinary field that applies statistical and computational techniques to extract insights from data. This course introduces essential tools and methods for data collection, cleaning, exploration, visualization, statistical analysis, and machine learning. Delivered through lectures, labs, and industry guest lectures, students will gain practical skills to solve real-world data problems.

Course Outcomes

Upon completion, students will be able to:

1. Understand key concepts, principles, and techniques of data science.
2. Collect, clean, and preprocess data for analysis.
3. Perform statistical analysis and build models using appropriate methods.
4. Apply machine learning algorithms for classification and regression.
5. Create and interpret data visualizations to communicate findings.

Course Content

1. **Introduction to Data Science** **[6 Hours]**

1.1 Overview of Data Science
1.2 Importance of Data science
1.3 Concept of Big Data
1.4 Skills and Tools required for Data Science
1.5 Ethical Considerations in Data Science: Privacy and Security in Data Science, Bias and Fairness in Data Science, Transparency and Accountability in Data Science
1.6 Applications of Data Science: Business Analytics, Social Media Analysis, Healthcare Analytics, Environmental Analytics, and Local use cases

2. **Statistical Analysis** **[10 Hours]**

2.1 Random Sampling and Sample Bias: Bias, Random Selection, Size Versus Quality, Sample Mean Versus Population Mean, Mean versus Median
2.2 Inferential Statistics: Central Limit Theorem, and Standard Error, Confidence Intervals, Normal Distribution, Student's t-Distribution, and Binomial Distribution
2.3 Hypothesis Tests
2.4 Statistical Significance
2.5 Regression and multi-variate analysis
2.6 Time Series Analysis
2.7 Conditional Probability: Bayes Theorem

3. **Data Collection and Preprocessing** **[10 hours]**

3.1 Data Sources: Reading Files, Scraping the Web, and Using APIs
3.2 Data Cleaning: Handling Missing Data, Noisy Data, and Incomplete Data
3.3 Data Wrangling: Join, Combine, and Reshape
3.4 Data Integration
3.5 Data Transformation
3.6 Handling Categorical Data

4. **Data Exploration and Visualization** **[10 hours]**

4.1 Exploratory Data Analysis (EDA): Philosophy of Exploratory Data Analysis, Estimates of Location, Estimates of Variability, Exploring the Data Distribution, Exploring Binary and Categorical Data
4.2 Data Visualization Techniques: Percentiles and Boxplots, Frequency Tables, Histograms, Estimates Line Plots, Bar Plots, Density Plots, and Scatter or Point Plots

5. **Introduction to Data Science Tools** **[9 hours]**

5.1 Basic Concept of NumPy and Pandas
5.2 Basic Concept of Matplotlib and Seaborn
5.3 Machine Learning with scikit-learn

Practical

Using Python or R tools students should practice enough on real-world data intensive problems to analyze and extract insights from data using effective data visualizations to communicate insights and findings.

References

1. Wes McKinney - Python for Data Analysis - Data Wrangling with pandas, NumPy, and Jupyter - O'Reilly Media (2022)
2. Joel Grus - Data Science from Scratch First Principles with Python, Second Edition - O'Reilly Media (2019)
3. Fawcett Tom, Provost Foster - Data science for business - O'Reilly (2013)
4. Peter Bruce, Andrew Bruce and Peter Gedeck - Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python - O'Reilly Media, 2020
5. O'Neil, Cathy, Schutt, Rachel - Doing data science - O'Reilly Media (2014)

Machine Learning

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

- To understand the key concepts, principles, and techniques of Machine Learning.
- To learn fundamental principles of Neural Networks.
- To apply machine learning algorithms to solve real-world problems.
- To evaluate machine learning models using appropriate metrics.
- To gain hands-on experience with machine learning tools and frameworks.

Course Details:

1. **Introduction to Machine Learning** **7hrs**
   - 1.1 Introduction: What and Why ML?
   - 1.2 Types of Learning: Supervised, Unsupervised, and Reinforced
   - 1.3 Examples of Machine Learning Applications
   - 1.4 Introduction to Pattern Recognition
   - 1.5 Terminologies: Features, Feature Vectors, Patterns, and Classifiers
   - 1.6 Pattern Classification System
   - 1.7 Examples of Pattern Recognition Applications
   - 1.8 Terminology: Data Augmentation, Class, Test dataset, and Training dataset
   - 1.9 Introduction to Deep Learning (DL): AI vs ML vs DL
2. **Artificial Neural Networks** **10hrs**
   - 2.1 Overview of Artificial Neural Networks
   - 2.2 Neural Network Topologies
   - 2.3 How to Choose an Activation Function: Sign, Bipolar, Tanh, Sigmoid, SoftMax, ReLU, and Leaky ReLU
   - 2.4 The Perceptron: Training a Perceptron and Perceptron Convergence Theorem
   - 2.5 Multilayer Perceptron and Backpropagation Algorithm
   - 2.6 Learning Rate, Cost Function, Gradient Descent, and Delta Rule

   i. Types of Gradient Descent
   ii. Challenges with Gradient Descent
   iii. Impact of Learning Rate

2.7 Regularization Techniques: Dropout, L1 and L2 Regularization

3. **Classification and Clustering** **10hrs**

3.1 Why Classification and Clustering
3.2 Decision Tree and K-NN
3.3 Linear Regression and Logistic Regression
3.4 Support Vector Machine: Hyperplane and Support Vectors
   i. Linear SVM
   ii. Non-linear SVM and Kernel Tricks
   iii. SVM for Multiclassification
3.5 Dimension Reduction Method: Principal Component Analysis (PCA)
3.6 Clustering: K-means and DBSCAN
3.7 Cluster Validation: Sum of Square Error, Silhouette Coefficient, and Dunn Index

4. **Estimating and Measures for Performance Evaluation** **10hrs**

4.1 Basic Concepts: Bias, Variance, Underfitting and Overfitting
4.2 Cross-Validation and Resampling Methods
   i. The Holdout Method
   ii. k-Fold Cross-Validation
   iii. Bootstrap
4.3 Ensemble Methods: Bagging, Boosting (AdaBoost and XGBoost), and Random Forest
4.4 Measuring Classifier Performance: Confusion Matrix
   i. Classification Accuracy and Misclassification Costs
   ii. Recall, Precision and F1 Score
   iii. Sensitivity and Specificity
4.5 Comparing Classifiers
   i. ROC Curve
   ii. The Area Under ROC Curve

5. **Introduction to Recommendation Systems** **8hrs**

5.1 Overview of Recommendation Systems
5.2 Types of Recommendation Systems
   i. Collaborative Filtering: User-Based and Item-Based
   ii. Content-Based Filtering
5.3 Hybrid Recommendation Systems
5.4 Evaluation Metrics for Recommendation Systems
5.5 Applications of Recommendation Systems

Practical:

Using Python or any other machine learning tools, students should practice extensively on real-world data-intensive problems. They are expected to implement machine learning models using popular machine learning libraries and frameworks such as Scikit-learn, TensorFlow, Keras, and PyTorch.

References:

1. Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer.
2. Simon Rogers, Mark Girolami, *A First Course in Machine Learning*, Chapman and Hall/CRC.
3. Andrew R. Webb, *Statistical Pattern Recognition*, Wiley.
4. Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective*, The MIT Press.
5. Francesco Ricci, Lior Rokach, Bracha Shapira (eds), *Recommender Systems Handbook*.
6. Kim Falk, *Practical Recommender Systems*.
7. Toby Segaran, *Programming Collective Intelligence*.

Social Media Mining

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of social media platforms, their data and applications. This course focuses on real skill development by implementing theoretical learning using any programming language. The students are expected to gain experience with both the theoretical and practical aspects of social media mining techniques regarding network and text mining. They will learn to collect social media data and design, implement, and evaluate applications that can generate new knowledge.

Course Description:

This course will cover the major aspects for mining and analyzing social media to discover interesting insights and useful knowledge. This course is divided into five chapters focusing on the basics, principles, and applications. The concepts of Graph and Text Mining Techniques are covered in detail. The first chapter focuses on Introduction, second chapter focuses on Graph Mining Techniques, third chapter focuses on Text Mining Techniques, fourth chapter focuses on Connectivity, Influence and Centralization, fifth chapter focuses on Network Evolution and the last chapter focuses on case studies. Throughout the course, a strong emphasis is on implementation and real world applications.

Course Outcomes:

- Programming proficiency
- Understand and use various types of social media data and APIs
- Examine connectivity in social networks
- Quantify the importance of a node in a network
- Apply natural language processing methods
- Understand and use social media data for real world analysis such as topic modeling, sentiment analysis, etc.

Course Contents

1. **Introduction** **5 Hours**
   - Web history and trends, WWW, W3C, Social Media and its importance, Semantic web, social media platforms and Social Networks, Social media data as Big Data, Data Mining, Social media mining, Social media mining techniques, uses and applications of Social Media, Potentials and Pitfalls of Social Media Data
2. **Graph Mining Techniques** **10 Hours**
   - Introduction, Nodes, Degree, Edges, path and cycles, graph representation, graph types and graph connectivity and Traversability
   - Trees and Forests, Special Subgraphs, Complete Graphs, Planar Graphs, Bipartite Graphs, Regular Graphs, Bridges
   - Graph Traversal, Shortest Path Algorithms, Minimum Spanning Trees, Network Flow Algorithms, Maximum Bipartite Matching, Bridge Detection
3. **Text Mining Techniques** **10 Hours**
   - Text Mining Basics, Predictive vs. Exploratory Analysis of Text, Text Mining Pipeline, Pre-processing, Feature Extraction, Feature Selection, Document Representation, Text Mining using Classification, Text Mining using Clustering, N gram analysis, Topic Modeling, POS tagging, Aspect detection, Name entity recognition, Word embeddings and large language models, Sentiment Analysis
4. **Connectivity, Influence and Centralization** **5 Hours**
   - Centrality measures, differences and similarities and uses, Applying network centrality measures for real world applications
5. **Network Evolution** **5 Hours**
   - Characteristics of real, large networks e.g. power law like degree distribution, high clustering, and small average shortest paths, Preferential Attachment Model, Small World Model, Link prediction and its measures
6. **Case Study with platforms like Twitter, Facebook, TripAdvisor 10 Hours**
   - OAuth and Social Media APIs, Trending Topics, user and content (e.g. retweet Relationships), Data collection methods (Streaming APIs, Search APIs, Social Graph API, etc), Follower/Friendship relationship, Measuring Similarity (user and content), Frequency analysis, Sentiment analysis, influential user identification

Laboratory Work:

All topics in chapter 6 must be implemented as lab work.

Text Books and References:

1. Graph Theory, 5th Edition by Reinhard Diestel / 2017, Springer
2. Reda Alhajj, Social Network Analysis and Mining, Springer
3. Nathan Danneman, Richard Heimann, "Social Media Mining with R", PACKT Publishing
4. Sharan Kumar Ravindran, Vikram Garg, "Mastering Social Media Mining with R", PACKT Publishing
5. Charu C. Aggarwal, Social Network Data Analytics
6. Reza Zafarani, Mohammad Ali Abbasi, Huan Liu, Social Media Mining, an Introduction
7. Rizwana Irfan, Christine K.King, et.al., A Survey on Text Mining in Social Networks, Cambridge University Press
8. Matthew A. Russell, "21 Recipes for Mining Twitter", O'Reilly

Geographic Information System (GIS)

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of Geographic Information Systems to undergraduate students. Students will understand background concepts, principles, and applications of GIS. They will be able to collect, use, and interpret GIS images, and apply GIS concepts to real-world applications.

Course Description:

This course covers the fundamental concepts of Geographic Information Systems, divided into seven chapters: Introduction, Digital Mapping and Cartography, GIS Data Structures, GIS Data Management, Vector Data Analysis, Raster Data Analysis, and GIS Trends and Applications.

Course Outcomes:

- Evaluate and compare different kinds of GIS data and maps.
- Develop digital maps.
- Work with maps on the web and develop Web GIS Applications.
- Develop applications based on spatial and non-spatial data.
- Accomplish serious investigations for research as well as professional assignments.

Course Details:

1. **Introduction** **5 hrs**
   Geographic Information Systems, Historical Overview, Components of

GIS, Functional Elements of GIS, Remote Sensing Data in GIS, Scope and Application of GIS.

2. **Digital Mapping and Cartography** **8 hrs**
   Maps, Map Elements, Map Layers, Map Scales and Representation, Coordinate Systems, Map Projection, Properties and Types of Map Projections, Cartographic Design, Reference Surface, Universal Transverse Mercator (UTM) Projection.
3. **GIS Data Structures** **8 hrs**
   Data Structure and Its Need, Vector Data Structure, Raster Data Structures, Spatial Indexing, Topology and Spatial Relationships, Triangulated Irregular Network (TIN), Data Compression Techniques, Lossy vs. Lossless Compression, Run-Length Coding, Huffman Coding, Lempel-Ziv-Welch (LZW) Compression, Tiling and Pyramid Structures.
4. **GIS Data Management** **6 hrs**
   DBMS, Database Queries, Spatial vs. Non-Spatial Queries, Geodatabases, Spatial Data Infrastructures, GIS Data Formats (GeoJSON, Shapefiles, GeoTIFF, SQLite, Geodatabase, KML), Data Integration and Interoperability.
5. **Spatial Data Analysis** **8 hrs**
   Raster Data Analysis: Reclassifying, Resampling, Map Algebra.
   Vector Data Analysis: Geometric Operations (Overlay, Buffer, Clip, Union, Intersection), Spatial Query and Selection (Point-in-Polygon, Distance-Based, Spatial Joins, Attribute-Based Queries), Network Analysis, Spatial Autocorrelation.
6. **GIS Trends and Applications** **10 hrs**
   FOOS4G, OSGeo, Mobile GIS, Cloud-Based GIS, Big Data and Geospatial Analytics, Open Data and Open-Source GIS.
   Applications in: Disaster Management, Urbanization and Settlement Mapping, Agriculture and Forestry, Health and Environment.

Laboratory Work:

1. Introduction to GIS software: ArcGIS / QGIS – demonstration of structure, tools, management, formats.
2. GIS data collection from various sources – GPS, Field Survey, Remote Sensing, etc.
3. Digitization and Map Making using tools like QGIS, ArcGIS.
4. Working with spatial and non-spatial data in databases.
5. Publish a web map (WMS).
6. Raster analysis example.

Text Books and References:

1. Paul A. Longley, Michael F. Goodchild, et al., *Geographic Information Science and Systems*, Fourth Edition, Wiley, 2015.
2. Chang, K.T., *Introduction to Geographic Information Systems*, Sixth Edition, McGraw Hill, New York.
3. Liu, J.-G., & Mason, P.J., *Image Processing and GIS for Remote Sensing: Techniques and Applications*, Wiley-Blackwell, 2016.
4. Ian Heywood, Sarah Cornelius, and Steve Carver, *An Introduction to Geographic Information Systems*, Pearson, London, 2012.
5. M. Neteler and H. Mitasova, *Open Source GIS: A GRASS Approach*, Third Edition, Springer, USA, 2009.

Online Resources and Journals:

1. ESRI Online Resources: https://www.esri.com/en-us/arcgis/products/arcgis-online/resources
2. *ISPRS International Journal of Geo-Information*, MDPI.
3. *International Journal of GIS*, Taylor & Francis.
4. *International Journal of Geoinformatics*, Association of Geoinformation Technology.

Geo Spatial Data Processing and Management

Program: Bachelor of Information Technology (BIT)
Course Code : ELE
Year: 7/8
Semester: VII/VIII
Credit Hours : 3
Nature: TH + PR

Course Objectives:

The purpose of the course is to provide fundamental knowledge of geospatial data, its processing and management to the undergraduate level students. Students will be able to collect, use and interpret geospatial data. Also, they will be able to apply GIS concepts for real world applications.

Course Description:

This course will cover the fundamental concepts of Geo Spatial Data Processing and Management. It is divided into five chapters: Introduction, Geospatial Data Acquisition, Geospatial Data Processing and Analysis, Geospatial Data Visualization and a Case Study i.e. Volunteered Geographic Information.

Course Outcomes:

1. Comprehend and communicate basic idea of geospatial data.
2. Comprehend, process and analyze geospatial data for real world applications.
3. Accomplish serious investigations for research as well as professional assignments.

Course Contents:

1. **Introduction to Geospatial Data** **(7 hours)**
   1.1 Data, Spatial data, Geospatial Data, types of Geospatial Data
   1.2 Geospatial Data characteristics
   1.3 Volunteered Geographic Information and its types

1.4 Spatial Social Networks
1.5 GeoPrivacy
1.6 Challenges and applications of Geospatial Data processing

2. **Geospatial Data Acquisition** **(10 hours)**

2.1 Data Sources (Remote sensing, GPS, Field Surveying, LiDAR, Crowd-Sourced Data, etc.)
2.2 Data Collection
2.3 Data Fusion and Integration
2.4 Data Preprocessing
2.5 Data Validation and Quality Assurance

3. **Geospatial Data Processing and Analysis** **(10 hours)**

3.1 Spatial statistics and analysis techniques
3.2 Geospatial Databases
3.3 Geospatial database models
3.4 Geospatial Database example (e.g. PostGIS)
3.5 Spatial Data Modeling and representation
3.6 Spatial indexing
3.7 Geospatial Query Language
3.8 Data Backup and storage
3.9 Spatial Data Exchange (GeoJSON, geopackages, Shapefiles)

4. **Geospatial Data Visualization** **(8 hours)**

4.1 Data visualization
4.2 Cartography
4.3 Choropleth maps
4.4 Dot density maps
4.5 Heatmaps
4.6 Cluster maps
4.7 Thematic maps
4.8 Map visualizations using Geospatial Query Language
4.9 Recent trends in Geospatial Data Visualization

5. **Case Study: Volunteered Geographic Information** **(10 hours)**

5.1 Volunteered Geographic Information (VGI)
5.2 User-Generated Content
5.3 VGI Data collection via Crowdsourcing
5.4 VGI Data Preprocessing and Quality Assessment
5.5 VGI Data Analysis
5.6 VGI Data Visualization
5.7 VGI Data integration with other Datasets
5.8 Examples of VGI Applications: Public Health and Disease Tracking or Tourism or Transportation and Mobility or related application

Laboratory Work:

1. Example of Geospatial Data Acquisition
2. Example of Geospatial Data Processing and Analysis
3. Example of Geospatial Data Visualization
4. Concepts in Chapter 5 are to be implemented as Laboratory Work

Textbooks and References:

1. De Smith, Michael John, Michael F. Goodchild, and Paul Longley (2007): *Geospatial Analysis: A Comprehensive Guide to Principles, Techniques and Software Tools*, Troubador Publishing Ltd.
2. Wenzhong Shi, Michael Goodchild, Brian Lees, Yee Leung (2012): *Advances in Geo-Spatial Information Science*, CRC Press, Boca Raton, FL.
3. Sui, Daniel, Sarah Elwood, and Michael Goodchild (2012): *Crowdsourcing Geographic Knowledge: Volunteered Geographic Information (VGI) in Theory and Practice*, Springer Science & Business Media.

Online and Journal Resources:

1. *IEEE Transactions on Geoscience and Remote Sensing*, IEEE
2. *ISPRS International Journal of Geo-Information*, MDPI
3. *International Journal of Geographic Information Science*, Taylor & Francis
4. *GeoJournal: Spatially Integrated Social Sciences and Humanities*, Springer